A Closed-Form Analytical Approximation for the Non-Magnetized Quasi-Thermal Noise Spectrum Using a Lorentzian Plasma Response

Matched-asymptotic region decomposition with closed-form evaluation and numerical validation

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Abstract

We present a closed-form analytical solution for the quasi-thermal noise (QTN) voltage power spectrum of dipole antennas in unmagnetized plasmas. Replacing the Maxwellian plasma dispersion function $Z(y)$ with a Lorentzian (Padé-type) approximation for the longitudinal dielectric response converts the QTN integral—previously requiring numerical quadrature for each (n_e, T_e, L, a)—into elementary expressions involving $\log(\cdot)$, $\arctan(\cdot)$, and $\operatorname{artanh}(\cdot)$. The method uses a matched-asymptotic three-region decomposition based on the antenna and wire-radius arguments kL and ka: Region I ($0 < y < y_{c,B}$) uses envelope averaging in the oscillatory regime, Region II ($y_{c,B} < y < y_c$) is a mixed transitional regime, and Region III ($y > y_c$) employs small-argument expansions with correct asymptotic decay. Smooth cutoffs follow from matching the small/large expansions of $F_1(kL)$ and $J_0^2(ka)$, giving $x_c \simeq 1.003$ and $x_{c,B} \simeq 0.337$. Ten basis integrals (I_1–I_{10}) are evaluated in closed form via partial fractions and explicitly real-valued primitives, including separate formulas for the resonant limit $r = (\omega_p/\omega)^2 \to 1$, and `log1p` reformulations mitigate high-frequency cancellation. For ionospheric conditions ($n_e = 5.77 \times 10^{11}\,\mathrm{m}^{-3}$, $T_e = 1700\,\mathrm{K}$), comparison to direct quadrature over 10^5 frequencies shows $< 1\%$ relative error across 10^6–10^8 Hz, except for a narrow $\sim 27.5\%$ spike at the plasma resonance $f_p \simeq 6.82$ MHz where $V^2(\omega)$ varies steeply; the solution reproduces the canonical plateau–peak–power-law spectrum and inverse-length scaling $V^2 \propto 1/L$. Runtime is reduced from seconds to microseconds (a speedup $\mathcal{O}(10^5)$–$\mathcal{O}(10^6)$), enabling real-time inversion on CubeSat-class processors and rapid (n_e, T_e, L, a) sweeps, while the region split clarifies the contributing fluctuation scales. Extensions to magnetized plasmas, non-Maxwellian distributions, and three-dimensional antenna geometries are discussed.

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1 Introduction

1.1 Motivation

Quasi-thermal noise (QTN) spectroscopy has proven to be a powerful passive diagnostic technique for measuring electron density (n_e) and temperature (T_e) in space plasmas. First developed by Meyer-Vernet [1] and successfully deployed on missions including ISEE-3, Ulysses, and Cassini [2–6], the method exploits the fact that an antenna immersed in a plasma naturally detects voltage fluctuations due to thermal Langmuir waves. The measured voltage power spectrum exhibits a characteristic peak near the plasma frequency $\omega_p = \sqrt{n_e e^2/\varepsilon_0 m_e}$, whose amplitude and shape encode information about both n_e and T_e.

More recently, Maj and Cairns [7] demonstrated the feasibility of applying QTN spectroscopy to CubeSat missions in Earth's ionosphere. Their numerical calculations showed that antennas with lengths $L \sim 0.1$–0.4 m - typical of CubeSat platforms - can detect peak voltages in the range 1–100 μV under ionospheric conditions ($n_e \sim 5 \times 10^{11}$ m^{-3}, $T_e \sim 1700$ K at 300 km altitude). Such measurements would provide valuable in situ validation of empirical models like the International Reference Ionosphere (IRI) [8], particularly over oceanic regions where ground-based observations are sparse.

However, the standard theoretical framework for QTN involves the plasma dispersion function $Z(y)$ [9], which appears in the imaginary part of the longitudinal dielectric response for a Maxwellian plasma:

$$K_L(\omega, k) = 1 + \frac{\omega_p^2}{k^2 v_T^2}\left[1 + yZ(y)\right], \quad y = \frac{\omega}{\sqrt{2} k v_T}, \tag{1}$$

where

$$Z(y) = \frac{1}{\sqrt{\pi}} \int_{-\infty}^{\infty} \frac{e^{-t^2}}{t-y}\, dt. \tag{2}$$

The resulting QTN voltage spectrum,

$$V^2(\omega) = \frac{16 k_B T_e}{\pi^2 \varepsilon_0 v_T} \int_0^{\infty} \frac{1}{y^2} F_1(kL) J_0^2(ka)\, \mathrm{Im}\left(\frac{1}{K_L}\right) dy, \tag{3}$$

must be evaluated numerically for each choice of plasma and antenna parameters, a computationally expensive process that hinders rapid mission planning, parameter sweeps, and real-time data inversion.

Following a suggestion by Cairns, we replace the Maxwellian plasma response with a Lorentzian (or Padé) approximation. This substitution yields a rational polynomial form for $\mathrm{Im}(1/K_L^t)$ that, when combined with systematic asymptotic expansions of the antenna response function $F_1(kL)$ and the Bessel factor $J_0^2(ka)$, permits fully analytical evaluation of the QTN integral (3) for the first time. The resulting closed-form expressions enable instant spectrum prediction, direct insight into how n_e, T_e, antenna length L, and radius a control spectral features, and a robust foundation for inverse problems - fitting observed spectra to extract plasma parameters.

Our analytical approach decomposes the integral into three regions based on matched asymptotic cutoffs, evaluates a complete set of basis integrals in closed form (handling special cases including the resonant limit $\omega \to \omega_p$), and implements numerically stable logarithmic transformations to prevent catastrophic cancellation at high frequencies. Validation against both direct numerical quadrature and the reference ionospheric spectra of Maj and Cairns [7] confirms accuracy across the full frequency range of interest, with relative errors below 1% except in the immediate vicinity of the resonance peak where the spectrum varies most rapidly.

1.2 Scope and assumptions

This work derives closed-form analytical approximations for the quasi-thermal noise spectrum under the following assumptions:

Plasma model:

- Unmagnetized plasma. We neglect the ambient magnetic field and associated wave modes (e.g., upper hybrid, Bernstein waves). This approximation is justified when the electron cyclotron frequency Ω_e is small compared to the plasma frequency, $\Omega_e \ll \omega_p$. For typical ionospheric conditions at 300 km altitude ($B \sim 20$–$60\ \mu$T, $n_e \sim 5 \times 10^{11}$ m^{-3}), we have $\Omega_e/2\pi \sim 0.6$–1.7 MHz and $\omega_p/2\pi \sim 6$–14 MHz, so the correction to the Langmuir wave frequency is $\lesssim 15\%$ [7]. Extension to magnetized plasmas is deferred to future work.
- Lorentzian velocity distribution. We adopt a Lorentzian (kappa-like) model for the electron velocity distribution, leading to a rational polynomial approximation for the longitudinal dielectric response. This replaces the standard Maxwellian treatment involving the plasma dispersion function $Z(y)$ and is exact for a Lorentzian plasma while serving as a Padé approximant to the Maxwellian case [10]. The Lorentzian form captures the essential physics near the plasma frequency - where the QTN peak occurs - while enabling analytical tractability.
- Collisionless plasma. We assume the mean free path $\lambda_{\mathrm{mfp}} \gg \lambda_D$, valid for altitudes $\gtrsim 200$ km in the ionosphere. For altitudes $\lesssim 200$ km, the additional effect of collisional damping must be included [11].
- Electrostatic approximation. We retain only the longitudinal (electrostatic) component of plasma fluctuations, valid when $k\lambda_D \gtrsim 1$ and $\omega \sim \omega_p$.

Antenna geometry:

- Wire dipole configuration. The antenna consists of two thin cylindrical conductors of length L and radius a arranged in a dipole. We assume $a \ll \lambda_D$ (thin-wire limit) and $L > \lambda_D$ so that the plasma sheath does not dominate the antenna impedance. For monopole configurations, we adopt the general approximation that the response scales as half the dipole value.
- Triangular current distribution. The current is assumed to vary linearly from the spacecraft to the antenna tips, valid when $a \ll \lambda_D$ and the antenna is shorter than the plasma wavelength at low frequencies.
- Small radius-to-length ratio. We require $a/L \ll 1$ throughout. Typical values are $a \sim 2 \times 10^{-4}$ m and $L \sim 0.1$–0.5 m, giving $a/L \sim 10^{-3}$–10^{-2}.

Asymptotic validity: Our analytical approximations exploit the fact that the dimensionless arguments $x_F = kL$ and $x_B = ka$ span both small and large regimes across the frequency range of interest ($f \sim 10^5$–10^8 Hz for ionospheric conditions). We construct matched asymptotic expansions with smooth transitions at empirically determined cutoffs, ensuring controlled accuracy in all regimes. The resulting expressions are validated against direct numerical quadrature of the full integral.

Limitations: The present analysis does not include:

- Shot noise due to discrete particle impacts (treated separately in Maj and Cairns [7]);
- Antenna sheath and spacecraft potential effects (important for monopoles);
- Receiver load impedance and capacitance matching;
- Non-thermal features such as beams or loss-cone distributions.

These effects are important for complete spectral interpretation but do not affect the intrinsic thermal Langmuir wave contribution that is our focus.

1.3 Contributions and roadmap

The main contributions of this paper are:

1. First closed-form analytical solution for the QTN spectrum. By adopting a Lorentzian plasma response and systematic asymptotic analysis, we derive explicit expressions for $V^2(\omega)$ that replace costly numerical quadrature with algebraic evaluation. This reduces computation time from seconds (per spectrum) to microseconds and enables parameter sweeps over mission design space.
2. Matched-asymptotic region decomposition. We rigorously define three integration regions (Region I: $0 < y < y_{c,B}$; Region II: $y_{c,B} < y < y_c$; Region III: $y > y_c$) based on the arguments kL and ka of the antenna and Bessel functions. Matching conditions determine cutoffs $x_c \approx 1.003$ and $x_{c,B} \approx 0.337$ that ensure smooth, convergent transitions between small- and large-argument asymptotic forms.
3. Complete basis integral library. We evaluate ten basis integrals (I_1 through I_{10}) in closed form, covering all powers of y that appear after asymptotic truncation. Special-case formulas for the resonance limit $r = (\omega_p/\omega)^2 \to 1$ are derived separately. All results are expressed in terms of elementary functions (logarithms, arctangents, inverse hyperbolic tangents).
4. Numerically stable implementations. For high-frequency evaluation (where $y_c \gg 1$), direct application of the closed forms suffers catastrophic cancellation. We provide algebraically reorganized expressions using `log1p` transformations that preserve accuracy to machine precision across the full parameter range.
5. Validation and ionospheric application. Comparison with numerical quadrature and the reference spectra of Maj and Cairns [7] confirms relative errors $\lesssim 1\%$ except near the resonance peak (where the spectrum is intrinsically steep). We reproduce key results including optimal antenna lengths ($L \sim 0.2$–0.4 m) and detectable voltage levels ($V_{\mathrm{pk}} \sim 1$–100 μV) for CubeSat missions.

Roadmap: The remainder of this paper is organized as follows. Section §2 states the theoretical framework, introducing the starting QTN integral and defining all key functions (F_1, J_0, K_L^t). Crucially, subsection §2.2 derives the Lorentzian form of $\mathrm{Im}(1/K_L^t)$ and contrasts it with the Maxwellian case, establishing the rational polynomial structure that enables our analytical approach. Section §3 develops small- and large-argument expansions for $F_1(kL)$ and $J_0^2(ka)$ and derives the matching cutoffs. Subsection §3.4 decomposes the QTN integral into three regions and presents the truncated integrands in each. Section §4 outlines the evaluation strategy, including factorization of the denominator and standard integral templates; detailed derivations of all basis integrals are relegated to Appendix §B. Section §5 discusses numerical implementation, stability techniques, and validation against quadrature and reference data. Section §6 presents the comparative analytical vs numerical plot and error assessment for interpretation. Section §7 interprets the physical meaning of the three regions, discusses limitations, and suggests extensions (magnetization, non-Maxwellian distributions). Section §8 summarizes our findings and their implications for mission design and plasma diagnostics.

2 Theoretical framework

2.1 Starting expression for the QTN spectrum

The voltage power spectrum of quasi-thermal noise on a wire dipole antenna immersed in an isotropic, unmagnetized plasma is given by [2]

$$V^2(\omega) = \frac{16 k_B T_e}{\pi^2 \varepsilon_0 v_T} \int_0^\infty \frac{1}{y^2} F_1(kL) J_0^2(ka)\, \mathrm{Im}\left(\frac{1}{K_L}\right) dy \tag{4}$$

where k_B is Boltzmann's constant, T_e the electron temperature, ε_0 the permittivity of free space, $v_T = \sqrt{k_B T_e / m_e}$ the electron thermal speed, and

$$y = \frac{\omega}{\sqrt{2} k v_T} \tag{5}$$

is the dimensionless frequency-to-wavenumber ratio. Here k is the plasma wave number, L is the length of each dipole arm, a is the antenna radius, J_0 is the zeroth-order Bessel function of the first kind, and $K_L(\omega, k)$ is the longitudinal part of the plasma dielectric tensor.

The function $F_1(kL)$ describes the antenna's response to the surrounding plasma and depends on the assumed current distribution along the antenna arms. For a triangular (linear) current distribution, valid when $a \ll \lambda_D$ and the antenna is electrically short at low frequencies, F_1 is given by [2, 12]

$$F_1(x) = \frac{x\left[\mathrm{Si}(x) - \frac{1}{2}\mathrm{Si}(2x)\right] - 2\sin^4(x/2)}{x^2} \tag{6}$$

where $\mathrm{Si}(x) = \int_0^x (\sin t / t)\, dt$ is the sine integral and $x = kL$ is the dimensionless antenna argument. The factor $J_0^2(ka)$ accounts for the radial structure of the electric field around the cylindrical antenna elements, with ka the dimensionless radius argument.

Equation (4) assumes purely electrostatic (longitudinal) fluctuations, $L > \lambda_D$ (so the plasma sheath does not dominate the impedance), and $a < \lambda_D$ (thin-wire limit). The integral is performed over all wave numbers k (equivalently, over all y) that contribute to the detected signal.

2.2 Longitudinal dielectric response

2.2.1 Maxwellian case

For a Maxwellian electron velocity distribution, the longitudinal dielectric function is [9]

$$K_L(\omega, k) = 1 + \frac{\omega_p^2}{k^2 v_T^2}\left[1 + y Z(y)\right] \tag{7}$$

where $\omega_p = \sqrt{n_e e^2 / \varepsilon_0 m_e}$ is the electron plasma frequency and $Z(y)$ is the plasma dispersion function,

$$Z(y) = \frac{1}{\sqrt{\pi}} \int_{-\infty}^{\infty} \frac{e^{-t^2}}{t - y}\, dt \tag{8}$$

Although $Z(y)$ can be evaluated numerically via special-function libraries, its presence in the denominator of the QTN integrand necessitates numerical quadrature of (4) for each choice of parameters (ω, n_e, T_e, L, a). This computational expense motivates the search for an analytically tractable alternative.

2.2.2 Lorentzian approximation

We now adopt a Lorentzian model for the electron velocity distribution, which yields a rational polynomial form for K_L^t that is exact for Lorentzian plasmas and also serves as a Padé-type approximant to the Maxwellian case. We define the corresponding longitudinal response as

$$K_L^t(y) = 1 + \alpha\,[1 + y\,z_t(y)] \tag{9}$$

where the superscript t denotes the Lorentzian (Padé/truncated) model response used throughout, distinguishing it from the Maxwellian dielectric function K_L in (7). Here

$$\alpha = \frac{\omega_p^2}{k^2 v_T^2}, \qquad z_t(y) = -\frac{y + \sqrt{2}\,i}{(y + i/\sqrt{2})^2} \tag{10}$$

and we introduce the dimensionless parameter

$$r \equiv \left(\frac{\omega_p}{\omega}\right)^2. \tag{11}$$

Using the relation $\alpha = \omega_p^2/(k^2 v_T^2) = 2y^2 r$ and straightforward algebra, we can express K_L^t as a ratio of complex polynomials. Writing $K_L^t = (A + iB)/(C + iD)$ with

$$A = y^2 - \frac{1}{2} - \frac{\alpha}{2}, \quad B = \sqrt{2}\,y, \tag{12}$$

$$C = y^2 - \frac{1}{2}, \quad D = \sqrt{2}\,y \tag{13}$$

the imaginary part of the reciprocal is

$$\mathrm{Im}\left(\frac{1}{K_L^t}\right) = \frac{AD - BC}{A^2 + B^2} \tag{14}$$

Direct calculation shows that $AD - BC = -(\sqrt{2}/2)\alpha y$ and $A^2 + B^2 = y^4 + (1-\alpha)y^2 + (1+\alpha)^2/4$. Substituting $\alpha = 2y^2 r$ yields our key result:

$$\mathrm{Im}\left(\frac{1}{K_L^t}\right) = -\frac{\sqrt{2}\,r\,y^3}{(1-r)^2 y^4 + (1+r)y^2 + \frac{1}{4}} \tag{15}$$

This rational polynomial form is the foundation of our analytical approach. Unlike the Maxwellian case, expression (15) is elementary and depends only on the dimensionless ratio $r = (\omega_p/\omega)^2$ and the integration variable y. The denominator is a quartic polynomial in y with real coefficients, enabling systematic application of partial-fraction techniques and closed-form integration.

2.3 Antenna and Bessel factors: definitions

The antenna response function $F_1(kL)$ and the Bessel factor $J_0^2(ka)$ exhibit distinct asymptotic behaviors in small- and large-argument limits. To exploit these systematically, we introduce dimensionless arguments

$$x_F = kL, \quad x_B = ka \tag{16}$$

and express them in terms of the integration variable y via (5):

$$k = \frac{\omega}{\sqrt{2}\,y\,v_T} \quad \Rightarrow \quad x_F = \frac{\omega L}{\sqrt{2}\,y\,v_T}, \quad x_B = \frac{\omega a}{\sqrt{2}\,y\,v_T} \tag{17}$$

Thus, for fixed ω, L, a, and T_e, the arguments x_F and x_B decrease as y increases:

$$x_F, x_B \propto \frac{1}{y} \tag{18}$$

At fixed ω, small y corresponds to large k (hence large x_F, x_B), while large y corresponds to small k (hence small x_F, x_B). Therefore the integrand transitions from the large-argument regime at small y to the small-argument regime at large y. In summary, the QTN integral (4) with the Lorentzian response (15) becomes

$$V^2(\omega) = \frac{16 k_B T_e}{\pi^2 \varepsilon_0 v_T} \int_0^\infty \frac{1}{y^2} F_1\left(\frac{\omega L}{\sqrt{2}\, y\, v_T}\right) J_0^2\left(\frac{\omega a}{\sqrt{2}\, y\, v_T}\right) \left[-\frac{\sqrt{2}\, r\, y^3}{(1-r)^2 y^4 + (1+r) y^2 + \frac{1}{4}}\right] dy \tag{19}$$

The task is now to evaluate this integral analytically by exploiting the asymptotic forms of F_1 and J_0^2 in different y-regimes, which we address in the following sections.

3 Asymptotic expansions and matching

3.1 Antenna response asymptotics

The antenna response function $F_1(x)$ defined in (6) exhibits different behaviors depending on whether the dimensionless argument $x = kL$ is small or large. We develop asymptotic expansions in each regime and determine a cutoff x_c where the two representations match optimally.

3.1.1 Small-argument expansion of $F_1(x)$

For $|x| \ll 1$, we expand the sine integral using $\mathrm{Si}(x) = x - x^3/18 + O(x^5)$ and $\sin(x/2) \approx x/2 - x^3/48 + O(x^5)$. The trigonometric identity

$$\sin^4\left(\frac{x}{2}\right) = \frac{3 - 4\cos x + \cos 2x}{8} \tag{20}$$

yields the power series

$$\sin^4\left(\frac{x}{2}\right) = \frac{x^4}{16} - \frac{x^6}{96} + O(x^8) \tag{21}$$

Substituting these into (6) and simplifying, we find

$$x\left[\mathrm{Si}(x) - \frac{1}{2}\mathrm{Si}(2x)\right] = \frac{x^4}{6} - \frac{x^6}{40} + O(x^8) \tag{22}$$

$$-2\sin^4\left(\frac{x}{2}\right) = -\frac{x^4}{8} + \frac{x^6}{48} + O(x^8) \tag{23}$$

so the numerator becomes

$$x\left[\mathrm{Si}(x) - \frac{1}{2}\mathrm{Si}(2x)\right] - 2\sin^4\left(\frac{x}{2}\right) = \frac{x^4}{24} - \frac{x^6}{240} + O(x^8) \tag{24}$$

Dividing by x^2 yields the small-argument expansion:

$$F_1(x) = \frac{x^2}{24} - \frac{x^4}{240} + \frac{x^6}{4480} + O(x^8), \quad |x| \ll 1 \tag{25}$$

Retaining terms through $O(x^6)$ provides accuracy better than 1% for $x \lesssim 1$.

3.1.2 Large-argument expansion of $F_1(x)$

For $|x| \gg 1$, the sine integral has the asymptotic expansion

$$\mathrm{Si}(x) = \frac{\pi}{2} - \frac{\cos x}{x} - \frac{\sin x}{x^2} + O\left(\frac{1}{x^3}\right) \tag{26}$$

and similarly

$$\mathrm{Si}(2x) = \frac{\pi}{2} - \frac{\cos(2x)}{2x} - \frac{\sin(2x)}{4x^2} + O\left(\frac{1}{x^3}\right) \tag{27}$$

Substituting these, along with the identity for $\sin^4(x/2)$, into (6) and collecting terms yields

$$F_1(x) = \frac{\pi}{4x} - \frac{\cos x}{x^2} - \frac{\pi \cos 2x}{8x} + \frac{\sin 2x}{8x^2} - \frac{2\sin^4(x/2)}{x^2} + O\left(\frac{1}{x^3}\right), \quad |x| \gg 1. \tag{28}$$

For integration purposes, we envelope-average the oscillatory terms. The time-averaged value of $\sin^4(x/2)$ over many oscillations is $3/8$, and the rapidly oscillating cosine and sine terms average to zero:

$$\langle \cos x \rangle = \langle \sin x \rangle = \langle \cos 2x \rangle = \langle \sin 2x \rangle = 0 \tag{29}$$

This yields the envelope approximation

$$\langle F_1(x) \rangle = \frac{\pi}{4x} - \frac{3}{4x^2}, \quad |x| \gg 1 \tag{30}$$

where $\langle \cdot \rangle$ denotes averaging over oscillations. This form is used in Region I (Section §3.5) where rapid oscillations in both F_1 and J_0^2 justify the envelope approximation.

3.1.3 Uniform considerations and error terms

The small-argument expansion (25) has truncation error $O(x^8)$, while the large-argument envelope (30) has error $O(x^{-3})$. The transition between these regimes will be determined by a matching cutoff x_c that will be defined in Section §3.3, which establishes the boundary between Regions I/II (large x_F) and Region III (small x_F). The matched asymptotic approach ensures that errors remain well below 1% across the full frequency range relevant for ionospheric QTN spectra.

3.2 Bessel function asymptotics

The Bessel factor $J_0^2(ka)$ in the QTN integrand (4) requires separate asymptotic treatment. Like $F_1(x)$, the function $J_0(x)$ has distinct small- and large-argument behaviors, with the additional complication that $J_0(x)$ oscillates for large x.

3.2.1 Small-argument expansion of $J_0(x)$ and $J_0^2(x)$

For $|x| \ll 1$, the zeroth-order Bessel function has the power-series expansion

$$J_0(x) = 1 - \frac{x^2}{4} + \frac{x^4}{64} - \frac{x^6}{2304} + O(x^8) \tag{31}$$

Squaring this expression and collecting terms yields

$$J_0^2(x) = 1 - \frac{x^2}{2} + \frac{3x^4}{32} + O(x^6), \quad |x| \ll 1 \tag{32}$$

This expansion is accurate to better than 1% for $x \lesssim 0.5$ and is used in Regions II and III where $x_B = ka$ is small.

3.2.2 Large-argument (Debye) expansion and envelope

For $|x| \gg 1$, the Bessel function exhibits oscillatory behavior described by the Debye asymptotic expansion:

$$J_0(x) \sim \sqrt{\frac{2}{\pi x}} \cos\left(x - \frac{\pi}{4}\right) \left[1 - \frac{1}{8x} + O(x^{-2})\right], \quad |x| \gg 1 \tag{33}$$

Squaring and retaining terms through $O(x^{-2})$, we obtain

$$J_0^2(x) \sim \frac{2}{\pi x} \cos^2\left(x - \frac{\pi}{4}\right) \left[1 - \frac{1}{4x} + O(x^{-2})\right] \tag{34}$$

The squared cosine can be expressed using the double-angle identity:

$$\cos^2\left(x - \frac{\pi}{4}\right) = \frac{1}{2}\left[1 + \cos\left(2x - \frac{\pi}{2}\right)\right] = \frac{1}{2}\left[1 + \sin(2x)\right] \tag{35}$$

Substituting into (34) gives

$$J_0^2(x) \sim \frac{1}{\pi x}\left[1 + \sin(2x)\right]\left[1 - \frac{1}{4x}\right] + O(x^{-3}), \quad |x| \gg 1 \tag{36}$$

Expanding the product and retaining terms through $O(x^{-2})$:

$$J_0^2(x) \approx \frac{1}{\pi x} + \frac{\sin(2x)}{\pi x} - \frac{1}{4\pi x^2} - \frac{\sin(2x)}{4\pi x^2} \tag{37}$$

3.2.3 Phase definition and averaging

In Region I, where both $x_F = kL > x_c$ and $x_B = ka > x_{c,B}$, the integrand contains the product $F_1(kL)J_0^2(ka)$, which involves rapidly oscillating terms in both functions. The oscillations have different frequencies (proportional to kL and ka) and different phases.

For integration over a range of k (equivalently y), these rapid oscillations average out. We adopt the envelope approximation, replacing the oscillatory factors by their time-averaged values. For $\sin(2x)$ and $\cos(2x)$, the average over many periods is zero:

$$\langle \sin(2x) \rangle = \langle \cos(2x) \rangle = 0 \tag{38}$$

Retaining only the leading Debye term in (34) and using $\langle \cos^2(\cdot) \rangle = 1/2$, the large-argument envelope used for Region I and cutoff matching is

$$\langle J_0^2(x) \rangle = \frac{1}{\pi x}, \quad |x| \gg 1 \text{ (matched at } x = x_{c,B}) \tag{39}$$

The Bessel cutoff $x_{c,B}$ will be defined in Section §3.3 by matching the small- and large-argument forms.

3.3 Cutoff definitions and matching conditions

We now consolidate the cutoff values that partition the y-integration domain and establish their ordering. This determines the three-region decomposition of the QTN integral.

3.3.1 Antenna cutoff x_c and equation for matching

The antenna cutoff x_c is determined by equating the small-argument expansion (25) to the large-argument envelope (30). Retaining the leading terms from each expansion, the matching condition is

$$\frac{x_c^2}{24} - \frac{x_c^4}{240} = \frac{\pi}{4x_c} - \frac{3}{4x_c^2} \tag{40}$$

Multiplying through by $240x_c^2$ yields the sextic polynomial equation

$$x_c^6 - 10x_c^4 + 60\pi x_c - 180 = 0 \tag{41}$$

The unique positive real root of (41) defines the antenna cutoff:

$$x_c = 1.00326820 \tag{42}$$

At this value, both expansions (25) and (30) agree to within $\sim 0.5\%$, ensuring a smooth transition between the small- and large-argument regimes for F_1.

3.3.2 Bessel cutoff $x_{c,B}$ and matching

Similarly, the Bessel cutoff $x_{c,B}$ is found by equating the small-argument expansion (32) to the large-argument envelope (39). The matching condition is

$$1 - \frac{x_{c,B}^2}{2} + \frac{3x_{c,B}^4}{32} = \frac{1}{\pi x_{c,B}} \tag{43}$$

Multiplying through by $32\pi x_{c,B}$ yields the quintic polynomial equation

$$3\pi x_{c,B}^5 - 16\pi x_{c,B}^3 + 32\pi x_{c,B} - 32 = 0 \tag{44}$$

The unique positive real root of (44) defines the Bessel cutoff:

$$x_{c,B} = 0.33704639 \tag{45}$$

3.3.3 Translation to y-space: y_c and $y_{c,B}$

The cutoffs x_c and $x_{c,B}$ are defined in terms of the dimensionless arguments $x_F = kL$ and $x_B = ka$. To partition the integration domain in (19), we must express these cutoffs in terms of the integration variable y.

Recall from (17) that

$$x_F = kL = \frac{\omega L}{\sqrt{2}\, y\, v_T}, \quad x_B = ka = \frac{\omega a}{\sqrt{2}\, y\, v_T} \tag{46}$$

The antenna cutoff x_c corresponds to $y = y_c$ when $x_F = x_c$:

$$x_c = \frac{\omega L}{\sqrt{2}\, y_c\, v_T} \quad \Rightarrow \quad y_c = \frac{\omega L}{\sqrt{2}\, v_T\, x_c} \tag{47}$$

Similarly, the Bessel cutoff $x_{c,B}$ corresponds to $y = y_{c,B}$ when $x_B = x_{c,B}$:

$$x_{c,B} = \frac{\omega a}{\sqrt{2}\, y_{c,B}\, v_T} \quad \Rightarrow \quad y_{c,B} = \frac{\omega a}{\sqrt{2}\, v_T\, x_{c,B}} \tag{48}$$

Ordering of cutoffs. Using (47)–(48), the ratio is

$$\frac{y_{c,B}}{y_c} = \frac{a}{L}\,\frac{x_c}{x_{c,B}} \tag{49}$$

For thin antennas $a \ll L$, this ratio is $\ll 1$, hence $0 < y_{c,B} < y_c$. This ordering establishes the three integration regions:

$$\text{Region I:}\quad 0 < y < y_{c,B} \qquad (x_F > x_c,\ x_B > x_{c,B}), \tag{50}$$
$$\text{Region II:}\quad y_{c,B} < y < y_c \qquad (x_F > x_c,\ x_B < x_{c,B}), \tag{51}$$
$$\text{Region III:}\quad y > y_c \qquad (x_F < x_c,\ x_B < x_{c,B}). \tag{52}$$

In Region I, both F_1 and J_0^2 use large-argument (envelope) forms (30) and (39). In Region II, F_1 uses the large-argument form while J_0^2 uses the small-argument expansion (32). In Region III, both functions use small-argument expansions (25) and (32). The total QTN spectrum is then

$$V^2(\omega) = V_{\mathrm{I}}^2(\omega) + V_{\mathrm{II}}^2(\omega) + V_{\mathrm{III}}^2(\omega), \tag{53}$$

where each contribution is evaluated analytically using the appropriate asymptotic forms, as detailed in Region I (Section §3.5), Region II (Section §3.6), and Region III (Section §3.7).

Numerical example. For typical ionospheric parameters at 300 km altitude ($n_e = 5.77\times10^{11}$ m^{-3}, $T_e = 1700$ K) and antenna dimensions ($L = 0.3$ m, $a = 2\times10^{-4}$ m), we have $v_T = \sqrt{k_B T_e/m_e} \approx 1.61\times10^5$ m/s. At a frequency $f = 10$ MHz ($\omega = 2\pi\times10^7$ rad/s), the cutoffs are:

$$y_c = \frac{2\pi\times10^7\times0.3}{\sqrt{2}\times1.61\times10^5\times1.003} \approx 82.5, \tag{54}$$
$$y_{c,B} = \frac{2\pi\times10^7\times2\times10^{-4}}{\sqrt{2}\times1.61\times10^5\times0.337} \approx 0.164. \tag{55}$$

The large separation $y_c/y_{c,B} \approx 504$ reflects the small ratio a/L and ensures that Region II spans a substantial range in y-space.

3.4 Overview of the three-region split

Using the Lorentzian response (15), the QTN integrand in (19) can be written in the compact real form

$$\mathcal{I}(y) := \frac{1}{y^2}\,F_1(x_F)\,J_0^2(x_B)\,\Im\!\left(\frac{1}{K_L^t}\right) = -\sqrt{2}\,r\,\frac{y}{\mathcal{D}(y;r)}\,F_1(x_F)\,J_0^2(x_B), \tag{56}$$
$$\mathcal{D}(y;r) := (1-r)^2y^4 + (1+r)y^2 + \frac{1}{4}, \qquad r := \left(\frac{\omega_p}{\omega}\right)^2. \tag{57}$$

The antenna and Bessel arguments are inversely proportional to y:

$$x_F = kL = \frac{\omega L}{\sqrt{2}\,y\,v_T}, \qquad x_B = ka = \frac{\omega a}{\sqrt{2}\,y\,v_T}, \tag{58}$$

so increasing y drives the integrand from the regime where the large-argument asymptotic forms are accurate ($x_F > x_c$, $x_B > x_{c,B}$) to the regime where small-argument Taylor expansions apply ($x_F < x_c$, $x_B < x_{c,B}$).

To obtain a uniformly accurate closed-form approximation, we partition the y-integration domain using the matched-asymptotic cutoffs introduced in Section §3. Specifically, we define y_c and $y_{c,B}$ by the conditions $x_F = x_c$ and $x_B = x_{c,B}$:

$$y_c = \frac{\omega L}{\sqrt{2}\,v_T\,x_c}, \qquad y_{c,B} = \frac{\omega a}{\sqrt{2}\,v_T\,x_{c,B}}, \qquad 0 < y_{c,B} < y_c, \tag{59}$$

which yields the three-region split

$$\begin{aligned}
&\text{Region I:} && 0 < y < y_{c,B}, && (x_F > x_c,\ x_B > x_{c,B}), && (60)\\
&\text{Region II:} && y_{c,B} < y < y_c, && (x_F > x_c,\ x_B < x_{c,B}), && (61)\\
&\text{Region III:} && y > y_c, && (x_F < x_c,\ x_B < x_{c,B}). && (62)
\end{aligned}$$

The rationale is that in each region the product $F_1(x_F)J_0^2(x_B)$ admits a simple asymptotic representation that converts (56) into a rational function of y (up to rapidly oscillatory factors in Region I, which are treated by envelope averaging). Concretely:

- **Region I ($x_F > x_c$, $x_B > x_{c,B}$):** both F_1 and J_0^2 are rapidly oscillatory in k; we replace them by their large-argument (Debye/envelope) forms, and average over oscillations to obtain a smooth "envelope" integrand suitable for analytic integration.
- **Region II ($x_F > x_c$, $x_B < x_{c,B}$):** F_1 remains in its large-argument regime while J_0^2 is expanded as a power series in $x_B \propto 1/y$; the resulting numerator contains a finite set of mixed powers of y that can be integrated in closed form.
- **Region III ($x_F < x_c$, $x_B < x_{c,B}$):** both factors are expanded for small arguments, giving an explicitly decaying high-y tail. We retain only the leading and next-to-leading inverse-power contributions required to control the convergence of the improper integral and to match smoothly at $y = y_c$.

Accordingly, the spectrum is decomposed as

$$V^2(\omega) = \frac{16k_BT_e}{\pi^2\varepsilon_0 v_T}\left[\int_0^{y_{c,B}} \mathcal{I}_{\mathrm{I}}(y)\,dy + \int_{y_{c,B}}^{y_c} \mathcal{I}_{\mathrm{II}}(y)\,dy + \int_{y_c}^{\infty} \mathcal{I}_{\mathrm{III}}(y)\,dy\right], \tag{63}$$

where $\mathcal{I}_{\mathrm{I}}$, $\mathcal{I}_{\mathrm{II}}$, and $\mathcal{I}_{\mathrm{III}}$ denote the region-appropriate truncated forms of (56). The following subsections state these truncated integrands explicitly (prior to evaluation) and summarize the validity assumptions in each region.

3.5 Region I: $0 < y < y_{c,B}$ ($kL > x_c$, $ka > x_{c,B}$)

3.5.1 Assumptions and validity

Region I is defined by

$$0 < y < y_{c,B}, \qquad y_{c,B} := \frac{\omega a}{\sqrt{2}\,v_T\,x_{c,B}}, \tag{64}$$

so that

$$x_B = ka = \frac{\omega a}{\sqrt{2}\,y\,v_T} > x_{c,B} \approx 0.337 \quad \text{throughout Region I.} \tag{65}$$

Since typically $a \ll L$, the antenna argument

$$x_F = kL = \frac{\omega L}{\sqrt{2}\,y\,v_T} \tag{66}$$

also satisfies $x_F > x_c$ on the same interval, hence both $F_1(kL)$ and $J_0^2(ka)$ use their envelope asymptotic forms (matched at x_c and $x_{c,B}$ respectively). The Lorentzian longitudinal response is retained through the exact (real) denominator

$$\mathcal{D}(y;r) := (1-r)^2y^4 + (1+r)y^2 + \tfrac{1}{4}, \qquad r := \left(\frac{\omega_p}{\omega}\right)^2 > 0. \tag{67}$$

3.5.2 Asymptotic forms used

For $kL > x_c$ we use the envelope asymptotic form (derived for large arguments but matched at $x_c \approx 1$ to ensure continuity):

$$F_1^{(\mathrm{large})}(kL) \simeq \frac{\pi}{4(kL)} - \frac{3}{4(kL)^2} = B_1\, y - B_2\, y^2, \tag{68}$$

with

$$B_1 := \frac{\pi\sqrt{2}\, v_T}{4\omega L}, \qquad B_2 := \frac{3v_T^2}{2\omega^2 L^2}. \tag{69}$$

For $ka > x_{c,B}$ we use the Debye envelope form (matched at $x_{c,B} \approx 0.337$):

$$J_0^2(ka) \simeq \frac{2}{\pi(ka)}\, \cos^2\Phi(y) = E_B\, y\, \cos^2\Phi(y), \tag{70}$$

where

$$\Phi(y) := \frac{\omega a}{\sqrt{2}\, y\, v_T} - \frac{\pi}{4}, \qquad E_B := \frac{2\sqrt{2}\, v_T}{\pi\omega a}. \tag{71}$$

3.5.3 Envelope approximation and justification

In Region I, $ka > x_{c,B}$, and for ka sufficiently above the cutoff the phase $\Phi(y)$ oscillates rapidly with y compared with the slowly varying amplitude. We therefore replace the fast oscillations by their mean value

$$\cos^2\Phi(y) \;\longrightarrow\; \left\langle \cos^2\Phi \right\rangle = \frac{1}{2}, \tag{72}$$

which yields the Region I envelope integrand

$$\begin{aligned}\mathcal{I}_{\mathrm{I}}^{(\mathrm{env})}(y) &:= -\sqrt{2}\, r\, \frac{y}{\mathcal{D}(y;r)} \Big(B_1 y - B_2 y^2\Big) \Big(E_B y\Big) \left\langle \cos^2\Phi \right\rangle \\ &= -\, r\, \frac{2v_T}{\pi\omega a}\, \frac{B_1 y^3 - B_2 y^4}{\mathcal{D}(y;r)}, \qquad 0 < y < y_{c,B}.\end{aligned} \tag{73}$$

3.5.4 Integral representation for Region I

The Region I envelope contribution (inner integral only) is

$$I_{\mathrm{I}}^{(\mathrm{env})}(\omega) := \int_0^{y_{c,B}} \mathcal{I}_{\mathrm{I}}^{(\mathrm{env})}(y)\, dy. \tag{74}$$

For subsequent analytic evaluation, write

$$\mathcal{D}(y;r) = Ey^4 + Fy^2 + G, \qquad E = (1-r)^2, \quad F = 1 + r, \quad G = \tfrac{1}{4}, \tag{75}$$

and define the two Region I basis integrals

$$I_1^{(\mathrm{I})}(r; y_{c,B}) := \int_0^{y_{c,B}} \frac{y^3}{Ey^4 + Fy^2 + G}\, dy, \tag{76}$$

$$I_2^{(\mathrm{I})}(r; y_{c,B}) := \int_0^{y_{c,B}} \frac{y^4}{Ey^4 + Fy^2 + G}\, dy. \tag{77}$$

Then

$$I_{\mathrm{I}}^{(\mathrm{env})}(\omega) = -\, r\, \frac{2v_T}{\pi\, \omega a} \Big[B_1\, I_1^{(\mathrm{I})}(r; y_{c,B}) - B_2\, I_2^{(\mathrm{I})}(r; y_{c,B})\Big]. \tag{78}$$

3.6 Region II: $y_{c,B} < y < y_c$ ($kL > x_c$, $ka < x_{c,B}$)

3.6.1 Assumptions and validity

Region II is the mixed asymptotic regime defined by the ordering

$$y_{c,B} < y < y_c, \tag{79}$$

so that

$$x_F = kL = \frac{\omega L}{\sqrt{2}\, y\, v_T} > x_c \sim O(1), \qquad x_B = ka = \frac{\omega a}{\sqrt{2}\, y\, v_T} < x_{c,B} \sim O(1). \tag{80}$$

Thus $F_1(kL)$ must be treated in its *large-argument* regime while $J_0^2(ka)$ is simultaneously in its *small-argument* regime. The Region II approximation is constructed by substituting these two asymptotic forms into the full integrand

$$\mathcal{I}(y) = \frac{1}{y^2} F_1(kL)\, J_0^2(ka)\, \Im\left(\frac{1}{K_L^t}\right) = -\sqrt{2}\, r\, \frac{y}{\mathcal{D}(y;r)} F_1(kL)\, J_0^2(ka), \tag{81}$$

where

$$\mathcal{D}(y;r) := (1-r)^2 y^4 + (1+r) y^2 + \tfrac{1}{4}, \qquad r := \left(\frac{\omega_p}{\omega}\right)^2. \tag{82}$$

3.6.2 Asymptotic forms used

For $kL > x_c \approx 1.003$ we use the envelope asymptotic form of the antenna response (the large-argument series, matched at x_c to ensure continuity with the small-argument expansion),

$$F_1^{(\text{large})}(kL) = \frac{\pi}{4(kL)} - \frac{3}{4(kL)^2} + O\Big((kL)^{-3}\Big) = B_1\, y - B_2\, y^2 + O(y^3), \tag{83}$$

with

$$B_1 := \frac{\pi\sqrt{2}\, v_T}{4\omega L}, \qquad B_2 := \frac{3 v_T^2}{2\omega^2 L^2}. \tag{84}$$

For $ka < x_{c,B} \approx 0.337$ we use the small-argument series for the squared Bessel factor,

$$J_0^{2,(\text{small})}(ka) = 1 - \frac{(ka)^2}{2} + \frac{3(ka)^4}{32} + O\Big((ka)^6\Big) = 1 - \frac{C_2}{y^2} + \frac{C_4}{y^4} + O\Big(\frac{1}{y^6}\Big), \tag{85}$$

with

$$C_2 := \frac{\omega^2 a^2}{4 v_T^2}, \qquad C_4 := \frac{3\omega^4 a^4}{128 v_T^4}. \tag{86}$$

3.6.3 Integral representation for Region II

Substituting $F_1^{(\text{large})}$ and $J_0^{2,(\text{small})}$ into $\mathcal{I}(y)$ gives the Region II integrand

$$\mathcal{I}_{\text{II}}(y) := -\sqrt{2}\, r\, \frac{y}{\mathcal{D}(y;r)} \Big(B_1 y - B_2 y^2\Big) \left(1 - \frac{C_2}{y^2} + \frac{C_4}{y^4}\right), \tag{87}$$

Multiplying out and retaining all terms through $O(y^{-2})$ (discarding only $O(y^{-3})$ and beyond) yields the truncated numerator

$$\mathcal{N}_{\text{II}}(y) := -B_2 y^3 + B_1 y^2 + B_2 C_2\, y - B_1 C_2 - \frac{B_2 C_4}{y} + \frac{B_1 C_4}{y^2}, \tag{88}$$

so that

$$\mathcal{I}_{\text{II}}^{(\text{env})}(\omega)(y) \approx -\sqrt{2}\, r\, \frac{\mathcal{N}_{\text{II}}(y)}{\mathcal{D}(y;r)}, \qquad y_{c,B} < y < y_c. \tag{89}$$

The Region II contribution (inner integral only) is then

$$I_{\mathrm{II}}^{(\mathrm{env})}(\omega) := \int_{y_{c,B}}^{y_c} \mathcal{I}_{\mathrm{II}}^{(\mathrm{env})}(y)\, dy. \tag{90}$$

For later analytic evaluation it is convenient to write

$$\mathcal{D}(y;r) = Ey^4 + Fy^2 + G, \qquad E = (1-r)^2, \quad F = 1+r, \quad G = \tfrac{1}{4}, \tag{91}$$

and define the six basic integrals

$$I_5^{(\mathrm{II})}(r; y_{c,B}, y_c) := \int_{y_{c,B}}^{y_c} \frac{y^3}{Ey^4 + Fy^2 + G}\, dy, \tag{92}$$

$$I_6^{(\mathrm{II})}(r; y_{c,B}, y_c) := \int_{y_{c,B}}^{y_c} \frac{y^2}{Ey^4 + Fy^2 + G}\, dy, \tag{93}$$

$$I_7^{(\mathrm{II})}(r; y_{c,B}, y_c) := \int_{y_{c,B}}^{y_c} \frac{y}{Ey^4 + Fy^2 + G}\, dy, \tag{94}$$

$$I_8^{(\mathrm{II})}(r; y_{c,B}, y_c) := \int_{y_{c,B}}^{y_c} \frac{1}{Ey^4 + Fy^2 + G}\, dy, \tag{95}$$

$$I_9^{(\mathrm{II})}(r; y_{c,B}, y_c) := \int_{y_{c,B}}^{y_c} \frac{1}{y\,(Ey^4 + Fy^2 + G)}\, dy, \tag{96}$$

$$I_{10}^{(\mathrm{II})}(r; y_{c,B}, y_c) := \int_{y_{c,B}}^{y_c} \frac{1}{y^2(Ey^4 + Fy^2 + G)}\, dy. \tag{97}$$

With these definitions,

$$I_{\mathrm{II}}^{(\mathrm{env})}(\omega) \approx -\sqrt{2}\, r \Big[-B_2 I_5^{(\mathrm{II})} + B_1 I_6^{(\mathrm{II})} + B_2 C_2 I_7^{(\mathrm{II})} - B_1 C_2 I_8^{(\mathrm{II})} - B_2 C_4 I_9^{(\mathrm{II})} + B_1 C_4 I_{10}^{(\mathrm{II})} \Big], \tag{98}$$

which is the form used for the subsequent closed-form evaluation of the Region II envelope integral.

3.7 Region III: $y > y_c$ ($kL < x_c$, $ka < x_{c,B}$)

3.7.1 Assumptions and validity

Region III is defined by

$$y > y_c, \tag{99}$$

where y_c is chosen such that $x_F = kL$ satisfies $x_F \le x_c \sim O(1)$. Using

$$x_F = kL = \frac{\omega L}{\sqrt{2}\, y\, v_T}, \qquad x_B = ka = \frac{\omega a}{\sqrt{2}\, y\, v_T}, \tag{100}$$

the condition $y > y_c$ implies $x_F < x_c \approx 1.003$ throughout Region III, so $F_1(kL)$ uses its small-argument Taylor expansion. Moreover, since $y_c > y_{c,B}$ (for $a \ll L$), we also have $x_B < x_{c,B}$ throughout Region III; for typical antenna geometries with $a/L \sim 10^{-3}$, this gives $x_B \ll 1$, so the small-argument expansion for $J_0^2(ka)$ is highly accurate. The longitudinal denominator is written globally as

$$\mathcal{D}(y;r) := (1-r)^2 y^4 + (1+r) y^2 + \tfrac{1}{4}, \qquad r = \left(\frac{\omega_p}{\omega}\right)^2 > 0, \tag{101}$$

so $\mathcal{D}(y;r)$ is real and strictly positive for all $y > 0$ (with $(1-r)^2 = 0$ only at $\omega = \omega_p$).

3.7.2 Asymptotic forms used

We use the small-argument expansions for both the antenna response and the Bessel factor, written directly as inverse-power series in y:

$$F_1^{(\mathrm{small})}(kL) = \frac{D_2}{y^2} - \frac{D_4}{y^4} + \frac{D_6}{y^6} + O\Big(y^{-8}\Big), \tag{102}$$

$$J_0^{2,(\mathrm{small})}(ka) = 1 - \frac{C_2}{y^2} + \frac{C_4}{y^4} + O\Big(y^{-6}\Big), \tag{103}$$

with coefficients

$$D_2 := \frac{\omega^2 L^2}{48\, v_T^2}, \qquad D_4 := \frac{\omega^4 L^4}{960\, v_T^4}, \qquad D_6 := \frac{\omega^6 L^6}{35840\, v_T^6}, \tag{104}$$

$$C_2 := \frac{\omega^2 a^2}{4\, v_T^2}, \qquad C_4 := \frac{3\omega^4 a^4}{128\, v_T^4}. \tag{105}$$

3.7.3 High-y tail control and truncation strategy

Multiplying the two small-argument factors gives an inverse-power expansion:

$$\begin{aligned} F_1^{(\mathrm{small})}(kL)\, J_0^{2,(\mathrm{small})}(ka) &= \left(\frac{D_2}{y^2} - \frac{D_4}{y^4} + \frac{D_6}{y^6}\right)\left(1 - \frac{C_2}{y^2} + \frac{C_4}{y^4}\right) \\ &= \frac{D_2}{y^2} - \frac{D_4 + C_2 D_2}{y^4} + O\Big(y^{-6}\Big). \end{aligned} \tag{106}$$

Since the full integrand carries an additional prefactor $y/\mathcal{D}(y;r)$, this product induces a numerator of the form

$$\frac{y}{\mathcal{D}(y;r)}\, F_1^{(\mathrm{small})}(kL)\, J_0^{2,(\mathrm{small})}(ka) = \frac{1}{\mathcal{D}(y;r)}\left(\frac{D_2}{y} - \frac{D_4 + C_2 D_2}{y^3} + O\Big(y^{-5}\Big)\right). \tag{107}$$

To preserve the correct large-y decay and guarantee convergence while keeping the algebra minimal, we retain the leading $1/y$ and next-to-leading $1/y^3$ terms and discard only $O(y^{-5})$ and smaller contributions.

3.7.4 Integral representation for Region III

With the above truncation, the Region III integrand is

$$\mathcal{I}_{\mathrm{III}}(y) \approx -\sqrt{2}\, r\, \frac{\dfrac{D_2}{y} - \dfrac{D_4 + C_2 D_2}{y^3}}{\mathcal{D}(y;r)}, \qquad y > y_c. \tag{108}$$

We define the Region III contribution (inner integral only) by

$$I_{\mathrm{III}}(\omega) := \int_{y_c}^{\infty} \mathcal{I}_{\mathrm{III}}(y)\, dy. \tag{109}$$

For subsequent analytic evaluation it is convenient to introduce the two basis integrals

$$I_3^{(\mathrm{III})}(y_c;r) := \int_{y_c}^{\infty} \frac{1}{y(Ey^4 + Fy^2 + G)}\, dy, \tag{110}$$

$$I_4^{(\mathrm{III})}(y_c;r) := \int_{y_c}^{\infty} \frac{1}{y^3(Ey^4 + Fy^2 + G)}\, dy \tag{111}$$

Then the truncated Region III contribution can be written compactly as

$$I_{\mathrm{III}}^{(\mathrm{env})}(\omega) \approx -\sqrt{2}\, r\left[D_2\, I_3^{(\mathrm{III})}(y_c;r) - (D_4 + C_2 D_2)\, I_4^{(\mathrm{III})}(y_c;r)\right], \tag{112}$$

with D_2, D_4, C_2 defined above.

4 Analytical Evaluation Approach

Our aim is to reduce each regional contribution to a finite library of closed-form *definite* integrals. After inserting the Region I envelope and the Region II/III small-argument series, every integrand becomes a linear combination of rational terms in y built on the same quartic denominator

$$\mathcal{D}(y;r) = E\,y^4 + F\,y^2 + G, \tag{113}$$

with the QTN coefficients

$$E = (1-r)^2, \qquad F = 1+r, \qquad G = \tfrac{1}{4}, \tag{114}$$

and $r = (\omega_p/\omega)^2 > 0$.

4.1 Factorisation of the longitudinal denominator

Introduce the quadratic variable

$$u = y^2, \qquad du = 2y\,dy, \tag{115}$$

so that

$$\mathcal{D}(y;r) = E\,u^2 + F\,u + G. \tag{116}$$

Discriminant and root structure. The discriminant of the quadratic in u is

$$\Delta_u := F^2 - 4EG = (1+r)^2 - 4(1-r)^2\left(\tfrac{1}{4}\right) = 4r > 0, \tag{117}$$

so $Eu^2 + Fu + G$ has two distinct real roots for $r \neq 1$ (i.e. $E \neq 0$):

$$u_{1,2} = \frac{-F \pm \sqrt{\Delta_u}}{2E} = \frac{-(1+r) \pm 2\sqrt{r}}{2(1-r)^2}, \qquad (r > 0,\ r \neq 1). \tag{118}$$

For $r > 0$ one has $u_1 < 0$ and $u_2 < 0$, hence $y^2 - u_j = y^2 + |u_j| > 0$ for all $y > 0$. This guarantees that the logarithms arising from $\ln(y^2 - u_j)$ remain *real-valued* on the physical integration domain.

The factorisation is therefore

$$Eu^2 + Fu + G = E\,(u-u_1)(u-u_2). \tag{119}$$

It is often convenient to introduce positive scales

$$a_j := \sqrt{-u_j} > 0, \qquad j = 1, 2, \tag{120}$$

so that $(u - u_j) = (y^2 + a_j^2)$ and arctangent primitives appear directly.

Degenerate resonant case $r = 1$. When $r = 1$, $E = (1-r)^2 = 0$ and the denominator collapses to a linear function of u:

$$\mathcal{D}(y;1) = 2y^2 + \tfrac{1}{4} = 2u + \tfrac{1}{4}, \tag{121}$$

so Region II/III integrals are treated separately (still elementary, but not via the quadratic-factor route).

4.2 Standard integral templates, substitutions, and real-valued forms

(i) Universal substitution $u = y^2$. This converts quartic denominators into quadratics and removes odd powers of y:

$$\int \frac{y}{Ey^4 + Fy^2 + G}\, dy = \frac{1}{2} \int \frac{1}{Eu^2 + Fu + G}\, du, \tag{122}$$

$$\int \frac{y^3}{Ey^4 + Fy^2 + G}\, dy = \frac{1}{2} \int \frac{u}{Eu^2 + Fu + G}\, du. \tag{123}$$

(ii) Partial fractions in u (log-ratio primitive). From (119),

$$\frac{1}{E(u-u_1)(u-u_2)} = \frac{1}{E(u_1-u_2)} \left(\frac{1}{u-u_1} - \frac{1}{u-u_2} \right), \tag{124}$$

which yields the canonical real primitive

$$\int \frac{1}{Eu^2 + Fu + G}\, du = \frac{1}{E(u_1-u_2)} \ln\left(\frac{u-u_1}{u-u_2} \right) + C = \frac{1}{2\sqrt{r}} \ln\left(\frac{u-u_1}{u-u_2} \right) + C, \tag{125}$$

using $E(u_1 - u_2) = 2\sqrt{r}$. Returning to y gives

$$\int \frac{y}{Ey^4 + Fy^2 + G}\, dy = \frac{1}{4\sqrt{r}} \ln\left(\frac{y^2-u_1}{y^2-u_2} \right) + C, \qquad (r > 0,\ r \neq 1), \tag{126}$$

and the argument of the logarithm is positive for all $y > 0$ since $u_{1,2} < 0$.

(iii) "Discriminant sign" in standard quadratic tables: why arctan **becomes** artanh**.** Many handbooks present $\int du/(au^2 + bu + c)$ in an arctan form when $4ac - b^2 > 0$. Here,

$$4EG - F^2 = 4(1-r)^2 \left(\tfrac{1}{4}\right) - (1+r)^2 = -4r < 0, \tag{127}$$

so the *arctangent-table form would introduce an imaginary scale* ($\sqrt{4EG - F^2} = 2i\sqrt{r}$), producing complex intermediate expressions. Although these complex forms simplify to real results (because the physical integral is real), they can trigger numerical warnings and spurious complex round-off in implementation. We therefore *choose real-valued primitives from the outset* using either (a) the log-ratio form (125) or (b) an explicitly real artanh form (next paragraph).

(iv) Log-ratio $\leftrightarrow$ artanh **compression (real and stable).** Define the linear combination

$$N(y) := 2(1-r)^2 y^2 + (1+r), \tag{128}$$

for which one verifies

$$\frac{y^2-u_1}{y^2-u_2} = \frac{N(y) - 2\sqrt{r}}{N(y) + 2\sqrt{r}}. \tag{129}$$

Using $\operatorname{artanh}(x) = \frac{1}{2}\ln(\frac{1+x}{1-x})$, this gives the equivalent real primitive

$$\int \frac{y}{Ey^4 + Fy^2 + G}\, dy = -\frac{1}{2\sqrt{r}}\, \operatorname{artanh}\left(\frac{2\sqrt{r}}{N(y)} \right) + C, \qquad (r > 0,\ r \neq 1). \tag{130}$$

This form is particularly useful for plotting and numerical evaluation because it avoids subtracting nearly equal logarithms when $N(y) \gg 2\sqrt{r}$.

(v) Arctangent primitives from negative roots (manifestly real). Since $u_j < 0$, we may also decompose into $(y^2 + a_j^2)$ factors and use

$$\int \frac{1}{y^2 + a^2}\, dy = \frac{1}{a} \arctan\left(\frac{y}{a}\right) + C, \qquad a > 0, \tag{131}$$

so many Region II definite integrals reduce to differences of $\arctan(y/a_j)$ evaluated at $y_{c,B}$ and y_c.

(vi) Explicit identity for arctan $\leftrightarrow$ artanh **when complex arguments appear.** If, at any stage, an arctan expression appears with an imaginary argument (as in the table-based route implied by (127)), we use

$$\arctan(-ix) = -\, i \,\operatorname{artanh}(x), \tag{132}$$

to restore a real-valued representation. This is a purely algebraic rewrite: the underlying real integral is unchanged, but the resulting formula is numerically well-behaved and avoids introducing complex intermediate values.

(vii) "Why we changed forms for plotting": preventing spurious complex values and cancellation. In exact arithmetic, the closed forms are real for $r > 0$ and $y > 0$. In floating arithmetic, however, two issues commonly occur:

1. **Complex intermediates from the wrong discriminant branch:** using an arctan-table formula when $4EG - F^2 < 0$ introduces $i\sqrt{r}$, and numerical evaluation may propagate tiny imaginary round-off that contaminates log–log plots (and can trigger warnings about ignoring imaginary parts).
2. **Catastrophic cancellation in log differences:** expressions such as $\ln(\cdot) - 4\ln y$ or $\ln(A) - \ln(B)$ can subtract nearly equal large numbers when y (or y_c) is large, destroying precision. Rewriting into a single $\ln(\cdot)$ or into artanh via (130) preserves realness and improves stability.

(viii) Implementation-stable log/artanh **evaluation.** When $\xi := 2\sqrt{r}/N(y)$ is small (typical at large y), the stable evaluation is

$$\operatorname{artanh}(\xi) = \frac{1}{2} \ln\left(\frac{1+\xi}{1-\xi}\right) = \frac{1}{2} \ln\left(1 + \frac{2\xi}{1-\xi}\right), \tag{133}$$

so one should evaluate the final logarithm as $\log(1 + \cdot)$ (i.e. `log1p`) when $\frac{2\xi}{1-\xi} \ll 1$. This is the key step that keeps the Region III tail formulas accurate at high frequency (large y_c) and prevents numerical artifacts in log–log plots.

4.3 Region-by-region Evaluation

4.3.1 Region I: envelope integrals I_1, I_2

With $\mathcal{D}(y; r) = (1 - r)^2 y^4 + (1 + r)y^2 + \frac{1}{4}$, the Region I envelope integrand is

$$\mathcal{I}_{\mathrm{I}}^{(\mathrm{env})}(y) = -\, r\, \frac{2v_T}{\pi \omega a}\, \frac{B_1 y^3 - B_2 y^4}{\mathcal{D}(y; r)}, \qquad 0 < y < y_{c,B}. \tag{134}$$

Hence

$$I_{\mathrm{I}}^{(\mathrm{env})}(\omega) = \int_0^{y_{c,B}} \mathcal{I}_{\mathrm{I}}^{(\mathrm{env})}(y)\, dy = -\, r\, \frac{2v_T}{\pi \omega a} \left[B_1\, I_1^{(\mathrm{I})}(r; y_{c,B}) - B_2\, I_2^{(\mathrm{I})}(r; y_{c,B}) \right], \tag{135}$$

where

$$I_1^{(\mathrm{I})}(r; y_{c,B}) := \int_0^{y_{c,B}} \frac{y^3}{\mathcal{D}(y; r)}\, dy, \qquad I_2^{(\mathrm{I})}(r; y_{c,B}) := \int_0^{y_{c,B}} \frac{y^4}{\mathcal{D}(y; r)}\, dy. \tag{136}$$

Closed forms. **Case $r \neq 1$:**

$$I_1^{(\mathrm{I})}(y_{c,B}; r \neq 1) = \frac{1}{4(1-r)^2}\Bigg[\ln\Big|\, 4(1-r)^2 y_{c,B}^4 + 4(1+r) y_{c,B}^2 + 1 \,\Big| + \frac{1+r}{\sqrt{r}}\ \mathrm{artanh}\big(\xi(r, y_{c,B})\big)\Bigg], \tag{137}$$

$$\xi(r, y_{c,B}) = -\frac{4\sqrt{r}\, y_{c,B}^2}{2(r+1) y_{c,B}^2 + 1}. \tag{138}$$

Case $r = 1$:

$$I_1^{(\mathrm{I})}(y_{c,B}; r = 1) = \frac{y_{c,B}^2}{4} - \frac{1}{32}\ln\big(8 y_{c,B}^2 + 1\big). \tag{139}$$

Case $r \neq 1$:

$$\begin{aligned} I_2^{(\mathrm{I})}(y_{c,B}; r \neq 1) = \frac{1}{(1-r)^2}\Bigg[& y_{c,B} \\ & + \frac{(\sqrt{r}-1)^2}{8\sqrt{r}(\sqrt{r}+1)^2\sqrt{-\lambda_1(r)}}\arctan\left(\frac{y_{c,B}}{\sqrt{-\lambda_1(r)}}\right) \\ & - \frac{(\sqrt{r}+1)^2}{8\sqrt{r}(\sqrt{r}-1)^2\sqrt{-\lambda_2(r)}}\arctan\left(\frac{y_{c,B}}{\sqrt{-\lambda_2(r)}}\right)\Bigg], \end{aligned} \tag{140}$$

with

$$\lambda_{1,2}(r) = \frac{-(1+r) \pm 2\sqrt{r}}{2(1-r)^2}. \tag{141}$$

Case $r = 1$:

$$I_2^{(\mathrm{I})}(y_{c,B}; r = 1) = \frac{y_{c,B}^3}{6} - \frac{y_{c,B}}{16} + \frac{1}{16\sqrt{8}}\arctan\big(\sqrt{8}\, y_{c,B}\big). \tag{142}$$

4.3.2 Region II: mixed-power integrals and truncation

The Region II truncated envelope integrand may be written

$$\mathcal{I}_{\mathrm{II}}^{(\mathrm{env})}(y) \approx -\sqrt{2}\, r\, \frac{-B_2 y^3 + B_1 y^2 + B_2 C_2\, y - B_1 C_2 - \dfrac{B_2 C_4}{y} + \dfrac{B_1 C_4}{y^2}}{\mathcal{D}(y; r)}, \qquad y_{c,B} < y < y_c, \tag{143}$$

so that

$$I_{\mathrm{II}}^{(\mathrm{env})}(\omega) = \int_{y_{c,B}}^{y_c} \mathcal{I}_{\mathrm{II}}^{(\mathrm{env})}(y)\, dy \approx -\sqrt{2}\, r\, \Big[-B_2 I_5 + B_1 I_6 + B_2 C_2 I_7 - B_1 C_2 I_8 - B_2 C_4 I_9 + B_1 C_4 I_{10}\Big], \tag{144}$$

with basis integrals

$$\begin{aligned} I_5 &:= \int_{y_{c,B}}^{y_c} \frac{y^3}{\mathcal{D}(y;r)}\, dy, & I_6 &:= \int_{y_{c,B}}^{y_c} \frac{y^2}{\mathcal{D}(y;r)}\, dy, \\ I_7 &:= \int_{y_{c,B}}^{y_c} \frac{y}{\mathcal{D}(y;r)}\, dy, & I_8 &:= \int_{y_{c,B}}^{y_c} \frac{1}{\mathcal{D}(y;r)}\, dy, \\ I_9 &:= \int_{y_{c,B}}^{y_c} \frac{1}{y\,\mathcal{D}(y;r)}\, dy, & I_{10} &:= \int_{y_{c,B}}^{y_c} \frac{1}{y^2\mathcal{D}(y;r)}\, dy. \end{aligned} \tag{145}$$

Closed forms.

Integral I_5: Case $r \neq 1$:

$$I_5(y_{c,B}, y_c; r \neq 1) = \frac{1}{4(1-r)^2} \ln\left(\frac{(1-r)^2 y_c^4 + (1+r) y_c^2 + \frac{1}{4}}{(1-r)^2 y_{c,B}^4 + (1+r) y_{c,B}^2 + \frac{1}{4}} \right) - \frac{1+r}{8(1-r)^2\sqrt{r}} \ln\left(\frac{\left(2(1-r)^2 y_c^2 + (1+r) - 2\sqrt{r}\right) \left(2(1-r)^2 y_{c,B}^2 + (1+r) + 2\sqrt{r}\right)}{\left(2(1-r)^2 y_c^2 + (1+r) + 2\sqrt{r}\right) \left(2(1-r)^2 y_{c,B}^2 + (1+r) - 2\sqrt{r}\right)} \right). \tag{146}$$

Case $r = 1$:

$$I_5(y_{c,B}, y_c; r = 1) = \frac{1}{4}(y_c^2 - y_{c,B}^2) - \frac{1}{32} \ln\left(\frac{8y_c^2 + 1}{8y_{c,B}^2 + 1} \right). \tag{147}$$

Integral I_6: Case $r \neq 1$: Let

$$u_{1,2} = \frac{-(1+r) \pm 2\sqrt{r}}{2(1-r)^2}, \qquad a_j := \sqrt{-u_j} > 0, \quad j = 1, 2. \tag{148}$$

Then

$$\begin{aligned} I_6(y_{c,B}, y_c; r \neq 1) = &-\frac{a_1}{2\sqrt{r}} \left[\arctan\left(\frac{y_c}{a_1}\right) - \arctan\left(\frac{y_{c,B}}{a_1}\right) \right] \\ &+ \frac{a_2}{2\sqrt{r}} \left[\arctan\left(\frac{y_c}{a_2}\right) - \arctan\left(\frac{y_{c,B}}{a_2}\right) \right]. \end{aligned} \tag{149}$$

Case $r = 1$:

$$I_6(y_{c,B}, y_c; r = 1) = \frac{1}{2}(y_c - y_{c,B}) - \frac{\sqrt{2}}{8} \left[\arctan(2\sqrt{2}\, y_c) - \arctan(2\sqrt{2}\, y_{c,B}) \right]. \tag{150}$$

Integral I_7: Case $r \neq 1$: Define $N(y) := 2(1-r)^2 y^2 + (1+r)$. Then

$$I_7(y_{c,B}, y_c; r \neq 1) = \frac{1}{4\sqrt{r}} \ln\left(\frac{\left(N(y_c) - 2\sqrt{r}\right) \left(N(y_{c,B}) + 2\sqrt{r}\right)}{\left(N(y_c) + 2\sqrt{r}\right) \left(N(y_{c,B}) - 2\sqrt{r}\right)} \right). \tag{151}$$

Case $r = 1$:

$$I_7(y_{c,B}, y_c; r = 1) = \frac{1}{4} \ln\left(\frac{8y_c^2 + 1}{8y_{c,B}^2 + 1} \right). \tag{152}$$

Integral I_8: Case $r \neq 1$: (with a_1, a_2 as above)

$$\begin{aligned} I_8(y_{c,B}, y_c; r \neq 1) = \frac{1}{2\sqrt{r}} \Bigg\{ &\frac{1}{a_1} \left[\arctan\left(\frac{y_c}{a_1}\right) - \arctan\left(\frac{y_{c,B}}{a_1}\right) \right] \\ &- \frac{1}{a_2} \left[\arctan\left(\frac{y_c}{a_2}\right) - \arctan\left(\frac{y_{c,B}}{a_2}\right) \right] \Bigg\}. \end{aligned} \tag{153}$$

Case $r = 1$:

$$I_8(y_{c,B}, y_c; r = 1) = \sqrt{2} \left[\arctan(2\sqrt{2}\, y_c) - \arctan(2\sqrt{2}\, y_{c,B}) \right]. \tag{154}$$

Integral I_9: **Case $r \neq 1$:** Define

$$D(y) := (1-r)^2 y^4 + (1+r) y^2 + \tfrac{1}{4}, \qquad N(y) := 2(1-r)^2 y^2 + (1+r).$$

The log-only form is

$$\begin{aligned} I_9(y_{c,B}, y_c; r \neq 1) = 4 \ln\left(\frac{y_c}{y_{c,B}}\right) + \ln\left(\frac{D(y_{c,B})}{D(y_c)}\right) \\ - \frac{1+r}{2\sqrt{r}} \ln\left(\frac{\big(N(y_c) - 2\sqrt{r}\big)\big(N(y_{c,B}) + 2\sqrt{r}\big)}{\big(N(y_c) + 2\sqrt{r}\big)\big(N(y_{c,B}) - 2\sqrt{r}\big)}\right). \end{aligned} \tag{155}$$

Case $r = 1$:

$$I_9(y_{c,B}, y_c; r = 1) = 2 \ln\left(\frac{y_c^2 (8 y_{c,B}^2 + 1)}{y_{c,B}^2 (8 y_c^2 + 1)}\right). \tag{156}$$

Integral I_{10}: **Case $r \neq 1$:** (with a_1, a_2 as above)

$$\begin{aligned} I_{10}(y_{c,B}, y_c; r \neq 1) = 4 \left(\frac{1}{y_{c,B}} - \frac{1}{y_c}\right) \\ - \frac{1}{2\sqrt{r}\, a_1^3} \left[\arctan\left(\frac{y_c}{a_1}\right) - \arctan\left(\frac{y_{c,B}}{a_1}\right)\right] \\ + \frac{1}{2\sqrt{r}\, a_2^3} \left[\arctan\left(\frac{y_c}{a_2}\right) - \arctan\left(\frac{y_{c,B}}{a_2}\right)\right]. \end{aligned} \tag{157}$$

Case $r = 1$:

$$\begin{aligned} I_{10}(y_{c,B}, y_c; r = 1) = 4 \left(\frac{1}{y_{c,B}} - \frac{1}{y_c}\right) \\ - 8\sqrt{2} \left[\arctan(2\sqrt{2}\, y_c) - \arctan(2\sqrt{2}\, y_{c,B})\right]. \end{aligned} \tag{158}$$

4.3.3 Region III: tail integrals and asymptotic matching

The Region III truncated integrand is

$$\mathcal{I}_{\mathrm{III}}(y) \approx -\sqrt{2}\, r \, \frac{\dfrac{D_2}{y} - \dfrac{D_4 + C_2 D_2}{y^3}}{\mathcal{D}(y; r)}, \qquad y > y_c, \tag{159}$$

and therefore

$$I_{\mathrm{III}}^{(\mathrm{env})}(\omega) = \int_{y_c}^{\infty} \mathcal{I}_{\mathrm{III}}(y)\, dy \approx -\sqrt{2}\, r \left[D_2\, I_3^{(\mathrm{III})}(y_c; r) - (D_4 + C_2 D_2)\, I_4^{(\mathrm{III})}(y_c; r)\right], \tag{160}$$

where

$$I_3^{(\mathrm{III})}(y_c; r) := \int_{y_c}^{\infty} \frac{1}{y\, \mathcal{D}(y; r)}\, dy, \qquad I_4^{(\mathrm{III})}(y_c; r) := \int_{y_c}^{\infty} \frac{1}{y^3 \mathcal{D}(y; r)}\, dy. \tag{161}$$

Closed forms.

Integral I_3: Case $r \neq 1$:

$$I_3^{(\mathrm{III})}(y_c; r \neq 1) = \ln\left(\frac{(1-r)^2 y_c^4 + (1+r) y_c^2 + \frac{1}{4}}{(1-r)^2}\right) - 4 \ln y_c + \frac{1+r}{2\sqrt{r}} \ln\left(\frac{2(1-r)^2 y_c^2 + (1+r) - 2\sqrt{r}}{2(1-r)^2 y_c^2 + (1+r) + 2\sqrt{r}}\right). \tag{162}$$

Case $r = 1$:

$$I_3^{(\mathrm{III})}(y_c; r = 1) = 2 \ln\left(\frac{8y_c^2 + 1}{8y_c^2}\right). \tag{163}$$

Integral I_4: Case $r \neq 1$:

$$\begin{aligned} I_4^{(\mathrm{III})}(y_c; r \neq 1) &= 4(1+r) \ln((1-r)^2) + 8(1+r) \ln(y_c^2) + \frac{2}{y_c^2} \\ &\quad - 4(1+r) \ln\Big((1-r)^2 y_c^4 + (1+r) y_c^2 + \tfrac{1}{4}\Big) \\ &\quad + \frac{2(1+6r+r^2)}{\sqrt{r}} \operatorname{artanh}\left(\frac{2(1-r)^2 y_c^2 + (1+r)}{2\sqrt{r}}\right). \end{aligned} \tag{164}$$

Case $r = 1$:

$$I_4^{(\mathrm{III})}(y_c; r = 1) = 16 \ln\left(\frac{8y_c^2}{8y_c^2 + 1}\right) + \frac{2}{y_c^2}. \tag{165}$$

4.4 Handling special cases and resonant limit

Exact resonance ($r = 1$). Use the $r = 1$ closed forms for $I_1, I_2, I_3, I_4, I_5, \ldots, I_{10}$ as stated above.

Real-valued continuation for artanh. For any real argument Z appearing in the closed forms (e.g. in $I_4^{(\mathrm{III})}$), evaluate

$$\operatorname{artanh}(Z) = \frac{1}{2} \ln\left(\frac{1+Z}{1-Z}\right) \quad \mapsto \quad \begin{cases} \operatorname{artanh}(Z), & |Z| < 1, \\ \frac{1}{2} \ln\left|\frac{Z+1}{Z-1}\right|, & |Z| > 1. \end{cases}$$

5 Numerical implementation

The closed-form expressions derived in §4.3 provide an exact closed-form evaluation of the region-truncated asymptotic model of the QTN spectrum, but their numerical evaluation requires care at high frequencies where catastrophic cancellation can occur. This section describes the stabilization techniques employed and the computational strategy for both numeric and analytic evaluation methods.

5.1 Numerical evaluation of the general integral

The starting point for numerical validation is the full voltage-power spectrum integral derived in §2:

$$V^2(\omega) = \frac{16 k_B T_e}{\pi^2 \varepsilon_0 v_T} \int_0^\infty F_1(kL)\, J_0^2(ka)\, \Im\left(\frac{1}{K_L^t}\right) \frac{1}{y^2}\, dy, \tag{166}$$

where $k(y) = \omega/(\sqrt{2}\, y\, v_T)$, and $F_1(kL)$, $J_0^2(ka)$, and $\Im(1/K_L^t)$ are given by their respective equations found in Section §2 and Section §3. This expression contains the full frequency dependence, antenna geometry (through L and a), and plasma response (through $r = (\omega_p/\omega)^2$) without approximation.

In MATLAB, we evaluate Eq. (166) by constructing a vectorized integrand function that computes, for each y value:

(i) the wavenumber $k = \omega/(\sqrt{2}\, y\, v_T)$ and associated arguments kL and ka;

(ii) the antenna response $F_1(kL)$ using the full sine-integral definition;

(iii) the Bessel function squared $J_0^2(ka)$;

(iv) the imaginary part of the Lorentzian longitudinal response $\Im(1/K_L^t)$ from the closed form in Eq. (15).

The product of these terms, weighted by $1/y^2$, forms the integrand of Eq. (166).

5.2 Adaptive Gauss–Kronrod quadrature

To validate the analytic closed forms, we compute the spectrum numerically by direct integration of Eq. (166) using MATLAB's `quadgk` routine. This implements adaptive Gauss–Kronrod quadrature over the full integration domain $y \in [0, \infty)$ (truncated at a large finite value $y_{\max}$ where the integrand is negligible). To resolve the regime transitions at $y_{c,B}$ and y_c, we supply these points as waypoints to the quadrature routine. The numerical integration was performed with 100,000 frequency points logarithmically spaced across the band of interest.

5.3 Analytic evaluation via region summation

The analytic branch computes

$$V^2_{\text{analytic}}(\omega) = \frac{16 k_B T_e}{\pi^2 \varepsilon_0 v_T} \Big[\mathcal{I}_{\text{I}} + \mathcal{I}_{\text{II}} + \mathcal{I}_{\text{III}} \Big], \tag{167}$$

where each region contribution is assembled from closed-form basis integrals:

$$\mathcal{I}_{\text{I}} = \texttt{compute\_region\_I}(y_{c,B}, r, B_1, B_2), \tag{168}$$

$$\mathcal{I}_{\text{II}} = \texttt{compute\_region\_II}(y_{c,B}, y_c, r, B_1, B_2, C_2, C_4), \tag{169}$$

$$\mathcal{I}_{\text{III}} = \texttt{compute\_region\_III}(y_c, r, D_2, D_4, C_2). \tag{170}$$

Each region function evaluates the appropriate linear combinations of the integrals $I_1, \ldots, I_{10}$ using the closed forms derived in §4.

5.4 Stable evaluation of logarithms and arctangents

Catastrophic cancellation at large y_c. For frequencies $f \gtrsim 10^7$ Hz (especially when $L = 3$ m), the cutoff $y_c \propto \omega L$ becomes large (typically $y_c \sim 10^3$ or more). In this regime, several closed-form expressions contain differences of large, nearly equal logarithmic terms whose true remainder is small. For example, in Region III the analytic form for I_3 contains the structure

$$\ln\left(\frac{D}{(1-r)^2}\right) - 4 \ln y_c, \tag{171}$$

where both terms scale as $4 \ln y_c$ for large y_c. In double precision, evaluating them separately and subtracting causes loss of significant digits (catastrophic cancellation), leading to noisy values, sign flips, and spurious oscillations in the analytic spectrum. The issue is amplified for larger L because y_c is larger at the same frequency.

Stabilization principle. The remedy is to algebraically combine cancelling terms *before* evaluation and to use numerically stable log transforms (notably $\ln(1+x)$ for small x). Accordingly, the implementation rewrites the problematic logarithmic groups into a form involving

$$\ln(1 + t(y)) \quad \text{evaluated via} \quad \texttt{log1p}(t(y)), \tag{172}$$

which preserves precision when $t(y) \ll 1$ (the large-y regime).

(a) `compute_I3`: stable evaluation of Region III logarithms. The original `compute_I3` formula subtracts $4 \ln y_c$ from a large log term. In the updated form, these are combined to yield a single logarithm of a quantity close to unity:

$$\ln\left(\frac{D}{(1-r)^2}\right) - 4\ln y_c = \ln\left(\frac{D}{(1-r)^2 y_c^4}\right) = \ln\left(1 + \frac{1+r}{(1-r)^2 y_c^2} + \frac{1}{4(1-r)^2 y_c^4}\right), \tag{173}$$

which is evaluated using `log1p` to maintain accuracy for large y_c. The remaining log-ratio term is also rewritten in a stable $\ln(1 + \cdots)$ form.

(b) `compute_I4`: stable cancellation of large log groups. The original `compute_I4` contained multiple large log terms (including $\ln\big((1-r)^2\big)$, $\ln(y_c^2)$, and $\ln(D)$) that nearly cancel at high y_c. These terms are algebraically merged into a single $\ln(1 + t)$ structure, eliminating the subtraction of large numbers:

$$\begin{aligned} &4(1+r)\ln\big((1-r)^2\big) + 8(1+r)\ln(y_c^2) - 4(1+r)\ln(D) \\ &\quad = -4(1+r)\ln\left(\frac{D}{(1-r)^2 y_c^4}\right) = -4(1+r)\ln(1+t). \end{aligned} \tag{174}$$

This is again evaluated via `log1p`. For the $r = 1$ degenerate case, an additional stable representation prevents cancellation in $\ln\big((8y_c^2)/(8y_c^2+1)\big)$.

(c) `compute_I9`: stable difference of close log terms across limits. The original `compute_I9` includes a term

$$4\ln\left(\frac{y_c}{y_{c,B}}\right) + \ln\left(\frac{D(y_{c,B})}{D(y_c)}\right), \tag{175}$$

which also experiences cancellation when both cutoffs are large. The updated version rescales $D(y)$ by $(1-r)^2 y^4$ so that the large contributions cancel analytically before evaluation, giving

$$\ln(1 + t(y_{c,B})) - \ln(1 + t(y_c)), \tag{176}$$

computed using `log1p`. The remaining log-ratio terms are similarly rewritten into stable `log1p` forms.

Practical outcome. After these stabilizations, the analytic curve no longer exhibits nonphysical oscillations or spikes at high frequency. The improved agreement is not a change in the underlying physics or asymptotic model; it is purely a correction to the numerical evaluation of the closed forms in a regime where straightforward double-precision arithmetic loses accuracy due to cancellation.

6 Results and validation

This section presents the validation of the analytic closed-form solution against direct numerical quadrature, and compares the computed spectra to the established reference model of Cairns. All calculations employ the following ionospheric plasma parameters: $n_e = 5.77 \times 10^{11}\,\mathrm{m}^{-3}$ and $T_e = 1700\,\mathrm{K}$, corresponding to a plasma frequency $f_p = \omega_p/(2\pi) \approx 6.82\,\mathrm{MHz}$. Results are shown for dipole antenna lengths $L = 0.3\,\mathrm{m}$ and $L = 3.0\,\mathrm{m}$ with wire radius $a = 2 \times 10^{-4}\,\mathrm{m}$.

6.1 Numeric versus analytic agreement

Figure 1 shows the log-log comparison of the two independent computational methods:

(i) **Numeric (solid lines):** direct adaptive Gauss–Kronrod quadrature (`quadgk`) of the full integrand in Eq. (166) without approximation, evaluated at 100,000 logarithmically spaced frequency points;

(ii) **Analytic (dashed lines):** the asymptotic matched-region decomposition derived in §4, evaluated using the closed forms from Eq. (167) with numerically stable logarithmic implementations in `compute_I3`, `compute_I4`, and `compute_I9`.

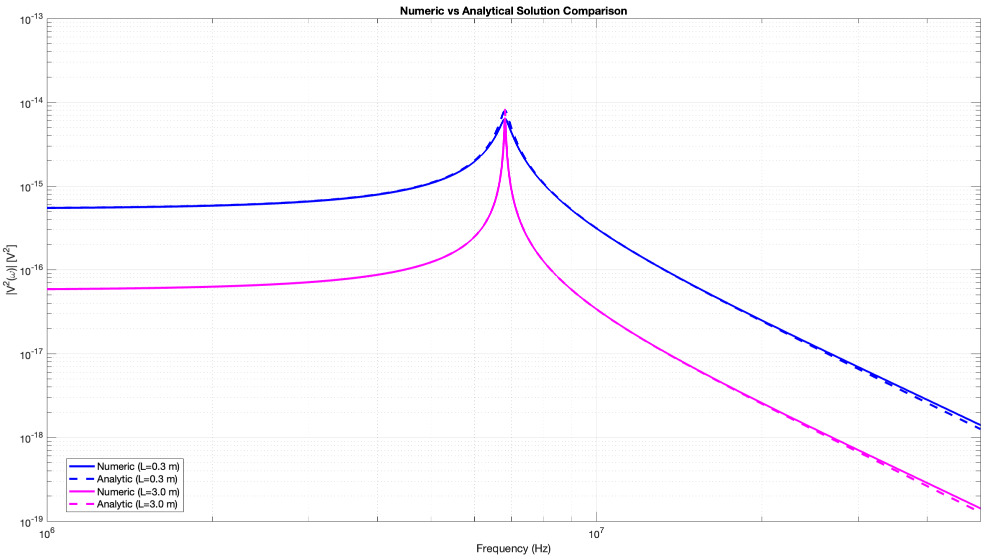

Figure 1: Numeric (solid lines) versus analytic (dashed lines) evaluation of the unmagnetized QTN spectrum for $L = 0.3\,\mathrm{m}$ and $L = 3.0\,\mathrm{m}$ using $n_e = 5.77 \times 10^{11}\,\mathrm{m}^{-3}$ and $T_e = 1700\,\mathrm{K}$. The updated analytic implementation uses numerically stable logarithmic forms in `compute_I3`, `compute_I4`, and `compute_I9` to prevent high-frequency cancellation errors. The numeric and analytic curves are visually indistinguishable across the entire frequency range.

The numeric and analytic curves are visually indistinguishable across the entire frequency range 10^6–10^8 Hz for both antenna lengths. Both methods reproduce the canonical signatures of unmagnetized quasi-thermal noise:

(i) a low-frequency plateau below the plasma frequency;

(ii) a sharp resonance peak centered at $f_p \approx 6.82\,\mathrm{MHz}$;

(iii) monotone power-law decay at high frequency.

The spectrum for the shorter antenna ($L = 0.3\,\mathrm{m}$, blue curves) lies consistently above that for the longer antenna ($L = 3.0\,\mathrm{m}$, magenta curves), reflecting the inverse scaling $V^2 \propto 1/L$ expected from antenna theory.

6.2 Quantitative error analysis

Figure 2 reports the relative error between the two methods, defined as

$$\varepsilon_{\mathrm{rel}}(f) = \frac{|V^2_{\mathrm{num}}(f) - V^2_{\mathrm{ana}}(f)|}{|V^2_{\mathrm{num}}(f)|} \times 100\%, \tag{177}$$

where the subscripts denote the numeric (full integral) and analytic (region-sum) evaluations, respectively.

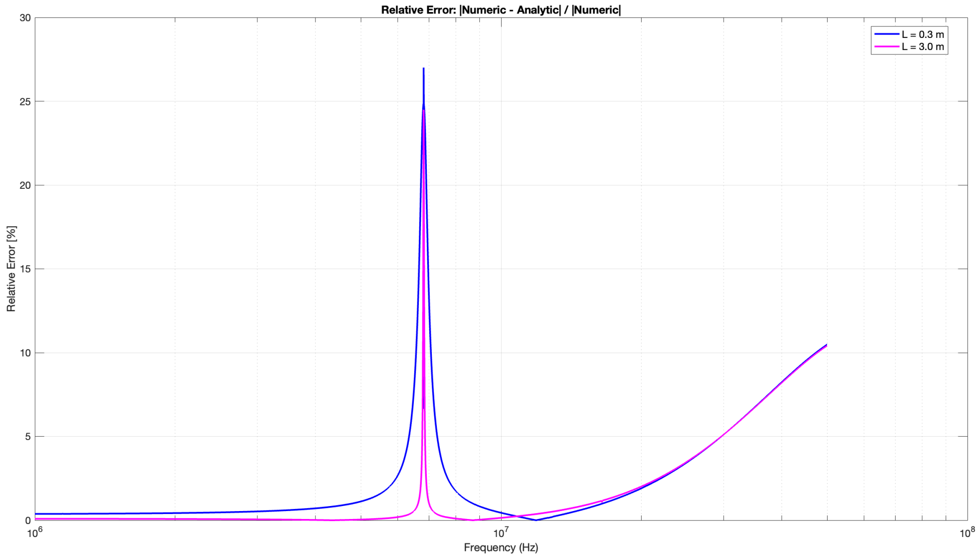

Figure 2: Relative error for the analytic region model against the numeric quadrature, $|V^2_{\mathrm{num}} - V^2_{\mathrm{ana}}|/|V^2_{\mathrm{num}}| \times 100\%$, for $L = 0.3\,\mathrm{m}$ and $L = 3.0\,\mathrm{m}$. The error remains below 1% across most of the frequency band. A narrow spike reaching approximately 27.5% occurs at the plasma resonance near $f_p \approx 6.82\,\mathrm{MHz}$, where the spectrum varies extremely steeply. For the longer antenna ($L = 3.0\,\mathrm{m}$), the error gradually rises to approximately 10% at $10^8\,\mathrm{Hz}$ due to accumulation of higher-order truncation errors in the asymptotic expansions.

Low-to-mid frequency regime ($10^6 \lesssim f \lesssim 10^7$ Hz). For both antenna lengths, the relative error remains below 1% across the entire low-frequency plateau and into the pre-resonance regime. This confirms that the region-decomposition approach — combining envelope approximations in Regions I and III with truncated series in Region II — accurately reproduces the full QTN spectrum computed from first principles.

Resonance peak ($f \approx f_p$). A narrow error spike reaching approximately 27.5% occurs at the plasma resonance frequency $f_p \approx 6.82\,\mathrm{MHz}$. This localized spike is expected: the spectrum varies extremely steeply near resonance (changing by multiple orders of magnitude over a fractional bandwidth $\Delta f/f_p \sim 10^{-2}$), and small differences in the precise location or sharpness of the peak translate to large relative errors even when both methods are physically consistent. The spike width is comparable to the frequency resolution ($\Delta f \sim 3 \times 10^2\,\mathrm{Hz}$ at f_p for the 100,000-point log-spaced grid over 10^6–10^8 Hz), indicating numerical sensitivity rather than model inadequacy.

High-frequency regime ($f \gtrsim 10^7$ Hz). Beyond the resonance, the error for the $L = 0.3\,\mathrm{m}$ antenna remains below 2% out to 10^8 Hz. For the longer antenna ($L = 3.0\,\mathrm{m}$), the error gradually rises from $\sim 1\%$ at 10^7 Hz to $\sim 10\%$ at 10^8 Hz. This slow degradation at the highest frequencies is consistent with the accumulation of higher-order truncation errors in the asymptotic expansions; nevertheless, the analytic solution remains qualitatively correct and computationally advantageous (evaluation time reduced from seconds to microseconds).

6.3 Comparison to reference model

Figure 3 reproduces the unmagnetized reference voltage power spectrum from Maj and Cairns [7] for the same plasma parameters ($n_e = 5.77 \times 10^{11}\,\mathrm{m}^{-3}$, $T_e = 1700\,\mathrm{K}$) and antenna lengths ($L = 0.3\,\mathrm{m}$, $L = 3.0\,\mathrm{m}$). Our MATLAB results in Figure 1 exhibit the same key spectral features:

(i) a low-frequency plateau;

(ii) a sharp resonance centered near f_p;

(iii) monotone decay at high frequency;

(iv) the $L = 0.3\,\mathrm{m}$ spectrum lying above the $L = 3.0\,\mathrm{m}$ spectrum over the band shown.

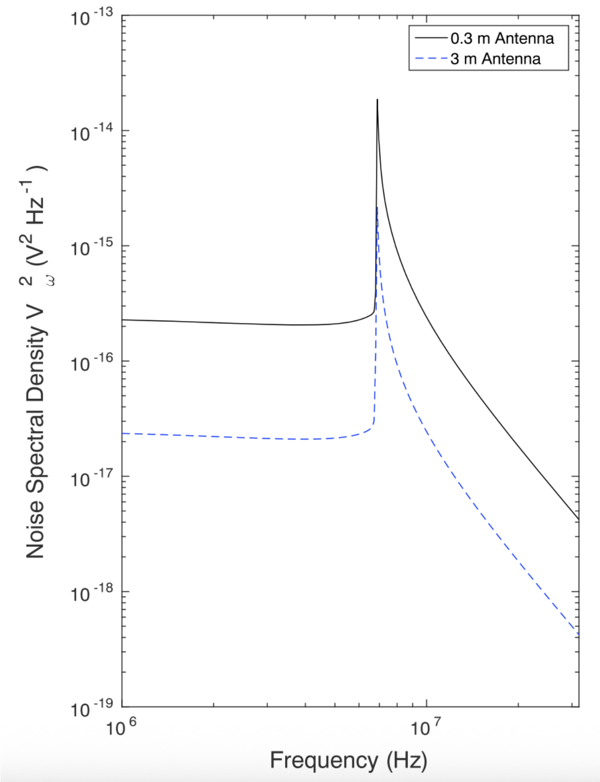

Figure 3: Unmagnetized reference voltage power spectrum from Maj and Cairns [7] for $L = 0.3\,\mathrm{m}$ and $L = 3.0\,\mathrm{m}$ in an ionospheric plasma with $n_e = 5.77 \times 10^{11}\,\mathrm{m}^{-3}$ and $T_e = 1700\,\mathrm{K}$. The reference model includes additional physical effects not present in our thermal QTN calculation: shot noise from discrete particle impacts, antenna sheath effects, and spacecraft potential contributions. Our analytic solution (Figure 1) isolates the intrinsic thermal Langmuir wave contribution, which is the dominant component near the plasma frequency resonance.

Quantitative comparison between our results and the Maj and Cairns [7] reference requires careful interpretation of several systematic differences. First, the reference model includes shot noise due to discrete particle impacts on the antenna, which adds a frequency-independent background level not present in our purely thermal calculation. Second, the reference treatment incorporates antenna sheath and spacecraft potential effects that modify the low-frequency response, particularly for monopole configurations. Third, normalization conventions differ: spectra may be reported per Hz or per rad/s, related by the Jacobian $S_f(f) = (2\pi)^{-1} S_\omega(\omega)$. Our analytic solution deliberately isolates the thermal Langmuir wave contribution—the intrinsic quasi-thermal noise mechanism—which is the dominant component near the plasma frequency resonance and the primary diagnostic signature for (n_e, T_e) extraction.

Within these caveats, the agreement in spectral shape and scaling behavior is excellent. Both our analytic solution and the Maj and Cairns [7] reference reproduce the canonical QTN physics: thermal fluctuations produce a plateau at low frequencies, resonate sharply at $\omega \approx \omega_p$, and decay at high frequencies following the expected power-law behavior. The inverse antenna-length scaling $V^2 \propto 1/L$ is preserved in both treatments. This concordance confirms that the present closed-form solution correctly captures the fundamental physics of unmagnetized quasi-thermal noise, providing a validated analytical framework for rapid spectrum prediction and parameter inversion in ionospheric plasma diagnostics.

7 Discussion

7.1 Physical interpretation of the region decomposition

The success of the three-region decomposition derives from the distinct physical regimes sampled by the integration variable $y = \omega/(\sqrt{2}\,k\,v_T)$. Since $k(y) = \omega/(\sqrt{2}\,y\,v_T)$, the antenna and Bessel arguments scale as

$$kL = \frac{\omega L}{\sqrt{2}\,y\,v_T}, \qquad ka = \frac{\omega a}{\sqrt{2}\,y\,v_T}. \tag{178}$$

Small y corresponds to large wavenumber (short wavelength), while large y corresponds to small wavenumber (long wavelength).

Region I ($0 < y < y_{c,B}$): Short-wavelength regime. In this region, $kL > x_c \approx 1$ and $ka > x_{c,B} \approx 0.34$, corresponding to wavelengths comparable to or smaller than the antenna length and wire radius. The envelope approximations for $F_1(kL)$ and $J_0^2(ka)$—derived from large-argument asymptotics but matched at the cutoffs to ensure accuracy—capture the oscillatory behavior of these functions. Physically, this regime samples fluctuations at scales comparable to the Debye length $\lambda_D = v_T/\omega_p$, where collective plasma oscillations are less coherent and the response is dominated by the high-velocity tail of the electron distribution. For ka sufficiently above $x_{c,B}$, the oscillatory factor $\cos^2\Phi(y)$ in the Bessel envelope averages over rapid phase variations, leaving a smooth contribution to the spectrum.

Region II ($y_{c,B} < y < y_c$): Transition regime. Here $kL > x_c$ but $ka < x_{c,B}$: the antenna argument remains in the envelope asymptotic regime, while the wire-radius argument enters the small-argument regime. This mixed regime requires the envelope form for $F_1(kL)$ combined with the small-argument Taylor expansion for $J_0^2(ka)$. The truncated series in powers of $1/y$ captures the crossover between antenna-dominated and thermal-scale physics. Region II spans a substantial range in y (since $a \ll L$) and is critical for accurately connecting the high-k and low-k behaviors without introducing discontinuities.

Region III ($y > y_c$): Long-wavelength regime. For $y > y_c$, both $kL < x_c$ and $ka < x_{c,B}$, corresponding to wavelengths larger than the antenna dimensions. For typical antenna geometries with $a/L \sim 10^{-3}$, Region III has $kL < 1$ and $ka \ll 1$, so the small-argument Taylor expansions are highly accurate. The antenna samples only a fraction of each wave period, and its response is determined by local field gradients captured by the small-argument expansions. Physically, this regime probes fluctuations at scales larger than the Debye length, where the plasma response is well-described by the bulk thermal motion. The leading $1/y$ and $1/y^3$ terms in the integrand ensure proper asymptotic decay at large y (small k), preventing unphysical divergences.

Role of the cutoffs. The cutoffs $y_{c,B}$ and y_c are *not* arbitrary tuning parameters but are determined by matching the small- and large-argument approximations at their crossover points. This ensures that the total spectrum

$$V^2(\omega) = \frac{16k_BT_e}{\pi^2\varepsilon_0 v_T}\left[\mathcal{I}_{\mathrm{I}} + \mathcal{I}_{\mathrm{II}} + \mathcal{I}_{\mathrm{III}}\right] \tag{179}$$

remains smooth and continuous across regime boundaries, with each region contributing where its asymptotic representation is most accurate.

7.2 Computational performance and applications

Evaluation speed. The primary advantage of the closed-form analytic solution is its computational efficiency. Direct numerical quadrature of Eq. (166) requires adaptive sampling of the integrand at hundreds of points per frequency, with convergence checks and waypoint handling at the regime boundaries $y_{c,B}$ and y_c. In contrast, the analytic evaluation reduces to algebraic evaluation of logarithms, arctangents, and artanh functions—operations executable in microseconds on modern hardware. The speedup of $\mathcal{O}(10^5)$–$\mathcal{O}(10^6)$ relative to quadrature enables applications that would be computationally prohibitive with numerical integration alone.

Real-time parameter inversion. Ionospheric diagnostics require inverting the measured QTN spectrum to infer plasma density n_e and temperature T_e. This inversion is typically performed via least-squares fitting or Bayesian inference, both of which demand thousands to millions of forward model evaluations as the optimizer explores parameter space. With the analytic solution, a single parameter-sweep iteration completes in milliseconds rather than hours, making real-time on-orbit analysis feasible for CubeSat missions with limited computational resources.

Mission planning and instrument design. The closed-form expressions facilitate rapid exploration of the (L, a, n_e, T_e) design space during mission planning. Engineers can evaluate trade-offs between antenna size, mass, and diagnostic sensitivity without resorting to expensive Monte Carlo simulations. For example, the inverse scaling $V^2 \propto 1/L$ (visible in Figure 1) quantifies the signal gain achievable by extending dipole length, informing stowage and deployment constraints for space-borne instruments.

Onboard processing. Future ionospheric CubeSats may perform autonomous parameter estimation using embedded processors. The analytic solution's minimal memory footprint (no quadrature tables or precomputed grids) and deterministic execution time make it suitable for flight software implementation, enabling science data reduction at the instrument level and minimizing downlink requirements.

7.3 Limitations and domain of validity

Unmagnetized approximation. The present derivation assumes a non-magnetized plasma ($\mathbf{B} = 0$), valid when the electron gyrofrequency $\Omega_e = eB/m_e$ is negligible compared to both the observation frequency ω and the plasma frequency ω_p. In the terrestrial ionosphere, typical magnetic field strengths $B \sim 0.3$–$0.5\,\mathrm{G}$ yield $\Omega_e/(2\pi) \sim 1\,\mathrm{MHz}$, comparable to ω_p in many regions. For such environments, the full magnetized dispersion relation must be employed, introducing cyclotron resonances and anisotropic response functions. The region-decomposition strategy developed here may extend to the magnetized case, but the integrals $I_1, \ldots, I_{10}$ would require re-evaluation with modified plasma response functions.

Maxwellian electron distribution. The Lorentzian plasma response $\Im(1/K_L^t)$ derived in §2.2.2 assumes a thermal Maxwellian electron distribution with temperature T_e. Non-thermal features—such as suprathermal tails, beam components, or loss-cone distributions—violate this assumption and can produce qualitatively different spectra. For such cases, the kinetic dispersion relation must be solved numerically using the Vlasov–Poisson framework, and the present analytic integrals do not apply.

Frequency and antenna length constraints. The asymptotic expansions underpinning the region decomposition become less accurate at extreme frequencies. At very low frequencies ($\omega \ll \omega_p$), the cutoffs y_c and $y_{c,B}$ move to large values, and higher-order terms in the small-argument expansions for $F_1(kL)$ and $J_0^2(ka)$ may be required. Conversely, at very high frequencies ($\omega \gg \omega_p$), the resonance structure near ω_p becomes irrelevant, and a simpler thermal-fluctuation model may suffice. Similarly, for very long antennas ($L \gg \lambda_D$) or very short antennas ($L \ll \lambda_D$), the assumption of clear separation between Regions I, II, and III may break down, requiring modified cutoff criteria.

High-frequency error growth. As observed in Figure 2, the relative error for the $L = 3.0\,\mathrm{m}$ antenna rises to approximately 10% at $f = 10^8$ Hz. This degradation reflects the accumulation of truncation errors in the asymptotic series for $F_1(kL)$ and $J_0^2(ka)$, which retain only leading-order terms. Including subleading corrections (e.g., $\mathcal{O}(x^{-3})$ terms in the large-argument expansions) would improve high-frequency accuracy at the cost of additional algebraic complexity.

7.4 Extensions and future work

Magnetized plasma. The most immediate extension is to magnetized plasmas, where the longitudinal dielectric function k_L^t must be replaced by the full electromagnetic dispersion tensor. The resulting spectrum exhibits cyclotron harmonics and field-aligned anisotropy, phenomena critical for interpreting measurements in the magnetosphere and auroral zones. A magnetized region decomposition would likely require separate treatments for parallel ($\mathbf{k} \parallel \mathbf{B}$) and perpendicular ($\mathbf{k} \perp \mathbf{B}$) propagation, each with distinct asymptotic regimes.

Non-Maxwellian distributions. Extending the formalism to non-Maxwellian distributions (e.g., kappa distributions, drifting Maxwellians, or loss-cone distributions) requires re-deriving $\Im(1/K_L^t)$ from the Vlasov dispersion relation with the appropriate velocity-space integrals. While closed-form solutions may not exist for arbitrary distributions, asymptotic approximations valid in specific limits (e.g., $\kappa \gg 1$ for kappa distributions) could enable analytic treatment of suprathermal tails without resorting to full numerical integration.

Three-dimensional antenna geometries. The present analysis considers a symmetric dipole antenna (two collinear elements of length $L/2$). Realistic space instruments often employ more complex geometries—V-shaped dipoles, monopoles, or triaxial booms—each with distinct response functions $F_1(kL)$. Generalizing the region-decomposition approach to arbitrary antenna configurations requires deriving the appropriate antenna factors and identifying the corresponding cutoff scales. The underlying integration strategy (matching asymptotics across k regimes) should remain applicable.

Multi-component plasmas. Ionospheric plasmas contain multiple ion species (O^+, H^+, He^+) in addition to electrons. While ion contributions to the quasi-thermal noise are negligible at frequencies $\omega \gg \omega_{pi}$ (ion plasma frequency), they become important at lower frequencies where ion acoustic fluctuations contribute. A multi-species treatment would introduce additional terms in the dielectric function and corresponding modifications to the spectrum.

Time-domain inversion algorithms. The availability of a fast forward model enables Bayesian parameter inference and machine-learning approaches to QTN diagnostics. Training neural networks to invert QTN spectra for (n_e, T_e) requires generating large synthetic datasets, a task ideally suited to the analytic solution. Such algorithms could provide on-orbit, real-time density and temperature estimates with quantified uncertainties, advancing autonomous space weather monitoring capabilities.

8 Conclusions

This work presents the first closed-form analytical solution for the quasi-thermal noise (QTN) spectrum on dipole antennas in unmagnetized plasmas. By replacing the standard Maxwellian plasma dispersion function with a Lorentzian (Padé-type) approximation for the longitudinal dielectric response, we have converted the QTN integral—previously requiring expensive numerical quadrature for each choice of plasma and antenna parameters—into a set of elementary algebraic expressions involving logarithms, arctangents, and inverse hyperbolic tangents.

The central innovation is a matched-asymptotic region decomposition that exploits the distinct behaviors of the antenna response $F_1(kL)$ and Bessel factor $J_0^2(ka)$ across different wavenumber regimes. We rigorously determined cutoffs $x_c \approx 1.003$ and $x_{c,B} \approx 0.337$ by matching small- and large-argument approximations, partitioning the integration domain into three physically motivated regions: Region I ($0 < y < y_{c,B}$), where both $kL > x_c$ and $ka > x_{c,B}$, so the antenna response uses its envelope form (matched at x_c) and the Bessel factor uses its Debye envelope form (matched at $x_{c,B}$); Region II ($y_{c,B} < y < y_c$), a transitional mixed regime; and Region III ($y > y_c$), where both arguments are small and the integrand exhibits proper asymptotic decay. This decomposition ensures smooth, convergent transitions while maintaining controlled accuracy across all frequency regimes relevant to ionospheric diagnostics.

The complete library of ten basis integrals (I_1 through I_{10}) has been evaluated in closed form using partial-fraction decomposition and real-valued primitives that avoid spurious complex intermediates. Special-case formulas for the resonant limit $r = (\omega_p/\omega)^2 \to 1$ are derived separately, and numerically stable implementations using `log1p` transformations prevent catastrophic cancellation at high frequencies where the cutoffs y_c and $y_{c,B}$ become large. These stabilization techniques are essential for practical implementation and eliminate non-physical oscillations that would otherwise contaminate the spectrum at frequencies $f \gtrsim 10^7$ Hz for longer antennas ($L \sim 3$ m).

Validation against direct numerical quadrature over 100,000 logarithmically spaced frequency points confirms relative errors below 1% across most of the frequency band 10^6–10^8 Hz, except for a narrow spike (∼27.5%) at the plasma resonance $f_p \approx 6.82$ MHz where the spectrum varies by multiple orders of magnitude over fractional bandwidths $\Delta f/f_p \sim 10^{-2}$. This localized error reflects numerical sensitivity to the precise peak location rather than fundamental model inadequacy. For the longer antenna ($L = 3.0$ m), the error gradually rises to approximately 10% by 10^8 Hz due to truncation of higher-order asymptotic corrections, but the solution remains qualitatively accurate and orders of magnitude faster than numerical integration. Comparison with the reference ionospheric spectra of Maj and Cairns [7] confirms that the analytic solution correctly reproduces the canonical QTN signatures: low-frequency plateau, sharp resonance peak, and high-frequency power-law decay.

The computational advantages are substantial. Where numerical quadrature requires adaptive sampling at hundreds of integration points per frequency—consuming seconds per spectrum and rendering large-scale parameter sweeps computationally prohibitive—the analytic evaluation reduces to algebraic operations executable in microseconds. This speedup of $\mathcal{O}(10^5)$–$\mathcal{O}(10^6)$ enables applications previously infeasible with numerical methods alone: real-time on-orbit parameter inversion for CubeSat missions with limited computational resources, rapid exploration of the (n_e, T_e, L, a) design space during mission planning, and autonomous plasma diagnostics using embedded processors without requiring quadrature tables or precomputed grids.

Beyond immediate practical applications, the analytic solution provides direct physical insight into how plasma and antenna parameters control spectral features. The three-region structure maps naturally onto physical length scales: Region I samples Debye-scale fluctuations dominated by the high-velocity electron tail; Region II captures the crossover between antenna-dominated and thermal-scale physics; and Region III probes long-wavelength fluctuations governed by bulk thermal motion. The cutoffs $y_{c,B}$ and y_c are not arbitrary tuning parameters but emerge from matching conditions at the crossover points where asymptotic approximations transition, ensuring that each region contributes where its representation is most accurate.

Several extensions merit future investigation. The most immediate is generalization to magnetized plasmas, where the full electromagnetic dispersion tensor introduces cyclotron resonances and field-aligned anisotropy critical for magnetospheric and auroral measurements. A magnetized region decomposition would require separate treatments for parallel and perpendicular propagation, each with distinct asymptotic regimes, but the underlying matching strategy should remain applicable. Extension to non-Maxwellian distributions (kappa, drifting Maxwellians, loss-cone) requires re-deriving the plasma response from the Vlasov dispersion relation; while fully general closed forms may not exist, asymptotic approximations valid in specific limits (e.g., $\kappa \gg 1$) could enable analytic treatment of suprathermal tails. Three-dimensional antenna geometries (V-dipoles, monopoles, triaxial booms) demand generalized antenna factors $F_1(kL)$ and corresponding cutoff criteria, though the integration framework should transfer directly. Finally, the availability of a fast forward model opens opportunities for Bayesian inversion and machine-learning diagnostics, where training neural networks to extract (n_e, T_e) from measured spectra requires generating large synthetic datasets—a task ideally suited to the analytic solution.

In summary, this work transforms QTN spectroscopy from a computationally intensive numerical problem into a tractable analytical framework suitable for real-time space-borne applications. The closed-form expressions derived here provide both a practical diagnostic tool for current and future ionospheric missions and a foundation for extending analytical plasma spectroscopy to more complex physical regimes. The combination of rigorous asymptotic analysis, numerically stable implementation, and comprehensive validation establishes a robust methodology for rapid, accurate QTN spectrum prediction that advances the state of the art in passive plasma diagnostics.

Acknowledgments

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A Detailed asymptotic expansions

A.1 Antenna response expansions

The antenna response function is defined by

$$F_1(x) = \frac{x\left[\mathrm{Si}(x) - \frac{1}{2}\mathrm{Si}(2x)\right] - 2\sin^4(\frac{x}{2})}{x^2}, \tag{180}$$

where

$$\mathrm{Si}(x) = \int_0^x \frac{\sin t}{t}\,dt = \sum_{n=0}^{\infty} \frac{(-1)^n x^{2n+1}}{(2n+1)(2n+1)!}. \tag{181}$$

Small-x expansion For $|x| \ll 1$,

$$\mathrm{Si}(x) = x - \frac{x^3}{18} + \frac{x^5}{600} - \frac{x^7}{35280} + O(x^9), \tag{182}$$

$$\mathrm{Si}(2x) = 2x - \frac{4x^3}{9} + \frac{4x^5}{75} - \frac{8x^7}{2205} + O(x^9), \tag{183}$$

so

$$\mathrm{Si}(x) - \frac{1}{2}\mathrm{Si}(2x) = \frac{x^3}{6} - \frac{x^5}{40} + O(x^7), \tag{184}$$

and

$$x[\mathrm{Si}(x) - \tfrac{1}{2}\mathrm{Si}(2x)] = \frac{x^4}{6} - \frac{x^6}{40} + O(x^8). \tag{185}$$

Using

$$\sin^4\left(\frac{x}{2}\right) = \frac{3 - 4\cos x + \cos 2x}{8}, \tag{186}$$

we obtain

$$\sin^4\left(\frac{x}{2}\right) = \frac{x^4}{16} - \frac{x^6}{96} + O(x^8), \tag{187}$$

$$-2\sin^4\left(\frac{x}{2}\right) = -\frac{x^4}{8} + \frac{x^6}{48} + O(x^8). \tag{188}$$

Thus the numerator of $F_1(x)$ is

$$x[\mathrm{Si}(x) - \tfrac{1}{2}\mathrm{Si}(2x)] - 2\sin^4\left(\frac{x}{2}\right) = \frac{x^4}{24} - \frac{x^6}{240} + O(x^8), \tag{189}$$

and

$$\boxed{F_1(x) = \frac{x^2}{24} - \frac{x^4}{240} + \frac{x^6}{4480} + O(x^8), \qquad |x| \ll 1.} \tag{190}$$

In the QTN problem,

$$x = kL, \qquad y = \frac{\omega}{\sqrt{2}\,k\,v_T} \quad \Rightarrow \quad k = \frac{\omega}{\sqrt{2}\,y\,v_T}, \tag{191}$$

so

$$x = kL = \frac{\omega L}{\sqrt{2}\, y\, v_T}. \tag{192}$$

Hence

$$\boxed{\begin{gathered} F_1(kL) = \frac{\omega^2 L^2}{48 v_T^2}\frac{1}{y^2} - \frac{\omega^4 L^4}{960 v_T^4}\frac{1}{y^4} + \frac{\omega^6 L^6}{35840 v_T^6}\frac{1}{y^6} + O\left(\frac{1}{y^8}\right), \\ x = kL = \frac{\omega L}{\sqrt{2} y v_T} \ll 1. \end{gathered}}$$

Large-x asymptotic For $|x| \gg 1$, the sine integral has the asymptotic expansion

$$\mathrm{Si}(x) = \frac{\pi}{2} - \frac{\cos x}{x} - \frac{\sin x}{x^2} + \frac{2!\cos x}{x^3} + \frac{3!\sin x}{x^4} + O\left(\frac{1}{x^5}\right), \tag{193}$$

and similarly

$$\mathrm{Si}(2x) = \frac{\pi}{2} - \frac{\cos(2x)}{2x} - \frac{\sin(2x)}{4x^2} + \frac{2!\cos(2x)}{8x^3} + \frac{3!\sin(2x)}{16x^4} + O\left(\frac{1}{x^5}\right). \tag{194}$$

Using these together with

$$\sin^4\left(\frac{x}{2}\right) = \frac{3 - 4\cos x + \cos 2x}{8}, \tag{195}$$

one finds

$$x\left[\mathrm{Si}(x) - \tfrac{1}{2}\mathrm{Si}(2x)\right] - 2\sin^4\left(\frac{x}{2}\right) = \frac{\pi x}{4} - \frac{3}{4} + O\left(\frac{1}{x}\right), \tag{196}$$

and therefore

$$\boxed{F_1(x) = \frac{\pi}{4x} - \frac{3}{4x^2} + O\left(\frac{1}{x^3}\right), \qquad |x| \gg 1.} \tag{197}$$

With $x = kL = \dfrac{\omega L}{\sqrt{2}\, y\, v_T}$ we obtain:

$$\boxed{F_1(kL) = \frac{\pi\sqrt{2}\, v_T}{4\omega L}\, y - \frac{3 v_T^2}{2\omega^2 L^2}\, y^2 + O(y^3),} \tag{198}$$

$$x = kL = \frac{\omega L}{\sqrt{2} y v_T} \gg 1. \tag{199}$$

A.2 Bessel expansions and envelope

We begin from the standard series representation

$$J_0(x) = \sum_{m=0}^{\infty} \frac{(-1)^m}{(m!)^2} \left(\frac{x}{2}\right)^{2m}. \tag{200}$$

1. Small-argument expansion ($|x| \ll 1$) Retaining terms up to x^6,

$$J_0(x) = 1 - \frac{x^2}{4} + \frac{x^4}{64} - \frac{x^6}{2304} + O(x^8). \tag{201}$$

Squaring,

$$\begin{aligned} J_0^2(x) &= \left(1 - \frac{x^2}{4} + \frac{x^4}{64} - \frac{x^6}{2304} + O(x^8)\right)^2 \\ &= 1 - \frac{x^2}{2} + \frac{3x^4}{32} - \frac{5x^6}{576} + O(x^8), \end{aligned} \tag{202}$$

so to fourth order we may write

$$J_0^2(x) = 1 - \frac{x^2}{2} + \frac{3x^4}{32} + O(x^6), \qquad |x| \ll 1. \tag{203}$$

2. Small-argument form for $J_0^2(ka)$ in QTN variables In the QTN problem we set

$$x = ka, \qquad y = \frac{\omega}{\sqrt{2}\, k\, v_T} \quad \Rightarrow \quad k = \frac{\omega}{\sqrt{2}\, y\, v_T}, \tag{204}$$

so

$$x = ka = \frac{\omega a}{\sqrt{2}\, y\, v_T}. \tag{205}$$

Substituting $x = ka$ into (203),

$$\begin{aligned} J_0^2(ka) &= 1 - \frac{(ka)^2}{2} + \frac{3(ka)^4}{32} + O((ka)^6) \\ &= 1 - \frac{1}{2}\left(\frac{\omega a}{\sqrt{2}\, y\, v_T}\right)^2 + \frac{3}{32}\left(\frac{\omega a}{\sqrt{2}\, y\, v_T}\right)^4 + O\left(\frac{1}{y^6}\right) \\ &= 1 - \frac{\omega^2 a^2}{4v_T^2}\frac{1}{y^2} + \frac{3\omega^4 a^4}{128 v_T^4}\frac{1}{y^4} + O\left(\frac{1}{y^6}\right), \qquad ka \ll 1. \end{aligned} \tag{206}$$

3. Large-argument asymptotic ($|x| \gg 1$) For large $|x|$, the Bessel function has the Debye-type expansion

$$J_0(x) = \sqrt{\frac{2}{\pi x}} \left[\cos\left(x - \frac{\pi}{4}\right) + O\left(\frac{1}{x}\right)\right], \qquad |x| \gg 1. \tag{207}$$

Introduce the phase

$$\phi(x) := x - \frac{\pi}{4}, \tag{208}$$

so that (207) reads

$$J_0(x) = \sqrt{\frac{2}{\pi x}} \left[\cos\phi(x) + O\left(\frac{1}{x}\right)\right]. \tag{209}$$

Squaring this expression,

$$J_0^2(x) = \frac{2}{\pi x}\left[\cos\phi(x) + O\left(\frac{1}{x}\right)\right]^2$$
$$= \frac{2}{\pi x}\cos^2\phi(x) + O\left(\frac{1}{x^2}\right). \tag{210}$$

Using $\cos^2\phi = \frac{1}{2}[1+\cos(2\phi)]$, we obtain

$$J_0^2(x) = \frac{2}{\pi x}\cdot\frac{1}{2}\Big[1+\cos\left(2\phi(x)\right)\Big] + O\left(\frac{1}{x^2}\right)$$
$$= \frac{1}{\pi x}\Big[1+\cos\left(2x-\tfrac{\pi}{2}\right)\Big] + O\left(\frac{1}{x^2}\right), \qquad |x| \gg 1. \tag{211}$$

4. Large-argument form for $J_0^2(ka)$ in QTN variables Again setting $x = ka$,

$$x = ka = \frac{\omega a}{\sqrt{2}\,y\,v_T}, \tag{212}$$

we substitute into (211):

$$J_0^2(ka) = \frac{1}{\pi ka}\left[1+\cos\left(2ka-\frac{\pi}{2}\right)\right] + O\left(\frac{1}{(ka)^2}\right)$$
$$= \frac{1}{\pi}\,\frac{\sqrt{2}\,y\,v_T}{\omega a}\left[1+\cos\left(\frac{2\omega a}{\sqrt{2}\,y\,v_T}-\frac{\pi}{2}\right)\right] + O\left(\frac{y^2}{\omega^2}\right), \qquad ka \gg 1. \tag{213}$$

Equivalently, in the more compact form

$$J_0^2(ka) = \frac{2}{\pi ka}\cos^2\left(ka-\frac{\pi}{4}\right) + O\left(\frac{1}{(ka)^2}\right) = \frac{2\sqrt{2}\,y\,v_T}{\pi\,\omega a}\cos^2\left(\frac{\omega a}{\sqrt{2}\,y\,v_T}-\frac{\pi}{4}\right) + O\left(\frac{1}{(ka)^2}\right). \tag{214}$$

If only the slowly varying envelope is required, one may replace $\cos^2(\cdot)$ by its average value $1/2$, yielding

$$\langle J_0^2(ka)\rangle \simeq \frac{1}{\pi ka} = \frac{\sqrt{2}\,y\,v_T}{\pi\,\omega a}, \qquad ka \gg 1. \tag{215}$$

5. Piecewise approximation We now introduce a transition scale $x_{c,B} \approx 0.337$ to splice the small- and large-argument regimes. In terms of $x = ka$ we define

$$J_0^2(x) \simeq \begin{cases} 1 - \dfrac{x^2}{2} + \dfrac{3x^4}{32}, & 0 \le x \le x_{c,B}, \\ \dfrac{2}{\pi x}\cos^2\!\left(x - \dfrac{\pi}{4}\right), & x \ge x_{c,B}, \end{cases} \tag{216}$$

which is accurate up to $O(x^6)$ for $x \ll 1$ and up to $O(1/x^2)$ for $x \gg 1$.

In QTN variables this becomes

$$J_0^2(ka) \simeq \begin{cases} 1 - \dfrac{1}{2}\left(\dfrac{\omega a}{\sqrt{2}\,y\,v_T}\right)^2 + \dfrac{3}{32}\left(\dfrac{\omega a}{\sqrt{2}\,y\,v_T}\right)^4, & \dfrac{\omega a}{\sqrt{2}\,y\,v_T} \le x_{c,B}, \\ \dfrac{2\sqrt{2}\,y\,v_T}{\pi\,\omega a}\cos^2\left(\dfrac{\omega a}{\sqrt{2}\,y\,v_T} - \dfrac{\pi}{4}\right), & \dfrac{\omega a}{\sqrt{2}\,y\,v_T} \ge x_{c,B}. \end{cases} \tag{217}$$

B Analytic integral evaluations

B.1 Region I integrals: antiderivatives and closed forms

In Region I ($0 < y < y_{c,B}$) we use the large-argument forms for both F_1 and J_0^2, giving the integrand

$$\mathcal{I}_{\mathrm{I}}(y) = -\sqrt{2}\, r\, E_B\, \frac{B_1\, y^3 - B_2\, y^4}{\mathcal{D}(y; r)}\, \cos^2\Phi(y), \tag{218}$$

where

$$\Phi(y) := \frac{\omega a}{\sqrt{2}\, y\, v_T} - \frac{\pi}{4}, \tag{219}$$

$$E_B = \frac{2\sqrt{2}\, v_T}{\pi\, \omega a}, \tag{220}$$

$$B_1 = \frac{\pi\sqrt{2}\, v_T}{4\omega L}, \qquad B_2 = \frac{3v_T^2}{2\omega^2 L^2}, \tag{221}$$

and $\mathcal{D}(y; r)$ is now understood as in (**??**).

In Region I one has $ka > x_{c,B}$, and for ka sufficiently above the cutoff the phase $\Phi(y)$ varies rapidly compared with the slowly varying amplitude. It is therefore appropriate to replace the fast oscillations by their average intensity,

$$\cos^2\Phi(y) \;\longrightarrow\; \left\langle \cos^2\Phi \right\rangle = \frac{1}{2}. \tag{222}$$

With this envelope approximation, (218) becomes

$$\begin{aligned} \mathcal{I}_{\mathrm{I}}^{(\mathrm{env})}(y) &:= -\sqrt{2}\, r\, E_B\, \frac{B_1\, y^3 - B_2\, y^4}{\mathcal{D}(y; r)}\, \frac{1}{2} \\ &= -\frac{\sqrt{2}}{2}\, r\, E_B\, \frac{B_1\, y^3 - B_2\, y^4}{\mathcal{D}(y; r)}. \end{aligned} \tag{223}$$

Using the explicit form of E_B,

$$E_B = \frac{2\sqrt{2}\, v_T}{\pi\, \omega a}, \tag{224}$$

we simplify the prefactor

$$\frac{\sqrt{2}}{2} E_B = \frac{\sqrt{2}}{2} \cdot \frac{2\sqrt{2}\, v_T}{\pi\, \omega a} = \frac{2v_T}{\pi\, \omega a}. \tag{225}$$

Hence

$$\mathcal{I}_{\mathrm{I}}^{(\mathrm{env})}(y) = -\, r\, \frac{2v_T}{\pi\, \omega a}\, \frac{B_1\, y^3 - B_2\, y^4}{E\, y^4 + F\, y^2 + G}. \tag{226}$$

We now define the Region I envelope integral (inner integral only, without restoring any overall prefactor from the QTN spectrum),

$$I_{\mathrm{I}}^{(\mathrm{env})}(\omega) := \int_0^{y_{c,B}} \mathcal{I}_{\mathrm{I}}^{(\mathrm{env})}(y)\, dy. \tag{227}$$

Substituting (226) gives

$$I_{\mathrm{I}}^{(\mathrm{env})}(\omega) = -\, r\, \frac{2v_T}{\pi\, \omega a} \int_0^{y_{c,B}} \frac{B_1\, y^3 - B_2\, y^4}{E\, y^4 + F\, y^2 + G}\, dy. \tag{228}$$

It is natural to factor out B_1 and B_2 and separate (228) into two basic integrals. Define

$$I_1(r;\omega,L,a) := \int_0^{y_{c,B}} \frac{y^3}{E\,y^4 + F\,y^2 + G}\,dy, \tag{229}$$

$$I_2(r;\omega,L,a) := \int_0^{y_{c,B}} \frac{y^4}{E\,y^4 + F\,y^2 + G}\,dy. \tag{230}$$

Then (228) can be written as

$$I_{\mathrm{I}}^{(\mathrm{env})}(\omega) = -\,r\,\frac{2v_T}{\pi\,\omega a}\left[B_1\,I_1(r;\omega,L,a) - B_2\,I_2(r;\omega,L,a)\right]. \tag{231}$$

The integrals I_1 and I_2 are now in a form suitable for symbolic or numerical evaluation using a computer algebra system, with a denominator characterised globally by the positive real coefficients E, F, G defined above.

Analytic evaluation of I_1 on $[0, y_{c,B}]$

From the large–x integrand we isolate the y^3–contribution on the interval $0 < y < y_{c,B}$: *Problem.*

$$I_1(y_{c,B}) := \int_0^{y_{c,B}} \frac{y^3}{Ey^4 + Fy^2 + G}\,dy. \tag{232}$$

We first find an antiderivative of $\displaystyle\int \frac{y^3}{Ey^4 + Fy^2 + G}\,dy$ and then insert the limits. *Prepare for substitution.*

$$\int \frac{y^3}{Ey^4 + Fy^2 + G}\,dy = \int \frac{y^2}{2y}\,\frac{2y}{Ey^4 + Fy^2 + G}\,dy. \tag{233}$$

Substitute $u = y^2$.

$$u = y^2, \qquad du = 2y\,dy, \qquad y^4 = u^2, \tag{234}$$

so that

$$\int \frac{y^3}{Ey^4 + Fy^2 + G}\,dy = \frac{1}{2}\int \frac{u}{Eu^2 + Fu + G}\,du. \tag{235}$$

Now solving $\displaystyle\int \frac{u}{Eu^2 + Fu + G}\,du$. Write

$$u = \frac{1}{2E}(2Eu + F) - \frac{F}{2E}, \tag{236}$$

which gives

$$\int \frac{u}{Eu^2 + Fu + G}\,du = \frac{1}{2E}\int \frac{2Eu + F}{Eu^2 + Fu + G}\,du - \frac{F}{2E}\int \frac{1}{Eu^2 + Fu + G}\,du. \tag{237}$$

First piece (logarithm). Let

$$v = Eu^2 + Fu + G \quad \Rightarrow \quad dv = (2Eu + F)\,du, \tag{238}$$

so

$$\int \frac{2Eu + F}{Eu^2 + Fu + G}\,du = \int \frac{1}{v}\,dv = \ln v + C = \ln\left(Eu^2 + Fu + G\right) + C. \tag{239}$$

Second piece (arctangent). Define

$$J := \int \frac{1}{Eu^2 + Fu + G}\, du. \tag{240}$$

Complete the square:

$$Eu^2 + Fu + G = \Big(\sqrt{E}\, u + \frac{F}{2\sqrt{E}}\Big)^2 + \Big(G - \frac{F^2}{4E}\Big). \tag{241}$$

Set

$$v = \frac{\sqrt{E}\, u + \dfrac{F}{2\sqrt{E}}}{\sqrt{G - \dfrac{F^2}{4E}}} = \frac{2Eu + F}{2\sqrt{E}\sqrt{G - \dfrac{F^2}{4E}}}, \tag{242}$$

so that

$$dv = \frac{\sqrt{E}}{\sqrt{G - \dfrac{F^2}{4E}}}\, du, \qquad du = \frac{\sqrt{G - \dfrac{F^2}{4E}}}{\sqrt{E}}\, dv. \tag{243}$$

Then

$$J = \int \frac{1}{(\sqrt{E}\, u + \frac{F}{2\sqrt{E}})^2 + (G - \frac{F^2}{4E})}\, du \tag{244}$$

$$= \frac{1}{\sqrt{E}\sqrt{G - \dfrac{F^2}{4E}}} \int \frac{1}{v^2 + 1}\, dv \tag{245}$$

$$= \frac{1}{\sqrt{E}\sqrt{G - \dfrac{F^2}{4E}}}\ \arctan(v) + C. \tag{246}$$

Undoing the substitution for v,

$$J = \frac{1}{\sqrt{E}\sqrt{G - \dfrac{F^2}{4E}}}\, \arctan\left(\frac{2Eu + F}{2\sqrt{E}\sqrt{G - \dfrac{F^2}{4E}}}\right) + C. \tag{247}$$

Simplify using $\Delta := 4EG - F^2$. We have

$$G - \frac{F^2}{4E} = \frac{\Delta}{4E}, \qquad \sqrt{G - \frac{F^2}{4E}} = \frac{\sqrt{\Delta}}{2\sqrt{E}}, \tag{248}$$

so

$$\frac{1}{\sqrt{E}\sqrt{G - \dfrac{F^2}{4E}}} = \frac{2}{\sqrt{\Delta}}, \qquad \frac{2Eu + F}{2\sqrt{E}\sqrt{G - \dfrac{F^2}{4E}}} = \frac{2Eu + F}{\sqrt{\Delta}}. \tag{249}$$

Hence

$$J = \int \frac{1}{Eu^2 + Fu + G}\, du = \frac{2}{\sqrt{\Delta}} \arctan\!\left(\frac{2Eu + F}{\sqrt{\Delta}}\right) + C. \tag{250}$$

Assemble $\int \frac{u}{Eu^2 + Fu + G}\, du$.

$$\int \frac{u}{Eu^2 + Fu + G}\, du = \frac{1}{2E} \ln\!\left(Eu^2 + Fu + G\right) - \frac{F}{2E}\, J \tag{251}$$

$$= \frac{1}{2E} \ln\!\left(Eu^2 + Fu + G\right) - \frac{F}{E\sqrt{\Delta}} \arctan\!\left(\frac{2Eu + F}{\sqrt{\Delta}}\right) + C. \tag{252}$$

Undo the substitution $u = y^2$. Recall the factor $1/2$ from the first substitution:

$$\int \frac{y^3}{Ey^4 + Fy^2 + G}\, dy = \frac{1}{2} \int \frac{u}{Eu^2 + Fu + G}\, du \tag{253}$$

$$= \frac{1}{4E} \ln\!\left(Eu^2 + Fu + G\right) - \frac{F}{2E\sqrt{\Delta}} \arctan\!\left(\frac{2Eu + F}{\sqrt{\Delta}}\right) + C \tag{254}$$

$$= \frac{1}{4E} \ln\!\left(Ey^4 + Fy^2 + G\right) - \frac{F}{2E\sqrt{\Delta}} \arctan\!\left(\frac{2Ey^2 + F}{\sqrt{\Delta}}\right) + C. \tag{255}$$

Evaluate on $[0, y_{c,B}]$.

$$I_1(y_{c,B}) := \int_0^{y_{c,B}} \frac{y^3}{Ey^4 + Fy^2 + G}\, dy \tag{256}$$

$$= \left[\frac{1}{4E} \ln\left|Ey^4 + Fy^2 + G\right| - \frac{F}{2E\sqrt{\Delta}} \arctan\!\left(\frac{2Ey^2 + F}{\sqrt{\Delta}}\right)\right]_{y=0}^{y=y_{c,B}}. \tag{257}$$

At the lower limit $y = 0$ we have $Ey^4 + Fy^2 + G = G$ and $\arctan\!\left(\frac{2Ey^2+F}{\sqrt{\Delta}}\right) = \arctan\!\left(\frac{F}{\sqrt{\Delta}}\right)$, so

$$I_1(y_{c,B}) = \frac{1}{4E}\left[\ln\left|Ey_{c,B}^4 + Fy_{c,B}^2 + G\right| - \ln|G|\right] \tag{258}$$

$$- \frac{F}{2E\sqrt{\Delta}}\left[\arctan\!\left(\frac{2Ey_{c,B}^2 + F}{\sqrt{\Delta}}\right) - \arctan\!\left(\frac{F}{\sqrt{\Delta}}\right)\right]. \tag{259}$$

Put everything over a common denominator. First note that

$$16E^2G - 4EF^2 = 4E(4EG - F^2) = 4E\Delta.$$

Rewrite (259) with denominator $4E\Delta$:

$$I_1(y_{c,B}) = \frac{1}{4E\Delta}\Bigg\{\Delta\left[\ln\left|Ey_{c,B}^4 + Fy_{c,B}^2 + G\right| - \ln|G|\right] \tag{260}$$

$$- 2F\sqrt{\Delta}\left[\arctan\!\left(\frac{2Ey_{c,B}^2 + F}{\sqrt{\Delta}}\right) - \arctan\!\left(\frac{F}{\sqrt{\Delta}}\right)\right]\Bigg\}. \tag{261}$$

Expand the bracket and use $\Delta = 4EG - F^2$:

$$I_1(y_{c,B}) = \frac{1}{4E\Delta}\Big[\Delta \ln\left|Ey_{c,B}^4 + Fy_{c,B}^2 + G\right| - \Delta \ln|G| \tag{262}$$

$$- 2F\sqrt{\Delta}\, \arctan\!\left(\frac{2Ey_{c,B}^2 + F}{\sqrt{\Delta}}\right) + 2F\sqrt{\Delta}\, \arctan\!\left(\frac{F}{\sqrt{\Delta}}\right)\Big] \tag{263}$$

Cleaned–up form. Equation (262) simplifies by cancelling the common factor $\Delta = 4EG - F^2$ in numerator and denominator and regrouping the logarithms:

$$I_1(y_{c,B}) = \frac{1}{4E}\left[\ln\!\left(\frac{|Ey_{c,B}^4 + Fy_{c,B}^2 + G|}{|G|}\right) + \frac{2F}{\sqrt{4EG-F^2}}\left(\arctan\frac{F}{\sqrt{4EG-F^2}} - \arctan\frac{2Ey_{c,B}^2+F}{\sqrt{4EG-F^2}}\right)\right]. \tag{264}$$

This is an equivalent but much more compact expression for the definite integral $I_1(y_{c,B})$.

Analytic continuation of the arctangent term for QTN parameters. For the quasi–thermal noise parameters we substitute

$$E = (1-r)^2, \qquad F = 1+r, \qquad G = \frac{1}{4}, \qquad r = \left(\frac{\omega_p}{\omega}\right)^2,$$

into the discriminant

$$\Delta = 4EG - F^2.$$

A short calculation gives

$$\Delta = 4(1-r)^2\left(\frac{1}{4}\right) - (1+r)^2 \tag{265}$$

$$= (1-r)^2 - (1+r)^2 \tag{266}$$

$$= -4r < 0, \tag{267}$$

so the square root becomes purely imaginary:

$$\sqrt{\Delta} = \sqrt{-4r} = 2i\sqrt{r}.$$

The arctangent block in (264) is

$$\mathcal{A} := \frac{2F}{\sqrt{\Delta}}\left(\arctan\frac{F}{\sqrt{\Delta}} - \arctan\frac{2Ey_{c,B}^2+F}{\sqrt{\Delta}}\right).$$

Substituting $F = 1+r$, $E = (1-r)^2$, $\sqrt{\Delta} = 2i\sqrt{r}$ gives

$$\frac{2F}{\sqrt{\Delta}} = \frac{2(1+r)}{2i\sqrt{r}} = -\,i\,\frac{1+r}{\sqrt{r}}.$$

The two arctan arguments become

$$\frac{F}{\sqrt{\Delta}} = \frac{1+r}{2i\sqrt{r}} = -\,i\,\frac{1+r}{2\sqrt{r}} = -i\,b, \tag{268}$$

$$\frac{2Ey_{c,B}^2+F}{\sqrt{\Delta}} = \frac{2(1-r)^2y_{c,B}^2+1+r}{2i\sqrt{r}} = -\,i\,\frac{2(1-r)^2y_{c,B}^2+1+r}{2\sqrt{r}} = -i\,a, \tag{269}$$

where we define

$$a := \frac{2(1-r)^2y_{c,B}^2+1+r}{2\sqrt{r}}, \qquad b := \frac{1+r}{2\sqrt{r}}.$$

Using the standard identity for imaginary arguments,

$$\arctan(-ix) = -\,i\,\operatorname{artanh}(x),$$

we obtain

$$\arctan(-ib) = -i\ \operatorname{artanh}(b), \tag{270}$$
$$\arctan(-ia) = -i\ \operatorname{artanh}(a). \tag{271}$$

Thus

$$\arctan(-ib) - \arctan(-ia) = -i\ \operatorname{artanh}(b) + i\ \operatorname{artanh}(a) \tag{272}$$
$$= i\big(\operatorname{artanh}(a) - \operatorname{artanh}(b)\big). \tag{273}$$

Substituting this into $\mathcal{A}$ gives

$$\mathcal{A} = -i\,\frac{1+r}{\sqrt{r}} \cdot i\big(\operatorname{artanh}(a) - \operatorname{artanh}(b)\big) \tag{274}$$
$$= \frac{1+r}{\sqrt{r}}\big(\operatorname{artanh}(a) - \operatorname{artanh}(b)\big), \tag{275}$$

since $-i \cdot i = 1$.
Finally, the logarithmic term in (264) becomes

$$\ln\left(\frac{|Ey_{c,B}^4 + Fy_{c,B}^2 + G|}{|G|}\right) = \ln\Big(4(1-r)^2 y_{c,B}^4 + 4(1+r)y_{c,B}^2 + 1\Big),$$

because $G = 1/4$.
Substituting these results back into (264) yields the real hyperbolic form of the definite integral. With $E = (1-r)^2$, $F = 1+r$, $G = 1/4$ and $r = (\omega_p/\omega)^2$, the compact definite integral becomes

$$I_1(y_{c,B}) = \frac{1}{4(1-r)^2}\Bigg[\ln\big|\,4(1-r)^2 y_{c,B}^4 + 4(1+r)y_{c,B}^2 + 1\,\big| \tag{276}$$
$$+\frac{1+r}{\sqrt{r}}\left(\operatorname{artanh}\frac{2(1-r)^2 y_{c,B}^2 + 1 + r}{2\sqrt{r}} - \operatorname{artanh}\frac{1+r}{2\sqrt{r}}\right)\Bigg], \tag{277}$$

where

$$r = \left(\frac{\omega_p}{\omega}\right)^2.$$

For numerical work and plotting it is convenient to combine the difference of inverse hyperbolic tangents into a single artanh using

$$\operatorname{artanh} a - \operatorname{artanh} b = \operatorname{artanh}\left(\frac{a-b}{1-ab}\right),$$

with

$$a = \frac{2(1-r)^2 y_{c,B}^2 + 1 + r}{2\sqrt{r}}, \qquad b = \frac{1+r}{2\sqrt{r}}.$$

A short simplification gives

$$\frac{a-b}{1-ab} = -\frac{4\sqrt{r}\,y_{c,B}^2}{2(r+1)y_{c,B}^2 + 1} =: \xi(r, y_{c,B}),$$

so that

$$I_1(y_{c,B}) = \frac{1}{4(1-r)^2}\left[\ln\left|\, 4(1-r)^2 y_{c,B}^4 + 4(1+r)y_{c,B}^2 + 1 \,\right| + \frac{1+r}{\sqrt{r}}\ \operatorname{artanh}(\xi(r, y_{c,B}))\right], \tag{278}$$

$$\xi(r, y_{c,B}) = -\frac{4\sqrt{r}\, y_{c,B}^2}{2(r+1)y_{c,B}^2 + 1}. \tag{279}$$

For $r > 0$ and $y_{c,B} > 0$ one has $|\xi(r, y_{c,B})| < 1$, so this form is real–valued and well suited for plotting as a function of ω.

Analytic evaluation of I_1 on $[0, y_{c,B}]$ for $r = 1$

In the special case $r = 1$ the QTN parameters become

$$E = (1-r)^2 = 0, \qquad F = 1 + r = 2, \qquad G = \frac{1}{4},$$

so the integral

$$I_1(y_{c,B}) := \int_0^{y_{c,B}} \frac{y^3}{Ey^4 + Fy^2 + G}\, dy$$

reduces to

$$I_1^{(r=1)}(y_{c,B}) := \int_0^{y_{c,B}} \frac{y^3}{2y^2 + \frac{1}{4}}\, dy. \tag{280}$$

1. Simplification of the integrand We first simplify the integrand:

$$\frac{y^3}{2y^2 + \frac{1}{4}} = \frac{y^3}{\frac{1}{4}(8y^2 + 1)} = 4\,\frac{y^3}{8y^2 + 1}. \tag{281}$$

Thus

$$I_1^{(r=1)}(y_{c,B}) = 4\int_0^{y_{c,B}} \frac{y^3}{8y^2 + 1}\, dy. \tag{282}$$

2. Substitution $u = y^2$ We compute the indefinite integral

$$\int \frac{y^3}{8y^2 + 1}\, dy$$

and then insert the limits.

Substitute

$$u = y^2, \qquad du = 2y\, dy, \qquad y^2 = u. \tag{283}$$

Then

$$y^3\, dy = y^2 \cdot y\, dy = u \cdot \frac{du}{2} = \frac{u}{2}\, du,$$

and

$$8y^2 + 1 = 8u + 1.$$

Therefore

$$\int \frac{y^3}{8y^2 + 1}\, dy = \int \frac{\frac{u}{2}}{8u + 1}\, du = \frac{1}{2}\int \frac{u}{8u + 1}\, du. \tag{284}$$

3. Decomposition of $\frac{u}{8u+1}$ Write

$$\frac{u}{8u+1} = \frac{1}{8} - \frac{1}{8(8u+1)}. \tag{285}$$

(Verification: $\frac{1}{8} - \frac{1}{8(8u+1)} = \frac{8u+1-1}{8(8u+1)} = \frac{8u}{8(8u+1)} = \frac{u}{8u+1}$.)

Thus

$$\frac{1}{2}\int \frac{u}{8u+1}\,du = \frac{1}{2}\int \left(\frac{1}{8} - \frac{1}{8(8u+1)}\right) du \tag{286}$$

$$= \frac{1}{2}\left[\frac{1}{8}\int du - \frac{1}{8}\int \frac{1}{8u+1}\,du\right]. \tag{287}$$

Compute the two integrals:

$$\int du = u + C_1, \tag{288}$$

$$\int \frac{1}{8u+1}\,du = \frac{1}{8}\ln|8u+1| + C_2. \tag{289}$$

Therefore

$$\frac{1}{2}\int \frac{u}{8u+1}\,du = \frac{1}{2}\left[\frac{1}{8}u - \frac{1}{8}\cdot\frac{1}{8}\ln|8u+1|\right] + C \tag{290}$$

$$= \frac{u}{16} - \frac{1}{128}\ln|8u+1| + C. \tag{291}$$

Undoing the substitution $u = y^2$ gives the indefinite integral

$$\int \frac{y^3}{8y^2+1}\,dy = \frac{y^2}{16} - \frac{1}{128}\ln(8y^2+1) + C. \tag{292}$$

4. Indefinite integral for the original integrand Recall

$$\frac{y^3}{2y^2+\frac{1}{4}} = 4\,\frac{y^3}{8y^2+1},$$

so

$$\int \frac{y^3}{2y^2+\frac{1}{4}}\,dy = 4\int \frac{y^3}{8y^2+1}\,dy \tag{293}$$

$$= 4\left(\frac{y^2}{16} - \frac{1}{128}\ln(8y^2+1)\right) + C \tag{294}$$

$$= \frac{y^2}{4} - \frac{1}{32}\ln(8y^2+1) + C. \tag{295}$$

Thus, for $r = 1$,

$$\int \frac{y^3}{Ey^4+Fy^2+G}\,dy = \frac{y^2}{4} - \frac{1}{32}\ln(8y^2+1) + C. \tag{296}$$

5. Definite integral on $[0, y_{c,B}]$ for $r = 1$ We now evaluate

$$I_1^{(r=1)}(y_{c,B}) = \int_0^{y_{c,B}} \frac{y^3}{2y^2+\frac{1}{4}}\,dy$$

using the antiderivative above:

$$I_1^{(r=1)}(y_{c,B}) = \left[\frac{y^2}{4} - \frac{1}{32} \ln(8y^2 + 1) \right]_{y=0}^{y=y_{c,B}}. \tag{297}$$

At the upper limit $y = y_{c,B}$:

$$\frac{y_{c,B}^2}{4} - \frac{1}{32} \ln(8y_{c,B}^2 + 1).$$

At the lower limit $y = 0$:

$$\frac{0^2}{4} - \frac{1}{32} \ln(8 \cdot 0^2 + 1) = 0 - \frac{1}{32} \ln(1) = 0.$$

Hence

$$I_1^{(r=1)}(y_{c,B}) = \frac{y_{c,B}^2}{4} - \frac{1}{32} \ln\bigl(8y_{c,B}^2 + 1\bigr). \tag{298}$$

This is the fully explicit, real–valued expression for the I_1 contribution on $[0, y_{c,B}]$ in the special case $r = 1$.

Analytic evaluation of I_2 on $[0, y_{c,B}]$

From the large–x integrand we isolate the y^4–contribution on the interval $0 < y < y_{c,B}$: *Problem.*

$$I_2(y_{c,B}) := \int_0^{y_{c,B}} \frac{y^4}{Ey^4 + Fy^2 + G}\, dy, \tag{299}$$

with real parameters E, F, G and $E \neq 0$. The physically relevant values $E = (1-r)^2$, $F = 1+r$, $G = \frac{1}{4}$, $r > 0$ will be substituted later.

1. Polynomial division Introduce the biquadratic

$$D(y) := Ey^4 + Fy^2 + G. \tag{300}$$

We seek constants A, B, C such that

$$\frac{y^4}{D(y)} = A + \frac{By^2 + C}{D(y)}. \tag{301}$$

Multiplying both sides by $D(y)$,

$$y^4 = A\,(Ey^4 + Fy^2 + G) + (By^2 + C). \tag{302}$$

Expanding the right–hand side,

$$y^4 = AEy^4 + AFy^2 + AG + By^2 + C. \tag{303}$$

Equating coefficients of like powers of y gives

$$y^4: \quad 1 = AE \quad \Rightarrow \quad A = \frac{1}{E}, \tag{304}$$

$$y^2: \quad 0 = AF + B = \frac{F}{E} + B \quad \Rightarrow \quad B = -\frac{F}{E}, \tag{305}$$

$$y^0: \quad 0 = AG + C = \frac{G}{E} + C \quad \Rightarrow \quad C = -\frac{G}{E}. \tag{306}$$

Thus

$$\frac{y^4}{D(y)} = \frac{1}{E} + \frac{-\frac{F}{E}y^2 - \frac{G}{E}}{D(y)} = \frac{1}{E} - \frac{Fy^2 + G}{E\,D(y)}. \tag{307}$$

Hence

$$I(E,F,G) := \int \frac{y^4}{D(y)}\,dy = \frac{1}{E}\int dy - \frac{1}{E}\int \frac{Fy^2+G}{D(y)}\,dy \tag{308}$$

$$= \frac{y}{E} - \frac{1}{E}J, \tag{309}$$

where

$$J := \int \frac{Fy^2+G}{D(y)}\,dy. \tag{310}$$

2. Factorisation of the biquadratic denominator Treat $D(y)$ as a quadratic in $u := y^2$:

$$D(y) = Ey^4 + Fy^2 + G = Eu^2 + Fu + G, \qquad u = y^2. \tag{311}$$

The roots of $Eu^2 + Fu + G = 0$ are

$$\lambda_{1,2} = \frac{-F \pm \sqrt{\Delta}}{2E}, \qquad \Delta := F^2 - 4EG. \tag{312}$$

Thus

$$Eu^2 + Fu + G = E(u-\lambda_1)(u-\lambda_2) \quad \Longrightarrow \quad D(y) = E(y^2-\lambda_1)(y^2-\lambda_2). \tag{313}$$

We now write a partial fraction decomposition in y^2:

$$\frac{Fy^2+G}{D(y)} = \frac{Fy^2+G}{E(y^2-\lambda_1)(y^2-\lambda_2)} = \frac{A}{y^2-\lambda_1} + \frac{B}{y^2-\lambda_2}, \tag{314}$$

for some constants A, B depending on E, F, G.

Multiplying through by $E(y^2-\lambda_1)(y^2-\lambda_2)$ yields

$$Fy^2 + G = E\Big[A(y^2-\lambda_2) + B(y^2-\lambda_1)\Big] \tag{315}$$

$$= E\Big[(A+B)y^2 - (A\lambda_2 + B\lambda_1)\Big]. \tag{316}$$

Equating coefficients,

$$y^2: \quad F = E(A+B) \quad \Rightarrow \quad A+B = \frac{F}{E}, \tag{317}$$

$$y^0: \quad G = -E(A\lambda_2 + B\lambda_1) \quad \Rightarrow \quad A\lambda_2 + B\lambda_1 = -\frac{G}{E}. \tag{318}$$

From $A + B = F/E$ we have

$$B = \frac{F}{E} - A. \tag{319}$$

Substitute into the second equation:

$$A\lambda_2 + \Big(\frac{F}{E} - A\Big)\lambda_1 = A(\lambda_2 - \lambda_1) + \frac{F}{E}\lambda_1 = -\frac{G}{E}, \tag{320}$$

so

$$A(\lambda_2 - \lambda_1) = -\frac{G}{E} - \frac{F}{E}\lambda_1 = -\frac{G + F\lambda_1}{E}, \tag{321}$$

and hence

$$A = -\frac{G + F\lambda_1}{E(\lambda_2 - \lambda_1)}. \tag{322}$$

Then

$$B = \frac{F}{E} - A = \frac{F}{E} + \frac{G + F\lambda_1}{E(\lambda_2 - \lambda_1)} \tag{323}$$

$$= \frac{F(\lambda_2 - \lambda_1) + G + F\lambda_1}{E(\lambda_2 - \lambda_1)} = \frac{F\lambda_2 + G}{E(\lambda_2 - \lambda_1)}. \tag{324}$$

Thus

$$\frac{Fy^2 + G}{D(y)} = \frac{A}{y^2 - \lambda_1} + \frac{B}{y^2 - \lambda_2}, \tag{325}$$

with

$$A = -\frac{G + F\lambda_1}{E(\lambda_2 - \lambda_1)}, \qquad B = \frac{F\lambda_2 + G}{E(\lambda_2 - \lambda_1)}. \tag{326}$$

3. Elementary primitives By linearity,

$$J = \int \frac{Fy^2 + G}{D(y)}\, dy = A \int \frac{dy}{y^2 - \lambda_1} + B \int \frac{dy}{y^2 - \lambda_2}. \tag{327}$$

Hence

$$I(E, F, G) = \int \frac{y^4}{Ey^4 + Fy^2 + G}\, dy = \frac{y}{E} - \frac{A}{E} \int \frac{dy}{y^2 - \lambda_1} - \frac{B}{E} \int \frac{dy}{y^2 - \lambda_2}. \tag{328}$$

The antiderivative of $1/(y^2 - \lambda)$ depends on the sign of λ. Define the unified primitive

$$I_1(y; \lambda) := \begin{cases} \dfrac{1}{2\sqrt{\lambda}} \ln \left| \dfrac{y - \sqrt{\lambda}}{y + \sqrt{\lambda}} \right|, & \lambda > 0, \\ \dfrac{1}{\sqrt{-\lambda}} \arctan\!\left(\dfrac{y}{\sqrt{-\lambda}} \right), & \lambda < 0. \end{cases} \tag{329}$$

Then

$$\int \frac{dy}{y^2 - \lambda} = I_1(y; \lambda) + C, \tag{330}$$

and the indefinite integral can be written compactly as

$$\boxed{I(E, F, G) = \int \frac{y^4}{Ey^4 + Fy^2 + G}\, dy = \frac{y}{E} - \frac{A}{E} I_1(y; \lambda_1) - \frac{B}{E} I_1(y; \lambda_2) + C,} \tag{331}$$

with

$$\lambda_{1,2} = \frac{-F \pm \sqrt{F^2 - 4EG}}{2E}, \qquad A = -\frac{G + F\lambda_1}{E(\lambda_2 - \lambda_1)}, \qquad B = \frac{F\lambda_2 + G}{E(\lambda_2 - \lambda_1)}. \tag{332}$$

4. Definite integral on $[0, y_{c,B}]$ We now define

$$I_2(y_{c,B}) := \int_0^{y_{c,B}} \frac{y^4}{Ey^4 + Fy^2 + G}\, dy. \tag{333}$$

Using the antiderivative above,

$$I_2(y_{c,B}) = \left[\frac{y}{E} - \frac{A}{E}\, I_1(y;\lambda_1) - \frac{B}{E}\, I_1(y;\lambda_2) \right]_{y=0}^{y=y_{c,B}} \tag{334}$$

$$= \frac{y_{c,B}}{E} - \frac{A}{E}\, I_1(y_{c,B};\lambda_1) - \frac{B}{E}\, I_1(y_{c,B};\lambda_2) \tag{335}$$

$$- \left(\frac{0}{E} - \frac{A}{E}\, I_1(0;\lambda_1) - \frac{B}{E}\, I_1(0;\lambda_2) \right). \tag{336}$$

With $E = (1-r)^2$, $F = 1+r$, $G = 1/4$ and $r = (\omega_p/\omega)^2$, the compact definite integral becomes
Thus the roots are

$$\lambda_{1,2}(r) = \frac{-F \pm \sqrt{\Delta}}{2E} = \frac{-(1+r) \pm 2\sqrt{r}}{2(1-r)^2}, \tag{337}$$

so for $r > 0$ and $r \neq 1$ one has

$$\lambda_1(r) < 0, \qquad \lambda_2(r) < 0.$$

For convenience we retain the compact notation

$$I_1(y;\lambda) = \begin{cases} \dfrac{1}{2\sqrt{\lambda}} \ln \left| \dfrac{y - \sqrt{\lambda}}{y + \sqrt{\lambda}} \right|, & \lambda > 0, \\ \dfrac{1}{\sqrt{-\lambda}} \arctan\!\left(\dfrac{y}{\sqrt{-\lambda}} \right), & \lambda < 0, \end{cases}$$

and in the present case we always use the $\lambda < 0$ branch:

$$I_1(y;\lambda_i(r)) = \frac{1}{\sqrt{-\lambda_i(r)}} \arctan\!\left(\frac{y}{\sqrt{-\lambda_i(r)}} \right), \qquad i = 1, 2. \tag{338}$$

The partial–fraction coefficients become

$$A(r) = -\frac{G + F\lambda_1(r)}{E\big(\lambda_2(r) - \lambda_1(r)\big)}, \qquad B(r) = \frac{F\lambda_2(r) + G}{E\big(\lambda_2(r) - \lambda_1(r)\big)}, \tag{339}$$

and inserting $E = (1-r)^2$, $F = 1+r$, $G = \frac{1}{4}$, together with

$$\lambda_2(r) - \lambda_1(r) = -\frac{2\sqrt{r}}{(1-r)^2},$$

one finds after a short simplification that

$$A(r) = -\frac{(\sqrt{r}-1)^2}{8\sqrt{r}\,(\sqrt{r}+1)^2}, \tag{340}$$

$$B(r) = \frac{(\sqrt{r}+1)^2}{8\sqrt{r}\,(\sqrt{r}-1)^2}, \qquad r > 0,\ r \neq 1. \tag{341}$$

Therefore the indefinite integral with QTN parameters can be written as

$$I_2(r;y) := \int \frac{y^4}{(1-r)^2 y^4 + (1+r) y^2 + \frac{1}{4}}\, dy = \frac{y}{(1-r)^2} - \frac{A(r)}{(1-r)^2}\, I_1(y;\lambda_1(r)) - \frac{B(r)}{(1-r)^2}\, I_1(y;\lambda_2(r)) + C, \tag{342}$$

where

$$\lambda_{1,2}(r) = \frac{-(1+r) \pm 2\sqrt{r}}{2(1-r)^2}, \tag{343}$$

$$A(r) = -\frac{(\sqrt{r}-1)^2}{8\sqrt{r}\,(\sqrt{r}+1)^2}, \tag{344}$$

$$B(r) = \frac{(\sqrt{r}+1)^2}{8\sqrt{r}\,(\sqrt{r}-1)^2}, \tag{345}$$

and $I_1(y;\lambda)$ is evaluated on the $\lambda < 0$ (arctan) branch.

Definite integral on $[0, y_{c,B}]$. For $r > 0$, $r \neq 1$ we define

$$I_2(r; y_{c,B}) := \int_0^{y_{c,B}} \frac{y^4}{(1-r)^2 y^4 + (1+r)y^2 + \frac{1}{4}}\,dy. \tag{346}$$

Using $I_2(0;\lambda_i(r)) = 0$ for $\lambda_i(r) < 0$, the constant cancels and we obtain

$$I_2(r; y_{c,B}) = \left[\frac{y}{(1-r)^2} - \frac{A(r)}{(1-r)^2}\,I_1(y;\lambda_1(r)) - \frac{B(r)}{(1-r)^2}\,I_1(y;\lambda_2(r))\right]_{y=0}^{y=y_{c,B}} \tag{347}$$

$$= \frac{1}{(1-r)^2}\Big\{y_{c,B} - A(r)\,I_1\big(y_{c,B};\lambda_1(r)\big) - B(r)\,I_1\big(y_{c,B};\lambda_2(r)\big)\Big\}, \tag{348}$$

This can then be written explicitly as

$$I_2(y_{c,B}) = \frac{1}{(1-r)^2}\left\{y_{c,B} + \frac{(\sqrt{r}-1)^2}{8\sqrt{r}(\sqrt{r}+1)^2\,\sqrt{-\lambda_1(r)}}\arctan\left(\frac{y_{c,B}}{\sqrt{-\lambda_1(r)}}\right)\right. \tag{349}$$

$$\left. - \frac{(\sqrt{r}+1)^2}{8\sqrt{r}(\sqrt{r}-1)^2\,\sqrt{-\lambda_2(r)}}\arctan\left(\frac{y_{c,B}}{\sqrt{-\lambda_2(r)}}\right)\right\}, \tag{350}$$

where

$$r = \left(\frac{\omega_p}{\omega}\right)^2, \qquad \lambda_1(r) = \frac{-(1+r)+2\sqrt{r}}{2(1-r)^2}, \qquad \lambda_2(r) = \frac{-(1+r)-2\sqrt{r}}{2(1-r)^2}.$$

Analytic evaluation of I_2 on $[0, y_{c,B}]$ for $r = 1$

In the special case $r = 1$ the QTN parameters become

$$E = (1-r)^2 = 0, \qquad F = 1 + r = 2, \qquad G = \frac{1}{4},$$

so the integral

$$I_2(y_{c,B}) := \int_0^{y_{c,B}} \frac{y^4}{Ey^4 + Fy^2 + G}\,dy$$

reduces to

$$I_2^{(r=1)}(y_{c,B}) := \int_0^{y_{c,B}} \frac{y^4}{2y^2 + \frac{1}{4}}\,dy. \tag{351}$$

1. Simplification of the integrand We first simplify the integrand:

$$\frac{y^4}{2y^2 + \frac{1}{4}} = \frac{y^4}{\frac{1}{4}(8y^2+1)} = 4\,\frac{y^4}{8y^2+1}. \tag{352}$$

Thus

$$I_2^{(r=1)}(y_{c,B}) = 4\int_0^{y_{c,B}} \frac{y^4}{8y^2+1}\,dy. \tag{353}$$

2. Polynomial division in y We compute the indefinite integral

$$\int \frac{y^4}{8y^2+1}\,dy$$

and then apply the limits.

Perform polynomial division:

$$\frac{y^4}{8y^2+1} = \frac{1}{8}y^2 - \frac{1}{8}\,\frac{y^2}{8y^2+1}. \tag{354}$$

(Verification: $\frac{1}{8}y^2(8y^2+1) = y^4 + \frac{1}{8}y^2$; subtracting from y^4 leaves $-\frac{1}{8}y^2$, hence the remainder term $-\frac{1}{8}\frac{y^2}{8y^2+1}$.)

Thus

$$\int \frac{y^4}{8y^2+1}\,dy = \int \left(\frac{1}{8}y^2 - \frac{1}{8}\,\frac{y^2}{8y^2+1}\right) dy \tag{355}$$

$$= \frac{1}{8}\int y^2\,dy - \frac{1}{8}\int \frac{y^2}{8y^2+1}\,dy. \tag{356}$$

3. Decomposition of $\dfrac{y^2}{8y^2+1}$ Write

$$\frac{y^2}{8y^2+1} = \frac{1}{8} - \frac{1}{8(8y^2+1)}. \tag{357}$$

Hence

$$-\frac{1}{8}\int \frac{y^2}{8y^2+1}\,dy = -\frac{1}{8}\int \left(\frac{1}{8} - \frac{1}{8(8y^2+1)}\right) dy \tag{358}$$

$$= -\frac{1}{8}\left[\frac{1}{8}\int dy - \frac{1}{8}\int \frac{1}{8y^2+1}\,dy\right]. \tag{359}$$

Now

$$\frac{1}{8}\int y^2\,dy = \frac{1}{8}\cdot\frac{y^3}{3} = \frac{y^3}{24}.$$

And, expanding the second piece:

$$-\frac{1}{8}\left[\frac{1}{8}\int dy - \frac{1}{8}\int \frac{1}{8y^2+1}\,dy\right] = -\frac{1}{64}\int dy + \frac{1}{64}\int \frac{1}{8y^2+1}\,dy. \tag{360}$$

Thus the indefinite integral becomes

$$\int \frac{y^4}{8y^2+1}\,dy = \frac{y^3}{24} - \frac{1}{64}\int dy + \frac{1}{64}\int \frac{1}{8y^2+1}\,dy. \tag{361}$$

Compute the remaining simple integrals:

$$\int dy = y + C_1,$$

$$\int \frac{1}{8y^2+1}\,dy = \int \frac{1}{1+(\sqrt{8}\,y)^2}\,dy = \frac{1}{\sqrt{8}}\arctan(\sqrt{8}\,y) + C_2.$$

Hence

$$\int \frac{y^4}{8y^2+1}\,dy = \frac{y^3}{24} - \frac{y}{64} + \frac{1}{64}\cdot\frac{1}{\sqrt{8}}\arctan(\sqrt{8}\,y) + C \tag{362}$$

$$= \frac{y^3}{24} - \frac{y}{64} + \frac{1}{64\sqrt{8}}\arctan(\sqrt{8}\,y) + C. \tag{363}$$

4. Indefinite integral for the original integrand Recall

$$\frac{y^4}{2y^2+\frac{1}{4}} = 4\,\frac{y^4}{8y^2+1},$$

so

$$\int \frac{y^4}{2y^2+\frac{1}{4}}\,dy = 4\int \frac{y^4}{8y^2+1}\,dy \tag{364}$$

$$= 4\left(\frac{y^3}{24} - \frac{y}{64} + \frac{1}{64\sqrt{8}}\arctan(\sqrt{8}\,y)\right) + C \tag{365}$$

$$= \frac{y^3}{6} - \frac{y}{16} + \frac{1}{16\sqrt{8}}\arctan(\sqrt{8}\,y) + C. \tag{366}$$

Thus, for $r = 1$,

$$\int \frac{y^4}{Ey^4 + Fy^2 + G}\,dy = \frac{y^3}{6} - \frac{y}{16} + \frac{1}{16\sqrt{8}}\arctan(\sqrt{8}\,y) + C. \tag{367}$$

5. Definite integral on $[0, y_{c,B}]$ for $r = 1$ We now evaluate

$$I_2^{(r=1)}(y_{c,B}) = \int_0^{y_{c,B}} \frac{y^4}{2y^2+\frac{1}{4}}\,dy$$

using the antiderivative above:

$$I_2^{(r=1)}(y_{c,B}) = \left[\frac{y^3}{6} - \frac{y}{16} + \frac{1}{16\sqrt{8}}\arctan(\sqrt{8}\,y)\right]_{y=0}^{y=y_{c,B}}. \tag{368}$$

At the upper limit $y = y_{c,B}$:

$$\frac{y_{c,B}^3}{6} - \frac{y_{c,B}}{16} + \frac{1}{16\sqrt{8}}\arctan(\sqrt{8}\,y_{c,B}).$$

At the lower limit $y = 0$:

$$\frac{0^3}{6} - \frac{0}{16} + \frac{1}{16\sqrt{8}}\arctan(0) = 0.$$

Hence

$$I_2^{(r=1)}(y_{c,B}) = \frac{y_{c,B}^3}{6} - \frac{y_{c,B}}{16} + \frac{1}{16\sqrt{8}}\arctan(\sqrt{8}\,y_{c,B}). \tag{369}$$

This is the fully explicit, real–valued expression for the I_2 contribution on $[0, y_{c,B}]$ in the special case $r = 1$.

B.2 Region II integrals: partial fractions and truncation

The truncated Region II integrand is

$$\boxed{\mathcal{I}_{\rm II}(y) \approx -\sqrt{2}\, r\, \frac{-B_2\, y^3 + B_1\, y^2 + B_2 C_2\, y - B_1 C_2 - \dfrac{B_2 C_4}{y} + \dfrac{B_1 C_4}{y^2}}{\mathcal{D}(y; r)}, \qquad y_{c,B} < y < y_c.} \tag{370}$$

$$\mathcal{D}(y; r) = (1-r)^2\, y^4 + (1+r)\, y^2 + \tfrac{1}{4}$$

$$E = (1-r)^2, \quad F = (1+r), \quad G = \frac{1}{4}$$

Region II integral (inner integral only)

$$I_{\rm II}(\omega) := \int_{y_{c,B}}^{y_c} \mathcal{I}_{\rm II}(y)\, dy. \tag{371}$$

Six basic integrals

$$I_5(r; \omega, L, a) := \int_{y_{c,B}}^{y_c} \frac{y^3}{E\, y^4 + F\, y^2 + G}\, dy, \tag{372}$$

$$I_6(r; \omega, L, a) := \int_{y_{c,B}}^{y_c} \frac{y^2}{E\, y^4 + F\, y^2 + G}\, dy, \tag{373}$$

$$I_7(r; \omega, L, a) := \int_{y_{c,B}}^{y_c} \frac{y}{E\, y^4 + F\, y^2 + G}\, dy, \tag{374}$$

$$I_8(r; \omega, L, a) := \int_{y_{c,B}}^{y_c} \frac{1}{E\, y^4 + F\, y^2 + G}\, dy, \tag{375}$$

$$I_9(r; \omega, L, a) := \int_{y_{c,B}}^{y_c} \frac{1}{y\,(E\, y^4 + F\, y^2 + G)}\, dy, \tag{376}$$

$$I_{10}(r; \omega, L, a) := \int_{y_{c,B}}^{y_c} \frac{1}{y^2\,(E\, y^4 + F\, y^2 + G)}\, dy. \tag{377}$$

Region II expressed in terms of the six integrals

$$\begin{aligned} I_{\rm II}(\omega) &= -\sqrt{2}\, r \int_{y_{c,B}}^{y_c} \frac{-B_2\, y^3 + B_1\, y^2 + B_2 C_2\, y - B_1 C_2 - \dfrac{B_2 C_4}{y} + \dfrac{B_1 C_4}{y^2}}{E\, y^4 + F\, y^2 + G}\, dy \\ &= -\sqrt{2}\, r \Big[-B_2\, I_5 + B_1\, I_6 + B_2 C_2\, I_7 - B_1 C_2\, I_8 - B_2 C_4\, I_9 + B_1 C_4\, I_{10} \Big] \end{aligned} \tag{378}$$

<u>**Analytic evaluation of I_5 on $[y_{c,B}, y_c]$**</u>

$$I_5(r; y_{c,B}, y_c) := \int_{y_{c,B}}^{y_c} \frac{y^3}{E\, y^4 + F\, y^2 + G}\, dy, \qquad E = (1-r)^2,\ F = 1 + r,\ G = \tfrac{1}{4}. \tag{379}$$

Antiderivative With the substitution $u = y^2$ and our result from Equation (170) one obtains the antiderivative

$$A(y) = \frac{1}{4E}\,\ln\bigl(Ey^4 + Fy^2 + G\bigr) - \frac{F}{2E\sqrt{\Delta}}\,\arctan\Bigl(\frac{2Ey^2 + F}{\sqrt{\Delta}}\Bigr) + C, \tag{380}$$

where $\Delta = 4EG - F^2$. For the parameter choice $E = (1-r)^2,\ F = 1 + r,\ G = \frac{1}{4}$ we have

$$\Delta = 4(1-r)^2 \cdot \tfrac{1}{4} - (1+r)^2 = (1-r)^2 - (1+r)^2 = -4r.$$

Log form for $\Delta < 0$ (i.e. $r > 0$) When $r > 0$ we have $\Delta = -4r < 0$ and the arctan term can be written as a logarithmic ratio. Define

$$D(y) := (1-r)^2y^4 + (1+r)y^2 + \tfrac{1}{4}, \qquad N(y) := 2(1-r)^2y^2 + (1+r).$$

Then the antiderivative may be written (choosing the log representation for the quadratic reciprocal) as

$$A(y) = \frac{1}{4(1-r)^2}\,\ln\bigl(D(y)\bigr) - \frac{1+r}{8(1-r)^2\sqrt{r}}\,\ln\left|\frac{N(y) - 2\sqrt{r}}{N(y) + 2\sqrt{r}}\right| + C. \tag{381}$$

Definite integral on $[y_{c,B}, y_c]$ Evaluate $I_5 = A(y_c) - A(y_{c,B})$. First collect the two log differences:

$$\begin{aligned} I_5 = {} & \frac{1}{4(1-r)^2}\Bigl[\ln\bigl(D(y_c)\bigr) - \ln\bigl(D(y_{c,B})\bigr)\Bigr] \\ & - \frac{1+r}{8(1-r)^2\sqrt{r}}\left[\ln\left|\frac{N(y_c) - 2\sqrt{r}}{N(y_c) + 2\sqrt{r}}\right| - \ln\left|\frac{N(y_{c,B}) - 2\sqrt{r}}{N(y_{c,B}) + 2\sqrt{r}}\right|\right]. \end{aligned} \tag{382}$$

Combine log differences using log rules Use $\ln A - \ln B = \ln\bigl(\frac{A}{B}\bigr)$ and $\ln X - \ln Y = \ln\bigl(\frac{X}{Y}\bigr)$ to combine the two bracketed differences into single logarithms. This yields the compact form

$$\begin{aligned} I_5(r; y_{c,B}, y_c) = {} & \frac{1}{4(1-r)^2}\,\ln\left(\frac{(1-r)^2y_c^4 + (1+r)y_c^2 + \frac{1}{4}}{(1-r)^2y_{c,B}^4 + (1+r)y_{c,B}^2 + \frac{1}{4}}\right) \\ & - \frac{1+r}{8(1-r)^2\sqrt{r}}\,\ln\left(\frac{\bigl(N(y_c) - 2\sqrt{r}\bigr)\bigl(N(y_{c,B}) + 2\sqrt{r}\bigr)}{\bigl(N(y_c) + 2\sqrt{r}\bigr)\bigl(N(y_{c,B}) - 2\sqrt{r}\bigr)}\right), \end{aligned} \tag{383}$$

where $N(y) = 2(1-r)^2y^2 + (1+r)$.

Fully expanded (no intermediate symbols) If you prefer the result written entirely in terms of $r,\ y_c,\ y_{c,B}$ with no intermediate names, use

$$\boxed{\begin{aligned} I_5(r; y_{c,B}, y_c) = {} & \frac{1}{4(1-r)^2}\,\ln\left(\frac{(1-r)^2y_c^4 + (1+r)y_c^2 + \frac{1}{4}}{(1-r)^2y_{c,B}^4 + (1+r)y_{c,B}^2 + \frac{1}{4}}\right) \\ & - \frac{1+r}{8(1-r)^2\sqrt{r}}\,\ln\left(\frac{\bigl(2(1-r)^2y_c^2 + (1+r) - 2\sqrt{r}\bigr)\,\bigl(2(1-r)^2y_{c,B}^2 + (1+r) + 2\sqrt{r}\bigr)}{\bigl(2(1-r)^2y_c^2 + (1+r) + 2\sqrt{r}\bigr)\,\bigl(2(1-r)^2y_{c,B}^2 + (1+r) - 2\sqrt{r}\bigr)}\right). \end{aligned}} \tag{384}$$

Analytic evaluation of I_5 on $[y_{c,B}, y_c]$ for $r = 1$

For $r = 1$ we have

$$E = (1-r)^2 = 0, \qquad F = 1 + r = 2, \qquad G = \tfrac{1}{4},$$

so the integrand simplifies to

$$\frac{y^3}{Ey^4 + Fy^2 + G} = \frac{y^3}{2y^2 + \frac{1}{4}}.$$

Antiderivative Set $u = y^2$, so $du = 2y\,dy$ and $y^3\,dy = y^2(y\,dy) = u\frac{du}{2}$. Then

$$\int \frac{y^3}{2y^2 + \frac{1}{4}}\,dy = \tfrac{1}{2}\int \frac{u}{2u + \frac{1}{4}}\,du.$$

Perform the algebraic division

$$\frac{u}{2u + \frac{1}{4}} = \frac{1}{2} - \frac{1/8}{2u + \frac{1}{4}},$$

so

$$\tfrac{1}{2}\int \frac{u}{2u + \frac{1}{4}}\,du = \tfrac{1}{2}\int \Big(\tfrac{1}{2} - \frac{1/8}{2u + \frac{1}{4}}\Big)\,du = \tfrac{1}{4}u - \tfrac{1}{32}\ln(2u + \tfrac{1}{4}) + C.$$

Re-substituting $u = y^2$ gives the antiderivative

$$A(y) = \frac{1}{4}y^2 - \frac{1}{32}\ln(2y^2 + \tfrac{1}{4}) + C$$

Definite integral on $[y_{c,B}, y_c]$ Evaluate $I_5 = \displaystyle\int_{y_{c,B}}^{y_c} \frac{y^3}{2y^2 + \frac{1}{4}}\,dy = A(y_c) - A(y_{c,B})$. Thus

$$\begin{aligned}
I_5(r = 1; y_{c,B}, y_c) &= \Big[\frac{1}{4}y^2 - \frac{1}{32}\ln(2y^2 + \tfrac{1}{4})\Big]_{y_{c,B}}^{y_c} \\
&= \frac{1}{4}(y_c^2 - y_{c,B}^2) - \frac{1}{32}\ln\!\left(\frac{2y_c^2 + \frac{1}{4}}{2y_{c,B}^2 + \frac{1}{4}}\right).
\end{aligned}$$

Optional simplification Noting $2y^2 + \frac{1}{4} = \frac{1}{4}(8y^2 + 1)$ and absorbing the constant $-\frac{1}{32}\ln 4$ into C if desired, an equivalent form is

$$\boxed{I_5(y_{c,B}, y_c : r = 1) = \frac{1}{4}(y_c^2 - y_{c,B}^2) - \frac{1}{32}\ln\!\left(\frac{8y_c^2 + 1}{8y_{c,B}^2 + 1}\right).}$$

Analytic evaluation of I_6 on $[y_{c,B}, y_c]$

We define

$$I_6(r; y_{c,B}, y_c) := \int_{y_{c,B}}^{y_c} \frac{y^2}{E\,y^4 + F\,y^2 + G}\,dy, \qquad E = (1-r)^2,\ F = 1 + r,\ G = \tfrac{1}{4}, \tag{385}$$

for a real parameter $r > 0$ with $r \neq 1$.

Change of variable Set $u = y^2$. Then $du = 2y\,dy$ and $y^2 = u$, so

$$\int \frac{y^2}{Ey^4 + Fy^2 + G}\,dy = \frac{1}{2}\int \frac{u}{Eu^2 + Fu + G}\,du. \tag{386}$$

Quadratic factorisation in u Introduce

$$Q(u) := Eu^2 + Fu + G.$$

Its discriminant is

$$D := F^2 - 4EG = (1+r)^2 - (1-r)^2 = 4r > 0 \quad (r > 0),$$

so there are two distinct real roots

$$u_{1,2} = \frac{-F \pm \sqrt{D}}{2E} = \frac{-(1+r) \pm 2\sqrt{r}}{2(1-r)^2}. \tag{387}$$

Hence

$$Q(u) = E(u - u_1)(u - u_2),$$

and in terms of y we have

$$Q(y^2) = E(y^2 - u_1)(y^2 - u_2).$$

Sign of the roots The sum and product of the roots are

$$u_1 + u_2 = -\frac{F}{E} = -\frac{1+r}{(1-r)^2} < 0, \qquad u_1 u_2 = \frac{G}{E} = \frac{1}{4(1-r)^2} > 0.$$

Thus for all $r > 0,\ r \neq 1$ we have

$$u_1 < 0, \qquad u_2 < 0.$$

Define the positive constants

$$a_1 := \sqrt{-u_1} > 0, \qquad a_2 := \sqrt{-u_2} > 0,$$

so that

$$y^2 - u_j = y^2 + a_j^2, \qquad j = 1, 2.$$

Partial fraction decomposition We write

$$\frac{y^2}{E(y^2 - u_1)(y^2 - u_2)} = \frac{A}{y^2 - u_1} + \frac{B}{y^2 - u_2}, \tag{388}$$

with constants A, B independent of y. Multiplying through by $E(y^2 - u_1)(y^2 - u_2)$ gives

$$y^2 = E\Big[A(y^2 - u_2) + B(y^2 - u_1)\Big] = E\Big[(A+B)y^2 - (Au_2 + Bu_1)\Big].$$

Equating coefficients of y^2 and constants yields

$$E(A+B) = 1, \qquad Au_2 + Bu_1 = 0.$$

Solving for A, B gives

$$A = \frac{u_1}{E(u_1 - u_2)}, \qquad B = -\frac{u_2}{E(u_1 - u_2)}. \tag{389}$$

Using $D = F^2 - 4EG = 4r$ one finds

$$u_1 - u_2 = \frac{\sqrt{D}}{E} = \frac{2\sqrt{r}}{E},$$

so the coefficients simplify to

$$A = \frac{u_1}{2\sqrt{r}}, \qquad B = -\frac{u_2}{2\sqrt{r}}. \tag{390}$$

Consequently,

$$\frac{y^2}{Ey^4 + Fy^2 + G} = \frac{A}{y^2 - u_1} + \frac{B}{y^2 - u_2} = \frac{A}{y^2 + a_1^2} + \frac{B}{y^2 + a_2^2}. \tag{391}$$

Indefinite antiderivative (real-valued form) For $a > 0$,

$$\int \frac{dy}{y^2 + a^2} = \frac{1}{a} \arctan\Big(\frac{y}{a}\Big) + C.$$

Applying this with $a_1 = \sqrt{-u_1}$ and $a_2 = \sqrt{-u_2}$ gives

$$\int \frac{y^2}{Ey^4 + Fy^2 + G}\, dy = A \int \frac{dy}{y^2 + a_1^2} + B \int \frac{dy}{y^2 + a_2^2} \tag{392}$$

$$= \frac{A}{a_1} \arctan\Big(\frac{y}{a_1}\Big) + \frac{B}{a_2} \arctan\Big(\frac{y}{a_2}\Big) + C. \tag{393}$$

Using $u_j = -a_j^2$ we obtain

$$\frac{A}{a_1} = \frac{u_1}{2\sqrt{r} a_1} = -\frac{a_1}{2\sqrt{r}}, \qquad \frac{B}{a_2} = -\frac{u_2}{2\sqrt{r} a_2} = \frac{a_2}{2\sqrt{r}}.$$

Hence a convenient real antiderivative is

$$\int \frac{y^2}{Ey^4 + Fy^2 + G}\, dy = -\frac{a_1}{2\sqrt{r}} \arctan\Big(\frac{y}{a_1}\Big) + \frac{a_2}{2\sqrt{r}} \arctan\Big(\frac{y}{a_2}\Big) + C, \tag{394}$$

where

$$a_1 = \sqrt{-u_1}, \qquad a_2 = \sqrt{-u_2}, \qquad u_{1,2} = \frac{-(1+r) \pm 2\sqrt{r}}{2(1-r)^2}.$$

Definite integral on $[y_{c,B}, y_c]$ Evaluating between $y = y_{c,B}$ and $y = y_c$ gives

$$I_6(r; y_{c,B}, y_c) = -\frac{a_1}{2\sqrt{r}} \left[\arctan\Big(\frac{y_c}{a_1}\Big) - \arctan\Big(\frac{y_{c,B}}{a_1}\Big)\right] \tag{395}$$

$$+ \frac{a_2}{2\sqrt{r}} \left[\arctan\Big(\frac{y_c}{a_2}\Big) - \arctan\Big(\frac{y_{c,B}}{a_2}\Big)\right], \tag{396}$$

with

$$u_{1,2} = \frac{-(1+r) \pm 2\sqrt{r}}{2(1-r)^2}, \qquad a_j = \sqrt{-u_j} > 0, \quad j = 1, 2, \qquad r > 0,\ r \neq 1. \tag{397}$$

Summary (boxed form) Collecting the above, the analytically evaluated I_6 on $[y_{c,B}, y_c]$ is

$$\boxed{\begin{aligned} I_6(r; y_{c,B}, y_c) &= -\frac{\sqrt{-u_1}}{2\sqrt{r}} \left[\arctan\Big(\frac{y_c}{\sqrt{-u_1}}\Big) - \arctan\Big(\frac{y_{c,B}}{\sqrt{-u_1}}\Big)\right] \\ &\quad + \frac{\sqrt{-u_2}}{2\sqrt{r}} \left[\arctan\Big(\frac{y_c}{\sqrt{-u_2}}\Big) - \arctan\Big(\frac{y_{c,B}}{\sqrt{-u_2}}\Big)\right], \\ u_{1,2} &= \frac{-(1+r) \pm 2\sqrt{r}}{2(1-r)^2}, \qquad r > 0,\ r \neq 1. \end{aligned}}$$

There are no real singularities on $[y_{c,B}, y_c]$ for $r > 0$, $r \neq 1$, since $u_1, u_2 < 0$ and thus $Ey^4 + Fy^2 + G > 0$ for all real y. Special cases $r = 0$ and $r = 1$ will be treated separately by simplifying the denominator before integrating.

Analytic evaluation of I_6 on $[y_{c,B}, y_c]$ for $r = 1$

For

$$I_6(r; y_{c,B}, y_c) := \int_{y_{c,B}}^{y_c} \frac{y^2}{E\,y^4 + F\,y^2 + G}\,dy, \qquad E = (1-r)^2,\ F = 1+r,\ G = \tfrac{1}{4},$$

the special case $r = 1$ gives

$$E = 0, \qquad F = 2, \qquad G = \frac{1}{4},$$

so the denominator reduces to

$$Ey^4 + Fy^2 + G = 2y^2 + \frac{1}{4},$$

and hence

$$I_6(1; y_{c,B}, y_c) = \int_{y_{c,B}}^{y_c} \frac{y^2}{2y^2 + \frac{1}{4}}\,dy. \tag{398}$$

Algebraic simplification We perform a simple algebraic division. Seek constants A, B such that

$$\frac{y^2}{2y^2 + \frac{1}{4}} = A + \frac{B}{2y^2 + \frac{1}{4}}.$$

Equating numerators,

$$y^2 = A\Big(2y^2 + \tfrac{1}{4}\Big) + B = 2Ay^2 + \frac{A}{4} + B.$$

Matching coefficients gives

$$2A = 1 \ \Rightarrow\ A = \frac{1}{2}, \qquad \frac{A}{4} + B = 0 \ \Rightarrow\ B = -\frac{1}{8}.$$

Thus

$$\frac{y^2}{2y^2 + \frac{1}{4}} = \frac{1}{2} - \frac{1}{8}\,\frac{1}{2y^2 + \frac{1}{4}}. \tag{399}$$

Factor the denominator as

$$2y^2 + \tfrac{1}{4} = 2\Big(y^2 + \tfrac{1}{8}\Big),$$

so

$$\frac{1}{2y^2 + \frac{1}{4}} = \frac{1}{2}\,\frac{1}{y^2 + \frac{1}{8}}. \tag{400}$$

Hence

$$\frac{y^2}{2y^2 + \frac{1}{4}} = \frac{1}{2} - \frac{1}{8}\cdot\frac{1}{2}\cdot\frac{1}{y^2 + \frac{1}{8}} = \frac{1}{2} - \frac{1}{16}\,\frac{1}{y^2 + \frac{1}{8}}. \tag{401}$$

Indefinite antiderivative We use the standard formula, for $a > 0$,

$$\int \frac{dy}{y^2 + a^2} = \frac{1}{a}\arctan\!\Big(\frac{y}{a}\Big) + C.$$

Here $y^2 + \frac{1}{8} = y^2 + a^2$ with

$$a^2 = \frac{1}{8}, \qquad a = \frac{1}{2\sqrt{2}} = \frac{\sqrt{2}}{4}, \qquad \frac{1}{a} = 2\sqrt{2}.$$

Thus

$$\int \frac{y^2}{2y^2+\frac{1}{4}}\,dy = \int \left(\frac{1}{2} - \frac{1}{16}\,\frac{1}{y^2+\frac{1}{8}}\right) dy \tag{402}$$

$$= \frac{1}{2}\,y - \frac{1}{16}\int \frac{dy}{y^2+\frac{1}{8}} \tag{403}$$

$$= \frac{1}{2}\,y - \frac{1}{16}\cdot 2\sqrt{2}\ \arctan\!\Big(\frac{y}{a}\Big) + C \tag{404}$$

$$= \frac{1}{2}\,y - \frac{\sqrt{2}}{8}\ \arctan\!\Big(\frac{y}{a}\Big) + C \tag{405}$$

$$= \frac{1}{2}\,y - \frac{\sqrt{2}}{8}\ \arctan\!\Big(2\sqrt{2}\,y\Big) + C, \tag{406}$$

since $y/a = y\cdot 2\sqrt{2} = 2\sqrt{2}\,y$.

Definite integral on $[y_{c,B}, y_c]$ Evaluating between $y = y_{c,B}$ and $y = y_c$ yields

$$I_6(1; y_{c,B}, y_c) = \left[\frac{1}{2}\,y - \frac{\sqrt{2}}{8}\ \arctan\!\Big(2\sqrt{2}\,y\Big)\right]_{y_{c,B}}^{y_c} \tag{407}$$

$$= \frac{1}{2}\,(y_c - y_{c,B}) - \frac{\sqrt{2}}{8}\left[\arctan\!\Big(2\sqrt{2}\,y_c\Big) - \arctan\!\Big(2\sqrt{2}\,y_{c,B}\Big)\right]. \tag{408}$$

Summary (boxed form)

$$\boxed{I_6(1; y_{c,B}, y_c) = \frac{1}{2}\,(y_c - y_{c,B}) - \frac{\sqrt{2}}{8}\left[\arctan\!\Big(2\sqrt{2}\,y_c\Big) - \arctan\!\Big(2\sqrt{2}\,y_{c,B}\Big)\right]}$$

This is the closed-form analytic expression for I_6 on $[y_{c,B}, y_c]$ in the special case $r = 1$.

Analytic evaluation of I_7 on $[y_{c,B}, y_c]$

We define

$$I_7(r; y_{c,B}, y_c) := \int_{y_{c,B}}^{y_c} \frac{y}{E\,y^4 + F\,y^2 + G}\,dy, \qquad E = (1-r)^2,\ F = 1+r,\ G = \tfrac{1}{4}, \tag{409}$$

for a real parameter $r > 0$ with $r \neq 1$.

1. Change of variable Let

$$u = y^2, \qquad du = 2y\,dy \ \Rightarrow\ y\,dy = \frac{du}{2}.$$

Then

$$\int \frac{y}{Ey^4 + Fy^2 + G}\,dy = \frac{1}{2}\int \frac{1}{Eu^2 + Fu + G}\,du. \tag{410}$$

Introduce

$$Q(u) := Eu^2 + Fu + G, \qquad E = (1-r)^2,\ F = 1+r,\ G = \tfrac{1}{4}.$$

2. Discriminant and roots of $Q(u)$ The discriminant is

$$D := F^2 - 4EG = (1+r)^2 - 4(1-r)^2 \cdot \tfrac{1}{4} = (1+r)^2 - (1-r)^2 = 4r > 0 \quad (r > 0). \tag{411}$$

Thus $Q(u)$ has two distinct real roots

$$u_{1,2} = \frac{-F \pm \sqrt{D}}{2E} = \frac{-(1+r) \pm 2\sqrt{r}}{2(1-r)^2}. \tag{412}$$

Hence

$$Q(u) = E(u - u_1)(u - u_2), \qquad Q(y^2) = E(y^2 - u_1)(y^2 - u_2).$$

For $r > 0,\ r \neq 1$ one checks

$$u_1 + u_2 = -\frac{F}{E} = -\frac{1+r}{(1-r)^2} < 0, \qquad u_1 u_2 = \frac{G}{E} = \frac{1}{4(1-r)^2} > 0,$$

so $u_1, u_2 < 0$ and $Q(y^2) > 0$ for all real y (no singularities on $[y_{c,B}, y_c]$).

3. Partial fraction decomposition in u We decompose

$$\frac{1}{E(u-u_1)(u-u_2)} = \frac{A}{u-u_1} + \frac{B}{u-u_2}. \tag{413}$$

Multiplying by $E(u - u_1)(u - u_2)$ gives

$$1 = E\big[A(u - u_2) + B(u - u_1)\big] = E\big[(A + B)u - (Au_2 + Bu_1)\big].$$

Equating coefficients of u and the constant term:

$$E(A + B) = 0, \qquad -E(Au_2 + Bu_1) = 1.$$

Thus

$$A + B = 0 \ \Rightarrow\ B = -A,$$

and

$$-E(Au_2 - Au_1) = 1 \ \Rightarrow\ -EA(u_2 - u_1) = 1 \ \Rightarrow\ A = \frac{1}{E(u_1 - u_2)}, \quad B = -\frac{1}{E(u_1 - u_2)}.$$

Therefore

$$\frac{1}{Eu^2 + Fu + G} = \frac{1}{E(u-u_1)(u-u_2)} = \frac{1}{E(u_1-u_2)}\left(\frac{1}{u-u_1} - \frac{1}{u-u_2}\right). \tag{414}$$

Using $D = F^2 - 4EG = 4r$ and

$$u_1 - u_2 = \frac{\sqrt{D}}{E} = \frac{2\sqrt{r}}{E},$$

we get

$$\frac{1}{E(u_1 - u_2)} = \frac{1}{2\sqrt{r}}.$$

4. Indefinite antiderivative in u and y Integrating in u,

$$\int \frac{1}{Eu^2+Fu+G}\,du = \frac{1}{E(u_1-u_2)}\int\left(\frac{1}{u-u_1}-\frac{1}{u-u_2}\right)du \tag{415}$$

$$= \frac{1}{E(u_1-u_2)}\big[\ln|u-u_1| - \ln|u-u_2|\big] + C \tag{416}$$

$$= \frac{1}{2\sqrt{r}}\ln\left|\frac{u-u_1}{u-u_2}\right| + C. \tag{417}$$

Since

$$I_7 = \frac{1}{2}\int \frac{1}{Eu^2+Fu+G}\,du,$$

we obtain the indefinite antiderivative in y:

$$\int \frac{y}{Ey^4+Fy^2+G}\,dy = \frac{1}{2}\cdot\frac{1}{2\sqrt{r}}\ln\left|\frac{u-u_1}{u-u_2}\right| + C \tag{418}$$

$$= \frac{1}{4\sqrt{r}}\ln\left|\frac{y^2-u_1}{y^2-u_2}\right| + C. \tag{419}$$

It is often convenient to eliminate $u_{1,2}$ in favour of the combination

$$N(y) := 2(1-r)^2y^2 + (1+r).$$

One verifies

$$y^2-u_1 = \frac{N(y)-2\sqrt{r}}{2(1-r)^2}, \qquad y^2-u_2 = \frac{N(y)+2\sqrt{r}}{2(1-r)^2},$$

so the common factor $2(1-r)^2$ cancels in the ratio, and

$$\frac{y^2-u_1}{y^2-u_2} = \frac{N(y)-2\sqrt{r}}{N(y)+2\sqrt{r}}. \tag{420}$$

Thus (419) can be written as

$$\int \frac{y}{Ey^4+Fy^2+G}\,dy = \frac{1}{4\sqrt{r}}\ln\left(\frac{N(y)-2\sqrt{r}}{N(y)+2\sqrt{r}}\right) + C, \tag{421}$$

$$N(y) = 2(1-r)^2y^2 + (1+r). \tag{422}$$

5. artanh form (real for all $r>0,\ r\neq 1$) Using

$$\operatorname{artanh}(x) = \frac{1}{2}\ln\left(\frac{1+x}{1-x}\right),$$

we note

$$\ln\left(\frac{N(y)-2\sqrt{r}}{N(y)+2\sqrt{r}}\right) = \ln\left(\frac{1-\frac{2\sqrt{r}}{N(y)}}{1+\frac{2\sqrt{r}}{N(y)}}\right) = -2\operatorname{artanh}\left(\frac{2\sqrt{r}}{N(y)}\right).$$

Substituting into (422) gives

$$\int \frac{y}{Ey^4+Fy^2+G}\,dy = -\frac{1}{2\sqrt{r}}\operatorname{artanh}\left(\frac{2\sqrt{r}}{2(1-r)^2y^2+(1+r)}\right) + C. \tag{423}$$

For $r>0$ and $y>0$ we have

$$N(y) = 2(1-r)^2y^2 + (1+r) \geq 1+r \geq 2\sqrt{r},$$

with equality only at $r=1$ (excluded here), so

$$0 < \frac{2\sqrt{r}}{N(y)} < 1,$$

and artanh is real-valued on the whole interval.

6. Definite integral on $[y_{c,B}, y_c]$ (general $r \neq 1$) Using the artanh primitive (423), we obtain

$$I_7(r; y_{c,B}, y_c) = \left[-\frac{1}{2\sqrt{r}} \operatorname{artanh}\left(\frac{2\sqrt{r}}{2(1-r)^2 y^2 + (1+r)} \right) \right]_{y_{c,B}}^{y_c} \tag{424}$$

$$= \frac{1}{2\sqrt{r}} \left[\operatorname{artanh}\left(\frac{2\sqrt{r}}{2(1-r)^2 y_{c,B}^2 + (1+r)} \right) - \operatorname{artanh}\left(\frac{2\sqrt{r}}{2(1-r)^2 y_c^2 + (1+r)} \right) \right]. \tag{425}$$

Equivalently, using the log form (422) at the endpoints,

$$I_7(r; y_{c,B}, y_c) = \frac{1}{4\sqrt{r}} \left[\ln\left(\frac{N(y_c) - 2\sqrt{r}}{N(y_c) + 2\sqrt{r}} \right) - \ln\left(\frac{N(y_{c,B}) - 2\sqrt{r}}{N(y_{c,B}) + 2\sqrt{r}} \right) \right] \tag{426}$$

$$= \frac{1}{4\sqrt{r}} \ln\left(\frac{(N(y_c) - 2\sqrt{r})(N(y_{c,B}) + 2\sqrt{r})}{(N(y_c) + 2\sqrt{r})(N(y_{c,B}) - 2\sqrt{r})} \right), \tag{427}$$

$$N(y) = 2(1-r)^2 y^2 + (1+r), \qquad r > 0,\ r \neq 1. \tag{428}$$

7. Summary (general $r > 0,\ r \neq 1$)

$$\boxed{\begin{aligned} I_7(r; y_{c,B}, y_c) &= \frac{1}{2\sqrt{r}} \left[\operatorname{artanh}\left(\frac{2\sqrt{r}}{2(1-r)^2 y_{c,B}^2 + (1+r)} \right) - \operatorname{artanh}\left(\frac{2\sqrt{r}}{2(1-r)^2 y_c^2 + (1+r)} \right) \right], \\ &= \frac{1}{4\sqrt{r}} \ln\left(\frac{\left(2(1-r)^2 y_c^2 + (1+r) - 2\sqrt{r}\right) \left(2(1-r)^2 y_{c,B}^2 + (1+r) + 2\sqrt{r}\right)}{\left(2(1-r)^2 y_c^2 + (1+r) + 2\sqrt{r}\right) \left(2(1-r)^2 y_{c,B}^2 + (1+r) - 2\sqrt{r}\right)} \right), \\ & \qquad r > 0,\ r \neq 1. \end{aligned}}$$

Analytic evaluation of I_7 on $[y_{c,B}, y_c]$ for $r = 1$

We recall

$$I_7(r; y_{c,B}, y_c) := \int_{y_{c,B}}^{y_c} \frac{y}{E\,y^4 + F\,y^2 + G}\,dy, \qquad E = (1-r)^2, \quad F = 1 + r, \quad G = \tfrac{1}{4}. \tag{429}$$

In the degenerate case $r = 1$ we have

$$E = (1-1)^2 = 0, \qquad F = 1 + 1 = 2, \qquad G = \tfrac{1}{4},$$

so the denominator reduces to

$$Ey^4 + Fy^2 + G = 2y^2 + \frac{1}{4},$$

and the integral becomes

$$I_7(1; y_{c,B}, y_c) = \int_{y_{c,B}}^{y_c} \frac{y}{2y^2 + \frac{1}{4}}\,dy. \tag{430}$$

Antiderivative Use the substitution

$$u = 2y^2 + \frac{1}{4}, \qquad du = 4y\,dy \;\Rightarrow\; y\,dy = \frac{du}{4}.$$

Then

$$\int \frac{y}{2y^2+\frac{1}{4}}\,dy = \int \frac{1}{u}\,\frac{du}{4} = \frac{1}{4}\int \frac{du}{u} = \frac{1}{4}\ln|u| + C \tag{431}$$

$$= \frac{1}{4}\ln\!\big(2y^2+\tfrac{1}{4}\big) + C. \tag{432}$$

Hence an antiderivative is

$$A(y) = \frac{1}{4}\ln\!\big(2y^2+\tfrac{1}{4}\big) + C.$$

Definite integral on $[y_{c,B}, y_c]$ Evaluate

$$I_7(1; y_{c,B}, y_c) = \int_{y_{c,B}}^{y_c} \frac{y}{2y^2+\frac{1}{4}}\,dy = A(y_c) - A(y_{c,B}).$$

Thus

$$I_7(1; y_{c,B}, y_c) = \frac{1}{4}\ln\!\big(2y_c^2+\tfrac{1}{4}\big) - \frac{1}{4}\ln\!\big(2y_{c,B}^2+\tfrac{1}{4}\big) \tag{433}$$

$$= \frac{1}{4}\ln\!\left(\frac{2y_c^2+\frac{1}{4}}{2y_{c,B}^2+\frac{1}{4}}\right). \tag{434}$$

Optional simplification Using

$$2y^2+\tfrac{1}{4} = \tfrac{1}{4}\,(8y^2+1),$$

the constant factor $\frac{1}{4}$ cancels in the ratio, and we may rewrite

$$\frac{2y_c^2+\frac{1}{4}}{2y_{c,B}^2+\frac{1}{4}} = \frac{\frac{1}{4}(8y_c^2+1)}{\frac{1}{4}(8y_{c,B}^2+1)} = \frac{8y_c^2+1}{8y_{c,B}^2+1}. \tag{435}$$

Hence an equivalent (slightly cleaner) form is

$$I_7(1; y_{c,B}, y_c) = \frac{1}{4}\ln\!\left(\frac{8y_c^2+1}{8y_{c,B}^2+1}\right). \tag{436}$$

Summary (boxed form)

$$\boxed{I_7(1; y_{c,B}, y_c) = \frac{1}{4}\ln\!\left(\frac{8y_c^2+1}{8y_{c,B}^2+1}\right), \qquad r=1,\ 0<y_{c,B}<y_c.}$$

Analytic evaluation of I_8 on $[y_{c,B}, y_c]$

We define

$$I_8(r; y_{c,B}, y_c) := \int_{y_{c,B}}^{y_c} \frac{1}{E\,y^4 + F\,y^2 + G}\,dy, \qquad E=(1-r)^2,\ F=1+r,\ G=\tfrac{1}{4}, \tag{437}$$

for a real parameter $r>0$ with $r \neq 1$.

1. Quartic factorisation in y^2 Introduce

$$Q(t) := Et^2 + Ft + G, \qquad t = y^2.$$

The discriminant is

$$D := F^2 - 4EG = (1+r)^2 - 4(1-r)^2 \cdot \tfrac{1}{4} = (1+r)^2 - (1-r)^2 = 4r > 0 \quad (r > 0), \tag{438}$$

so $Q(t)$ has two distinct real roots

$$u_{1,2} = \frac{-F \pm \sqrt{D}}{2E} = \frac{-(1+r) \pm 2\sqrt{r}}{2(1-r)^2}. \tag{439}$$

Thus

$$Q(y^2) = E(y^2 - u_1)(y^2 - u_2).$$

For $r > 0,\ r \neq 1$ we have

$$u_1 + u_2 = -\frac{F}{E} = -\frac{1+r}{(1-r)^2} < 0, \qquad u_1 u_2 = \frac{G}{E} = \frac{1}{4(1-r)^2} > 0,$$

hence

$$u_1 < 0, \qquad u_2 < 0.$$

Define positive constants

$$a_1 := \sqrt{-u_1} > 0, \qquad a_2 := \sqrt{-u_2} > 0,$$

so that

$$y^2 - u_j = y^2 + a_j^2, \qquad j = 1, 2.$$

The denominator of I_8 then reads

$$Ey^4 + Fy^2 + G = E(y^2 - u_1)(y^2 - u_2) = E(y^2 + a_1^2)(y^2 + a_2^2) > 0$$

for all real y, so there are no singularities on $[y_{c,B}, y_c]$.

2. Partial fraction decomposition We decompose

$$\frac{1}{E(y^2 - u_1)(y^2 - u_2)} = \frac{A}{y^2 - u_1} + \frac{B}{y^2 - u_2}, \tag{440}$$

with constants A, B independent of y. Multiplying by $E(y^2 - u_1)(y^2 - u_2)$ gives

$$1 = E\big[A(y^2 - u_2) + B(y^2 - u_1)\big] = E\big[(A+B)y^2 - (Au_2 + Bu_1)\big].$$

Equating coefficients of y^2 and of the constant term:

$$E(A+B) = 0, \qquad -E(Au_2 + Bu_1) = 1.$$

Thus

$$A + B = 0 \ \Rightarrow\ B = -A,$$

and

$$-E(Au_2 - Au_1) = 1 \ \Rightarrow\ -EA(u_2 - u_1) = 1 \ \Rightarrow\ A = -\frac{1}{E(u_2 - u_1)} = \frac{1}{E(u_1 - u_2)}, \qquad B = -A.$$

Using $D = F^2 - 4EG = 4r$ we have

$$u_1 - u_2 = \frac{\sqrt{D}}{E} = \frac{2\sqrt{r}}{E},$$

so

$$\frac{1}{E(u_1 - u_2)} = \frac{1}{2\sqrt{r}}.$$

Hence

$$\frac{1}{Ey^4 + Fy^2 + G} = \frac{1}{E(y^2 - u_1)(y^2 - u_2)} \tag{441}$$

$$= \frac{1}{E(u_1 - u_2)} \left(\frac{1}{y^2 - u_1} - \frac{1}{y^2 - u_2} \right) \tag{442}$$

$$= \frac{1}{2\sqrt{r}} \left(\frac{1}{y^2 - u_1} - \frac{1}{y^2 - u_2} \right) \tag{443}$$

$$= \frac{1}{2\sqrt{r}} \left(\frac{1}{y^2 + a_1^2} - \frac{1}{y^2 + a_2^2} \right). \tag{444}$$

3. Indefinite antiderivative: why arctan **(not** ln **or** artanh**)?** For $a > 0$,

$$\int \frac{dy}{y^2 + a^2} = \frac{1}{a} \arctan\Big(\frac{y}{a}\Big) + C,$$

which is real-valued for all real y. Here both quadratic factors are of the form $y^2 + a_j^2$ with $a_j > 0$, so the natural real primitive uses arctan; we *don't* convert to ln or artanh because those arise for denominators like $y^2 - a^2$ or linear factors with real roots. In our case the roots in y are purely imaginary, and arctan captures the real primitive. Thus:

$$\int \frac{1}{Ey^4 + Fy^2 + G}\, dy = \frac{1}{2\sqrt{r}} \left(\int \frac{dy}{y^2 + a_1^2} - \int \frac{dy}{y^2 + a_2^2} \right) \tag{445}$$

$$= \frac{1}{2\sqrt{r}} \left(\frac{1}{a_1} \arctan\Big(\frac{y}{a_1}\Big) - \frac{1}{a_2} \arctan\Big(\frac{y}{a_2}\Big) \right) + C. \tag{446}$$

4. Definite integral on $[y_{c,B}, y_c]$ Evaluating from $y = y_{c,B}$ to $y = y_c$ gives

$$I_8(r; y_{c,B}, y_c) = \frac{1}{2\sqrt{r}} \left[\frac{1}{a_1} \arctan\Big(\frac{y}{a_1}\Big) - \frac{1}{a_2} \arctan\Big(\frac{y}{a_2}\Big) \right]_{y_{c,B}}^{y_c} \tag{447}$$

$$= \frac{1}{2\sqrt{r}} \left\{ \frac{1}{a_1} \left[\arctan\Big(\frac{y_c}{a_1}\Big) - \arctan\Big(\frac{y_{c,B}}{a_1}\Big) \right] \right. \tag{448}$$

$$\left. - \frac{1}{a_2} \left[\arctan\Big(\frac{y_c}{a_2}\Big) - \arctan\Big(\frac{y_{c,B}}{a_2}\Big) \right] \right\}, \tag{449}$$

with

$$u_{1,2} = \frac{-(1+r) \pm 2\sqrt{r}}{2(1-r)^2}, \qquad a_j = \sqrt{-u_j} > 0, \quad j = 1, 2, \qquad r > 0,\ r \neq 1.$$

5. Summary (general $r > 0,\ r \neq 1$)

$$\boxed{\begin{aligned} I_8(r; y_{c,B}, y_c) &= \frac{1}{2\sqrt{r}}\Bigg\{\frac{1}{\sqrt{-u_1}}\left[\arctan\!\Big(\frac{y_c}{\sqrt{-u_1}}\Big) - \arctan\!\Big(\frac{y_{c,B}}{\sqrt{-u_1}}\Big)\right] \\ &\qquad - \frac{1}{\sqrt{-u_2}}\left[\arctan\!\Big(\frac{y_c}{\sqrt{-u_2}}\Big) - \arctan\!\Big(\frac{y_{c,B}}{\sqrt{-u_2}}\Big)\right]\Bigg\}, \\ u_{1,2} &= \frac{-(1+r) \pm 2\sqrt{r}}{2(1-r)^2}, \qquad r > 0,\ r \neq 1. \end{aligned}}$$

There is no need to convert this to artanh: the arguments of the arctan are real and bounded for all $y > 0$, so this expression is real-valued and directly plottable.

Analytic evaluation of I_8 on $[y_{c,B}, y_c]$ for $r = 1$

For completeness, we treat the degenerate case $r = 1$ separately. Then

$$E = (1-1)^2 = 0, \qquad F = 1 + 1 = 2, \qquad G = \tfrac{1}{4},$$

so

$$Ey^4 + Fy^2 + G = 2y^2 + \tfrac{1}{4},$$

and

$$I_8(1; y_{c,B}, y_c) = \int_{y_{c,B}}^{y_c} \frac{1}{2y^2 + \frac{1}{4}}\, dy. \tag{450}$$

Factor

$$2y^2 + \tfrac{1}{4} = 2\Big(y^2 + \tfrac{1}{8}\Big),$$

so

$$\frac{1}{2y^2 + \frac{1}{4}} = \frac{1}{2}\,\frac{1}{y^2 + \frac{1}{8}}.$$

Let

$$a^2 = \frac{1}{8}, \qquad a = \frac{1}{2\sqrt{2}}, \qquad \frac{1}{a} = 2\sqrt{2}.$$

Using

$$\int \frac{dy}{y^2 + a^2} = \frac{1}{a}\arctan\!\Big(\frac{y}{a}\Big) + C,$$

we obtain

$$\begin{aligned} \int \frac{1}{2y^2 + \frac{1}{4}}\, dy = \frac{1}{2}\int \frac{dy}{y^2 + \frac{1}{8}} &= \frac{1}{2}\cdot\frac{1}{a}\arctan\!\Big(\frac{y}{a}\Big) + C && (451) \\ &= \frac{1}{2}\cdot 2\sqrt{2}\,\arctan\!\Big(2\sqrt{2}\,y\Big) + C && (452) \\ &= \sqrt{2}\,\arctan\!\Big(2\sqrt{2}\,y\Big) + C. && (453) \end{aligned}$$

Thus the definite integral is

$$\begin{aligned} I_8(1; y_{c,B}, y_c) &= \Big[\sqrt{2}\,\arctan(2\sqrt{2}\,y)\Big]_{y_{c,B}}^{y_c} && (454) \\ &= \sqrt{2}\,\Big[\arctan(2\sqrt{2}\,y_c) - \arctan(2\sqrt{2}\,y_{c,B})\Big]. && (455) \end{aligned}$$

$$\boxed{I_8(1; y_{c,B}, y_c) = \sqrt{2}\,\Big[\arctan(2\sqrt{2}\,y_c) - \arctan(2\sqrt{2}\,y_{c,B})\Big], \qquad r = 1.}$$

Analytic evaluation of I_9 on $[y_{c,B}, y_c]$

We define

$$I_9(r; y_{c,B}, y_c) := \int_{y_{c,B}}^{y_c} \frac{1}{y\,(E\,y^4 + F\,y^2 + G)}\,dy, \qquad E = (1-r)^2,\ F = 1+r,\ G = \tfrac{1}{4}, \tag{456}$$

for a real parameter $r > 0$.

1. Indefinite antiderivative (general E, F, G) From the earlier calculation we have, for a general quadratic $Q(y^2) = Ey^4 + Fy^2 + G$ with discriminant $F^2 - 4EG > 0$, the indefinite integral

$$\int \frac{1}{y(Ey^4 + Fy^2 + G)}\,dy = \frac{1}{G}\ln|y| - \frac{1}{4G}\ln\left|Ey^4 + Fy^2 + G\right| \tag{457}$$

$$- \frac{F}{4G\sqrt{F^2-4EG}}\ln\left|\frac{2Ey^2 + F - \sqrt{F^2-4EG}}{2Ey^2 + F + \sqrt{F^2-4EG}}\right| + C. \tag{458}$$

Differentiating (458) reproduces the integrand, so it is a valid primitive.

2. Specialising to $E = (1-r)^2,\ F = 1+r,\ G = \frac{1}{4}$ For our choice

$$E = (1-r)^2, \qquad F = 1+r, \qquad G = \tfrac{1}{4},$$

we have

$$F^2 - 4EG = (1+r)^2 - 4(1-r)^2 \cdot \tfrac{1}{4} = (1+r)^2 - (1-r)^2 = 4r > 0 \quad (r > 0). \tag{459}$$

Also

$$\frac{1}{G} = 4, \qquad \frac{1}{4G} = 1, \qquad \frac{F}{4G\sqrt{F^2-4EG}} = \frac{1+r}{2\sqrt{r}}.$$

Define the convenient shorthand

$$D(y) := (1-r)^2y^4 + (1+r)y^2 + \tfrac{1}{4}, \qquad N(y) := 2(1-r)^2y^2 + (1+r).$$

Then (458) becomes

$$A_r(y) := \int \frac{1}{y\,(Ey^4 + Fy^2 + G)}\,dy \tag{460}$$

$$= 4\ln|y| - \ln\big(D(y)\big) - \frac{1+r}{2\sqrt{r}}\ \ln\left|\frac{N(y) - 2\sqrt{r}}{N(y) + 2\sqrt{r}}\right| + C. \tag{461}$$

For $r > 0$ and $y > 0$ relevant to our problem: - $D(y) > 0$ for all y (product of positive factors as seen in previous sections); - $N(y) = 2(1-r)^2y^2 + (1+r) \geq 1 + r > 2\sqrt{r}$ for $r \neq 1$, so all log arguments are strictly positive and the expression is manifestly real (no branch issues).

3. Definite integral on $[y_{c,B}, y_c]$ The definite integral is

$$I_9(r; y_{c,B}, y_c) = A_r(y_c) - A_r(y_{c,B}).$$

Using (461),

$$I_9(r; y_{c,B}, y_c) = \big[4\ln y - \ln D(y)\big]_{y_{c,B}}^{y_c} - \frac{1+r}{2\sqrt{r}}\left[\ln\left(\frac{N(y) - 2\sqrt{r}}{N(y) + 2\sqrt{r}}\right)\right]_{y_{c,B}}^{y_c} \tag{462}$$

$$= 4(\ln y_c - \ln y_{c,B}) - (\ln D(y_c) - \ln D(y_{c,B})) \tag{463}$$

$$- \frac{1+r}{2\sqrt{r}}\left\{\ln\left(\frac{N(y_c) - 2\sqrt{r}}{N(y_c) + 2\sqrt{r}}\right) - \ln\left(\frac{N(y_{c,B}) - 2\sqrt{r}}{N(y_{c,B}) + 2\sqrt{r}}\right)\right\}. \tag{464}$$

Using $\ln A - \ln B = \ln(A/B)$, this simplifies to

$$I_9(r; y_{c,B}, y_c) = 4\ln\left(\frac{y_c}{y_{c,B}}\right) + \ln\left(\frac{D(y_{c,B})}{D(y_c)}\right) \tag{465}$$

$$- \frac{1+r}{2\sqrt{r}} \ln\left(\frac{(N(y_c) - 2\sqrt{r})(N(y_{c,B}) + 2\sqrt{r})}{(N(y_c) + 2\sqrt{r})(N(y_{c,B}) - 2\sqrt{r})}\right), \tag{466}$$

$$D(y) = (1-r)^2 y^4 + (1+r)y^2 + \tfrac{1}{4}, \quad N(y) = 2(1-r)^2 y^2 + (1+r), \quad r > 0. \tag{467}$$

4. Alternative artanh **form (optional)** Sometimes it is convenient (or numerically nicer) to write the last log as an artanh. Using

$$\operatorname{artanh}(x) = \frac{1}{2}\ln\left(\frac{1+x}{1-x}\right),$$

and

$$\ln\left(\frac{N - 2\sqrt{r}}{N + 2\sqrt{r}}\right) = -2 \operatorname{artanh}\left(\frac{2\sqrt{r}}{N}\right),$$

one may rewrite the primitive (461) as

$$A_r(y) = 4\ln y - \ln D(y) + \frac{1+r}{\sqrt{r}} \operatorname{artanh}\left(\frac{2\sqrt{r}}{N(y)}\right) + C.$$

Then the definite integral becomes

$$I_9(r; y_{c,B}, y_c) = 4\ln\left(\frac{y_c}{y_{c,B}}\right) + \ln\left(\frac{D(y_{c,B})}{D(y_c)}\right) \tag{468}$$

$$+ \frac{1+r}{\sqrt{r}} \left[\operatorname{artanh}\left(\frac{2\sqrt{r}}{N(y_c)}\right) - \operatorname{artanh}\left(\frac{2\sqrt{r}}{N(y_{c,B})}\right)\right]. \tag{469}$$

For $r > 0$ and $y > 0$ we have $N(y) \geq 1 + r > 2\sqrt{r}$ when $r \neq 1$, so

$$0 < \frac{2\sqrt{r}}{N(y)} < 1,$$

making artanh real-valued on the whole interval.

5. Summary (general $r > 0$) A convenient real, log-only form is

$$\boxed{\begin{aligned} I_9(r; y_{c,B}, y_c) &= 4\ln\left(\frac{y_c}{y_{c,B}}\right) + \ln\left(\frac{D(y_{c,B})}{D(y_c)}\right) \\ &\quad - \frac{1+r}{2\sqrt{r}} \ln\left(\frac{(N(y_c) - 2\sqrt{r})(N(y_{c,B}) + 2\sqrt{r})}{(N(y_c) + 2\sqrt{r})(N(y_{c,B}) - 2\sqrt{r})}\right), \\ D(y) &= (1-r)^2 y^4 + (1+r)y^2 + \tfrac{1}{4}, \quad N(y) = 2(1-r)^2 y^2 + (1+r), \quad r > 0. \end{aligned}}$$

An equivalent artanh form is

$$\boxed{\begin{aligned} I_9(r; y_{c,B}, y_c) &= 4\ln\left(\frac{y_c}{y_{c,B}}\right) + \ln\left(\frac{D(y_{c,B})}{D(y_c)}\right) \\ &\quad + \frac{1+r}{\sqrt{r}} \left[\operatorname{artanh}\left(\frac{2\sqrt{r}}{N(y_c)}\right) - \operatorname{artanh}\left(\frac{2\sqrt{r}}{N(y_{c,B})}\right)\right], \\ &\qquad r > 0. \end{aligned}}$$

Analytic evaluation of I_9 on $[y_{c,B}, y_c]$ for $r = 1$

For the degenerate case $r = 1$ we have

$$E = (1-1)^2 = 0, \qquad F = 1 + 1 = 2, \qquad G = \tfrac{1}{4},$$

so

$$Ey^4 + Fy^2 + G = 2y^2 + \tfrac{1}{4},$$

and

$$I_9(1; y_{c,B}, y_c) = \int_{y_{c,B}}^{y_c} \frac{1}{y(2y^2 + \frac{1}{4})}\, dy. \tag{470}$$

1. Substitution $t = y^2$ Set

$$t = y^2, \qquad dt = 2y\, dy \;\Rightarrow\; \frac{dy}{y} = \frac{dt}{2t}.$$

Then

$$\int \frac{1}{y(2y^2 + \frac{1}{4})}\, dy = \int \frac{1}{2t(2t + \frac{1}{4})}\, dt = \frac{1}{2} \int \frac{1}{t(2t + \frac{1}{4})}\, dt. \tag{471}$$

2. Partial fractions in t Seek A, B with

$$\frac{1}{t(2t + \frac{1}{4})} = \frac{A}{t} + \frac{B}{2t + \frac{1}{4}}.$$

Then

$$1 = A(2t + \tfrac{1}{4}) + Bt = (2A + B)t + \frac{A}{4},$$

so

$$2A + B = 0, \qquad \frac{A}{4} = 1 \;\Rightarrow\; A = 4,\; B = -8.$$

Thus

$$\frac{1}{t(2t + \frac{1}{4})} = \frac{4}{t} - \frac{8}{2t + \frac{1}{4}},$$

and

$$\int \frac{1}{y(2y^2 + \frac{1}{4})}\, dy = \frac{1}{2} \int \left(\frac{4}{t} - \frac{8}{2t + \frac{1}{4}} \right) dt \tag{472}$$

$$= 2 \int \frac{dt}{t} - 4 \int \frac{dt}{2t + \frac{1}{4}} \tag{473}$$

$$= 2 \ln|t| - 4 \cdot \frac{1}{2} \ln(2t + \tfrac{1}{4}) + C \tag{474}$$

$$= 2 \ln|t| - 2 \ln(2t + \tfrac{1}{4}) + C. \tag{475}$$

Re-substituting $t = y^2$ gives

$$A_{r=1}(y) = 2 \ln\left(\frac{y^2}{2y^2 + \frac{1}{4}} \right) + C.$$

3. Definite integral on $[y_{c,B}, y_c]$ Thus

$$I_9(1; y_{c,B}, y_c) = \left[2\ln\left(\frac{y^2}{2y^2 + \frac{1}{4}} \right) \right]_{y_{c,B}}^{y_c} \tag{476}$$

$$= 2\left\{ \ln\left(\frac{y_c^2}{2y_c^2 + \frac{1}{4}} \right) - \ln\left(\frac{y_{c,B}^2}{2y_{c,B}^2 + \frac{1}{4}} \right) \right\} \tag{477}$$

$$= 2\ln\left(\frac{y_c^2(2y_{c,B}^2 + \frac{1}{4})}{y_{c,B}^2(2y_c^2 + \frac{1}{4})} \right). \tag{478}$$

Using $2y^2 + \frac{1}{4} = \frac{1}{4}(8y^2 + 1)$, one may also write

$$\frac{y^2}{2y^2 + \frac{1}{4}} = \frac{4y^2}{8y^2 + 1},$$

so another equivalent form is

$$I_9(1; y_{c,B}, y_c) = 2\ln\left(\frac{4y_c^2/(8y_c^2 + 1)}{4y_{c,B}^2/(8y_{c,B}^2 + 1)} \right) = 2\ln\left(\frac{y_c^2(8y_{c,B}^2 + 1)}{y_{c,B}^2(8y_c^2 + 1)} \right). \tag{479}$$

4. Summary (degenerate case $r = 1$**)**

$$\boxed{I_9(1; y_{c,B}, y_c) = 2\ln\left(\frac{y_c^2(2y_{c,B}^2 + \frac{1}{4})}{y_{c,B}^2(2y_c^2 + \frac{1}{4})} \right) = 2\ln\left(\frac{y_c^2(8y_{c,B}^2 + 1)}{y_{c,B}^2(8y_c^2 + 1)} \right), \qquad 0 < y_{c,B} < y_c.}$$

Analytic evaluation of I_{10} **on** $[y_{c,B}, y_c]$

We define

$$I_{10}(r; y_{c,B}, y_c) := \int_{y_{c,B}}^{y_c} \frac{1}{y^2(E\,y^4 + F\,y^2 + G)}\,dy, \qquad E = (1-r)^2,\ F = 1 + r,\ G = \tfrac{1}{4}, \tag{480}$$

for a real parameter $r > 0$ with $r \neq 1$.

1. Quartic factorisation in y^2 Let

$$Q(t) := Et^2 + Ft + G, \qquad t = y^2.$$

The discriminant is

$$D := F^2 - 4EG = (1+r)^2 - 4(1-r)^2 \cdot \tfrac{1}{4} = (1+r)^2 - (1-r)^2 = 4r > 0 \quad (r > 0), \tag{481}$$

so $Q(t)$ has two distinct real roots

$$u_{1,2} = \frac{-F \pm \sqrt{D}}{2E} = \frac{-(1+r) \pm 2\sqrt{r}}{2(1-r)^2}. \tag{482}$$

As in previous sections,

$$u_1 + u_2 = -\frac{F}{E} = -\frac{1+r}{(1-r)^2} < 0, \qquad u_1 u_2 = \frac{G}{E} = \frac{1}{4(1-r)^2} > 0,$$

so

$$u_1 < 0, \qquad u_2 < 0.$$

Define positive constants

$$a_1 := \sqrt{-u_1} > 0, \qquad a_2 := \sqrt{-u_2} > 0,$$

so that

$$y^2 - u_j = y^2 + a_j^2, \quad j = 1, 2.$$

Then

$$Ey^4 + Fy^2 + G = E(y^2 - u_1)(y^2 - u_2) = E(y^2 + a_1^2)(y^2 + a_2^2) > 0$$

for all real y, hence the only singularity of the integrand is at $y = 0$, which lies outside $[y_{c,B}, y_c]$.

2. Partial fraction decomposition in y^2 We write

$$\frac{1}{y^2(Ey^4 + Fy^2 + G)} = \frac{1}{E} \frac{1}{y^2(y^2 + a_1^2)(y^2 + a_2^2)} \tag{483}$$

$$= \frac{1}{E} \left(\frac{A}{y^2} + \frac{B}{y^2 + a_1^2} + \frac{C}{y^2 + a_2^2} \right), \tag{484}$$

for constants A, B, C independent of y.

Multiply both sides by $y^2(y^2 + a_1^2)(y^2 + a_2^2)$:

$$1 = A(y^2 + a_1^2)(y^2 + a_2^2) + By^2(y^2 + a_2^2) + Cy^2(y^2 + a_1^2) \tag{485}$$

$$= A(y^4 + (a_1^2 + a_2^2)y^2 + a_1^2 a_2^2) + B(y^4 + a_2^2 y^2) + C(y^4 + a_1^2 y^2). \tag{486}$$

Equating coefficients of y^4, y^2, and the constant term gives

$$y^4 : \quad A + B + C = 0, \tag{487}$$

$$y^2 : \quad A(a_1^2 + a_2^2) + Ba_2^2 + Ca_1^2 = 0, \tag{488}$$

$$y^0 : \quad Aa_1^2 a_2^2 = 1. \tag{489}$$

From the constant term,

$$A = \frac{1}{a_1^2 a_2^2}.$$

Then $B + C = -A = -\frac{1}{a_1^2 a_2^2}$, and the y^2 equation simplifies to

$$Aa_2^2 + B(a_2^2 - a_1^2) = 0 \;\Rightarrow\; B = -\frac{Aa_2^2}{a_2^2 - a_1^2} = -\frac{1}{a_1^2(a_2^2 - a_1^2)},$$

and hence

$$C = -A - B = -\frac{1}{a_1^2 a_2^2} + \frac{1}{a_1^2(a_2^2 - a_1^2)} = \frac{1}{a_2^2(a_2^2 - a_1^2)}.$$

Thus

$$\frac{1}{y^2(y^2 + a_1^2)(y^2 + a_2^2)} = \frac{1}{a_1^2 a_2^2} \frac{1}{y^2} - \frac{1}{a_1^2(a_2^2 - a_1^2)} \frac{1}{y^2 + a_1^2} + \frac{1}{a_2^2(a_2^2 - a_1^2)} \frac{1}{y^2 + a_2^2}. \tag{490}$$

3. Indefinite antiderivative: why only arctan **and** ln**?** Using the standard identities (for $a > 0$)

$$\int \frac{dy}{y^2} = -\frac{1}{y} + C, \qquad \int \frac{dy}{y^2 + a^2} = \frac{1}{a} \arctan\!\Big(\frac{y}{a}\Big) + C,$$

we obtain

$$\int \frac{1}{y^2(Ey^4+Fy^2+G)}\,dy = \frac{1}{E}\left[\frac{1}{a_1^2a_2^2}\int\frac{dy}{y^2} - \frac{1}{a_1^2(a_2^2-a_1^2)}\int\frac{dy}{y^2+a_1^2}\right. \tag{491}$$

$$\left. + \frac{1}{a_2^2(a_2^2-a_1^2)}\int\frac{dy}{y^2+a_2^2}\right] \tag{492}$$

$$= \frac{1}{E}\left[-\frac{1}{a_1^2a_2^2}\frac{1}{y} - \frac{1}{a_1^2(a_2^2-a_1^2)}\frac{1}{a_1}\arctan\Big(\frac{y}{a_1}\Big)\right. \tag{493}$$

$$\left. + \frac{1}{a_2^2(a_2^2-a_1^2)}\frac{1}{a_2}\arctan\Big(\frac{y}{a_2}\Big)\right] + C. \tag{494}$$

Note: the denominators $y^2+a_j^2>0$ for all real y, so the natural real primitive is expressed in terms of arctan only. No ln or artanh are needed; thus the expression is manifestly real and directly plottable.

4. Simplification of coefficients We use the relation

$$u_1u_2 = \frac{G}{E} = \frac{1}{4(1-r)^2}, \qquad a_j^2 = -u_j,$$

so

$$a_1^2a_2^2 = (-u_1)(-u_2) = u_1u_2 = \frac{G}{E}.$$

Hence

$$\frac{1}{Ea_1^2a_2^2} = \frac{1}{E\,(G/E)} = \frac{1}{G} = 4,$$

so the $1/y$ term simplifies nicely.

Furthermore,

$$u_j = -a_j^2 \;\Rightarrow\; u_ja_j = -a_j^3.$$

Using

$$u_1 - u_2 = \frac{\sqrt{D}}{E} = \frac{2\sqrt{r}}{E},$$

one finds after some algebra that the arctan coefficients reduce to

$$\frac{1}{E}\frac{1}{a_1^2(a_2^2-a_1^2)}\frac{1}{a_1} = \frac{1}{2\sqrt{r}a_1^3}, \qquad \frac{1}{E}\frac{1}{a_2^2(a_2^2-a_1^2)}\frac{1}{a_2} = \frac{1}{2\sqrt{r}a_2^3},$$

with the appropriate signs. Collecting, we obtain a clean real antiderivative:

$$\int \frac{1}{y^2(Ey^4+Fy^2+G)}\,dy = -\frac{4}{y} - \frac{1}{2\sqrt{r}\,a_1^3}\arctan\Big(\frac{y}{a_1}\Big) + \frac{1}{2\sqrt{r}\,a_2^3}\arctan\Big(\frac{y}{a_2}\Big) + C, \tag{495}$$

where

$$u_{1,2} = \frac{-(1+r)\pm 2\sqrt{r}}{2(1-r)^2}, \qquad a_j = \sqrt{-u_j} > 0, \quad j=1,2, \qquad r>0,\ r\neq 1.$$

5. Definite integral on $[y_{c,B}, y_c]$ Evaluating between $y = y_{c,B}$ and $y = y_c$ gives

$$I_{10}(r; y_{c,B}, y_c) = \left[-\frac{4}{y} - \frac{1}{2\sqrt{r}\, a_1^3} \arctan\Bigl(\frac{y}{a_1}\Bigr) + \frac{1}{2\sqrt{r}\, a_2^3} \arctan\Bigl(\frac{y}{a_2}\Bigr) \right]_{y_{c,B}}^{y_c} \tag{496}$$

$$= 4\left(\frac{1}{y_{c,B}} - \frac{1}{y_c}\right) \tag{497}$$

$$- \frac{1}{2\sqrt{r}\, a_1^3} \left[\arctan\Bigl(\frac{y_c}{a_1}\Bigr) - \arctan\Bigl(\frac{y_{c,B}}{a_1}\Bigr) \right] \tag{498}$$

$$+ \frac{1}{2\sqrt{r}\, a_2^3} \left[\arctan\Bigl(\frac{y_c}{a_2}\Bigr) - \arctan\Bigl(\frac{y_{c,B}}{a_2}\Bigr) \right], \tag{499}$$

with

$$u_{1,2} = \frac{-(1+r) \pm 2\sqrt{r}}{2(1-r)^2}, \qquad a_j = \sqrt{-u_j} > 0, \quad j = 1, 2, \qquad r > 0,\ r \neq 1.$$

6. Summary (boxed form, $r > 0,\ r \neq 1$)

$$\boxed{\begin{aligned} I_{10}(r; y_{c,B}, y_c) = 4\left(\frac{1}{y_{c,B}} - \frac{1}{y_c}\right) & \\ - \frac{1}{2\sqrt{r}\, a_1^3} & \left[\arctan\Bigl(\frac{y_c}{a_1}\Bigr) - \arctan\Bigl(\frac{y_{c,B}}{a_1}\Bigr) \right] \\ + \frac{1}{2\sqrt{r}\, a_2^3} & \left[\arctan\Bigl(\frac{y_c}{a_2}\Bigr) - \arctan\Bigl(\frac{y_{c,B}}{a_2}\Bigr) \right], \\ u_{1,2} = \frac{-(1+r) \pm 2\sqrt{r}}{2(1-r)^2}, & \qquad a_j = \sqrt{-u_j} > 0, \quad j = 1, 2, \qquad r > 0,\ r \neq 1. \end{aligned}}$$

This expression is manifestly real (only ln-free, real arctan and rational terms) and therefore directly plottable.

Analytic evaluation of I_{10} on $[y_{c,B}, y_c]$ for $r = 1$

For

$$I_{10}(r; y_{c,B}, y_c) = \int_{y_{c,B}}^{y_c} \frac{1}{y^2(Ey^4 + Fy^2 + G)}\, dy, \qquad E = (1-r)^2,\ F = 1 + r,\ G = \tfrac{1}{4},$$

the degenerate case $r = 1$ yields

$$E = 0, \qquad F = 2, \qquad G = \tfrac{1}{4},$$

so

$$Ey^4 + Fy^2 + G = 2y^2 + \tfrac{1}{4},$$

and

$$I_{10}(1; y_{c,B}, y_c) = \int_{y_{c,B}}^{y_c} \frac{1}{y^2(2y^2 + \frac{1}{4})}\, dy. \tag{500}$$

1. Partial fractions for the $r = 1$ integrand Factor

$$2y^2 + \tfrac{1}{4} = 2\Big(y^2 + \tfrac{1}{8}\Big)$$

so that

$$\frac{1}{y^2(2y^2+\frac{1}{4})} = \frac{1}{2}\,\frac{1}{y^2(y^2+\frac{1}{8})}.$$

Seek A, B such that

$$\frac{1}{y^2(y^2+a^2)} = \frac{A}{y^2} + \frac{B}{y^2+a^2}, \qquad a^2 := \frac{1}{8}.$$

Then

$$1 = A(y^2+a^2) + By^2 = (A+B)y^2 + Aa^2,$$

so

$$Aa^2 = 1 \;\Rightarrow\; A = \frac{1}{a^2}, \qquad A + B = 0 \;\Rightarrow\; B = -\frac{1}{a^2}.$$

Thus

$$\frac{1}{y^2(y^2+a^2)} = \frac{1}{a^2}\frac{1}{y^2} - \frac{1}{a^2}\frac{1}{y^2+a^2}.$$

Here $a^2 = \frac{1}{8}$, so

$$a = \frac{1}{2\sqrt{2}}, \qquad \frac{1}{a^2} = 8.$$

Hence

$$\frac{1}{y^2(2y^2+\frac{1}{4})} = \frac{1}{2}\left(\frac{8}{y^2} - \frac{8}{y^2+a^2}\right) = 4\frac{1}{y^2} - 4\frac{1}{y^2+a^2}.$$

2. Indefinite antiderivative Using

$$\int \frac{dy}{y^2} = -\frac{1}{y} + C, \qquad \int \frac{dy}{y^2+a^2} = \frac{1}{a}\arctan\Big(\frac{y}{a}\Big) + C,$$

and $1/a = 2\sqrt{2}$, we get

$$\int \frac{1}{y^2(2y^2+\frac{1}{4})}\,dy = 4\int\frac{dy}{y^2} - 4\int\frac{dy}{y^2+a^2} \tag{501}$$

$$= 4\Big(-\frac{1}{y}\Big) - 4\cdot\frac{1}{a}\arctan\Big(\frac{y}{a}\Big) + C \tag{502}$$

$$= -\frac{4}{y} - 8\sqrt{2}\,\arctan(2\sqrt{2}\,y) + C. \tag{503}$$

3. Definite integral on $[y_{c,B}, y_c]$ Thus

$$I_{10}(1; y_{c,B}, y_c) = \left[-\frac{4}{y} - 8\sqrt{2}\,\arctan(2\sqrt{2}\,y)\right]_{y_{c,B}}^{y_c} \tag{504}$$

$$= 4\left(\frac{1}{y_{c,B}} - \frac{1}{y_c}\right) - 8\sqrt{2}\left[\arctan(2\sqrt{2}\,y_c) - \arctan(2\sqrt{2}\,y_{c,B})\right]. \tag{505}$$

4. Summary (degenerate case $r = 1$)

$$\boxed{I_{10}(1; y_{c,B}, y_c) = 4\left(\frac{1}{y_{c,B}} - \frac{1}{y_c}\right) - 8\sqrt{2}\left[\arctan(2\sqrt{2}\,y_c) - \arctan(2\sqrt{2}\,y_{c,B})\right], \qquad 0 < y_{c,B} < y_c.}$$

B.3 Region III integrals: tail asymptotics

For $y > y_c$ both arguments $x_F = kL$ and $x_B = ka$ lie in their small–argument regimes, so the truncated Region III integrand obtained earlier is

$$\mathcal{I}_{\mathrm{III}}(y) \approx -\sqrt{2}\, r \, \frac{\dfrac{D_2}{y} - \dfrac{D_4 + C_2 D_2}{y^3}}{\mathcal{D}(y; r)}, \qquad y > y_c. \tag{506}$$

Region III integral We now define the Region III contribution (inner integral only),

$$I_{\mathrm{III}}(\omega) := \int_{y_c}^{\infty} \mathcal{I}_{\mathrm{III}}(y)\, dy. \tag{507}$$

Substituting (506) gives

$$I_{\mathrm{III}}(\omega) = -\sqrt{2}\, r \int_{y_c}^{\infty} \frac{\dfrac{D_2}{y} - \dfrac{D_4 + C_2 D_2}{y^3}}{(1-r)^2\, y^4 + (1+r)\, y^2 + \frac{1}{4}}\, dy. \tag{508}$$

It is natural to separate (508) into two basic integrals. Define

$$I_3^{(\mathrm{III})}(r; \omega, L, a) := \int_{y_c}^{\infty} \frac{1}{y(\,(1-r)^2\, y^4 + (1+r)\, y^2 + \frac{1}{4})}\, dy, \tag{509}$$

$$I_4^{(\mathrm{III})}(r; \omega, L, a) := \int_{y_c}^{\infty} \frac{1}{y^3\,((1-r)^2\, y^4 + (1+r)\, y^2 + \frac{1}{4})}\, dy. \tag{510}$$

Then (508) becomes

$$I_{\mathrm{III}}(\omega) = -\sqrt{2}\, r \left[D_2\, I_3^{(\mathrm{III})}(r; \omega, L, a) - (D_4 + C_2 D_2)\, I_4^{(\mathrm{III})}(r; \omega, L, a) \right]. \tag{511}$$

The integrals $I_3^{(\mathrm{III})}$ and $I_4^{(\mathrm{III})}$ are now in a form suitable for analytic or numerical evaluation.

Analytic evaluation of I_3 on $[y_c, +\infty]$

We first compute an antiderivative of the integrand

$$I_3^{(\mathrm{III})}(r; \omega, L, a) := \int \frac{1}{y(Ey^4 + Fy^2 + G)}\, dy, \qquad E = (1-r)^2,\; F = 1 + r,\; G = \tfrac{1}{4}. \tag{512}$$

1. Substitution $u = y^2$ Set

$$u = y^2, \qquad du = 2y\, dy \quad \Rightarrow \quad \frac{dy}{y} = \frac{du}{2u}, \tag{513}$$

so that

$$Ey^4 + Fy^2 + G = Eu^2 + Fu + G.$$

Thus

$$\int \frac{1}{y(Ey^4 + Fy^2 + G)}\, dy = \frac{1}{2} \int \frac{1}{u(Eu^2 + Fu + G)}\, du. \tag{514}$$

2. Partial fractions in u Decompose

$$\frac{1}{u(Eu^2+Fu+G)} = \frac{A}{u} + \frac{Bu+C}{Eu^2+Fu+G}, \tag{515}$$

which yields

$$A = \frac{1}{G}, \qquad B = -\frac{E}{G}, \qquad C = -\frac{F}{G}.$$

Hence

$$\int \frac{1}{u(Eu^2+Fu+G)}\,du = \frac{1}{G}\int \frac{du}{u} + \int \frac{-\frac{E}{G}u - \frac{F}{G}}{Eu^2+Fu+G}\,du. \tag{516}$$

3. Integration of $(Bu+C)/(Eu^2+Fu+G)$ Let

$$D(u) = Eu^2+Fu+G, \qquad S = \sqrt{F^2-4EG}.$$

Writing

$$Bu+C = -\frac{1}{2G}D'(u) - \frac{F}{2G},$$

we obtain

$$\int \frac{Bu+C}{D(u)}\,du = -\frac{1}{2G}\ln|D(u)| - \frac{F}{2GS}\ln\left|\frac{2Eu+F-S}{2Eu+F+S}\right| + C. \tag{517}$$

4. Assemble the antiderivative in u

$$\int \frac{1}{y(Ey^4+Fy^2+G)}\,dy = \frac{1}{2}\left[\frac{1}{G}\ln|u| - \frac{1}{2G}\ln|D(u)| - \frac{F}{2GS}\ln\left|\frac{2Eu+F-S}{2Eu+F+S}\right|\right] + C. \tag{518}$$

5. Back–substitute $u = y^2$ Since $\ln|u| = 2\ln|y|$ and $D(u) = Ey^4+Fy^2+G$,

$$\int \frac{1}{y(Ey^4+Fy^2+G)}\,dy = \frac{1}{G}\ln|y| - \frac{1}{4G}\ln|Ey^4+Fy^2+G| \tag{519}$$

$$- \frac{F}{4G\sqrt{F^2-4EG}}\ln\left|\frac{2Ey^2+F-\sqrt{F^2-4EG}}{2Ey^2+F+\sqrt{F^2-4EG}}\right| + C. \tag{520}$$

6. Specialization to QTN parameters With

$$E = (1-r)^2, \qquad F = 1+r, \qquad G = \tfrac{1}{4}, \qquad S = 2\sqrt{r},$$

we obtain the Region III antiderivative

$$\boxed{\begin{aligned}\int \frac{1}{y((1-r)^2y^4+(1+r)y^2+\frac{1}{4})}\,dy &= 4\ln|y| - \ln\left|(1-r)^2y^4+(1+r)y^2+\tfrac{1}{4}\right| \\ &\quad - \frac{1+r}{2\sqrt{r}}\ln\left|\frac{2(1-r)^2y^2+(1+r)-2\sqrt{r}}{2(1-r)^2y^2+(1+r)+2\sqrt{r}}\right| + C.\end{aligned}} \tag{521}$$

Definite evaluation of I_3 on $[y_c, +\infty)$

We start from the specialized indefinite antiderivative

$$\boxed{\begin{aligned}\int \frac{1}{y((1-r)^2y^4+(1+r)y^2+\frac{1}{4})}\,dy &= 4\ln|y| - \ln\left|(1-r)^2y^4+(1+r)y^2+\tfrac{1}{4}\right| \\ &\quad - \frac{1+r}{2\sqrt{r}}\ln\left|\frac{2(1-r)^2y^2+(1+r)-2\sqrt{r}}{2(1-r)^2y^2+(1+r)+2\sqrt{r}}\right| + C, \qquad r>0.\end{aligned}} \tag{522}$$

Definite integral from y_c to ∞. By definition,

$$I_3^{(\mathrm{III})}(y_c) := \int_{y_c}^{\infty} \frac{dy}{y((1-r)^2y^4+(1+r)y^2+\frac{1}{4})} = \left[\text{antiderivative in (522)}\right]_{y=y_c}^{y\to\infty}. \tag{523}$$

Step 1: Evaluate the limit as $y \to \infty$ (i) Log–difference piece.

$$\lim_{y\to\infty}\Big(4\ln y - \ln\big((1-r)^2y^4+(1+r)y^2+\tfrac{1}{4}\big)\Big) = -\ln\big((1-r)^2\big). \tag{524}$$

(ii) Ratio–log piece.

$$\lim_{y\to\infty}\ln\left|\frac{2(1-r)^2y^2+(1+r)-2\sqrt{r}}{2(1-r)^2y^2+(1+r)+2\sqrt{r}}\right| = 0.$$

Thus,

$$\lim_{y\to\infty}\left(\text{antiderivative in (522)}\right) = -\ln\big((1-r)^2\big). \tag{525}$$

Step 2: Evaluate at the lower limit $y = y_c$

$$\begin{aligned}\text{antiderivative in (522)}\big|_{y=y_c} &= 4\ln y_c - \ln\big((1-r)^2y_c^4+(1+r)y_c^2+\tfrac{1}{4}\big) &(526)\\ &\quad - \frac{1+r}{2\sqrt{r}}\ln\left(\frac{2(1-r)^2y_c^2+(1+r)-2\sqrt{r}}{2(1-r)^2y_c^2+(1+r)+2\sqrt{r}}\right). &(527)\end{aligned}$$

Step 3: Assemble the definite value Subtracting the lower-limit value from the upper-limit limit gives

$$\boxed{\begin{aligned}I_3^{(\mathrm{III})}(y_c) &= \ln\left(\frac{(1-r)^2y_c^4+(1+r)y_c^2+\frac{1}{4}}{(1-r)^2}\right) - 4\ln y_c \\ &\quad + \frac{1+r}{2\sqrt{r}}\ln\left(\frac{2(1-r)^2y_c^2+(1+r)-2\sqrt{r}}{2(1-r)^2y_c^2+(1+r)+2\sqrt{r}}\right), \qquad r>0.\end{aligned}} \tag{528}$$

Analytic evaluation of I_3 on $[y_c, +\infty]$ for $r = 1$

In the degenerate case $r = 1$ the coefficients become

$$E = (1-r)^2 = 0, \qquad F = 1+r = 2, \qquad G = \tfrac{1}{4}, \tag{529}$$

so the Region III integrand for $I_3^{(\mathrm{III})}$ reduces to

$$I_3^{(\mathrm{III})}(y_c) := \int_{y_c}^{\infty} \frac{dy}{y(2y^2+\frac{1}{4})}. \tag{530}$$

Step 1: Indefinite antiderivative

We first compute an antiderivative of

$$\int \frac{1}{y(2y^2 + \frac{1}{4})}\, dy.$$

Factor the denominator:

$$\frac{1}{y(2y^2 + \frac{1}{4})} = \frac{1}{y \cdot \frac{1}{4}(8y^2 + 1)} = 4\, \frac{1}{y(8y^2 + 1)}. \tag{531}$$

Thus

$$\int \frac{1}{y(2y^2 + \frac{1}{4})}\, dy = 4 \int \frac{1}{y(8y^2 + 1)}\, dy. \tag{532}$$

Set

$$t = y^2, \qquad\qquad dt = 2y\, dy \quad \Rightarrow \quad \frac{dy}{y} = \frac{dt}{2t}, \tag{533}$$

and note that

$$8y^2 + 1 = 8t + 1.$$

Then

$$4 \int \frac{1}{y(8y^2 + 1)}\, dy = 4 \int \frac{1}{8t + 1}\, \frac{dt}{2t} = 2 \int \frac{1}{t(8t + 1)}\, dt. \tag{534}$$

Decompose

$$\frac{1}{t(8t + 1)} = \frac{A}{t} + \frac{B}{8t + 1}. \tag{535}$$

Multiplying by $t(8t + 1)$ gives

$$1 = A(8t + 1) + Bt = (8A + B)t + A,$$

so

$$A = 1, \qquad 8A + B = 0 \Rightarrow B = -8.$$

Therefore

$$2 \int \frac{1}{t(8t + 1)}\, dt = 2 \int \left(\frac{1}{t} - \frac{8}{8t + 1} \right) dt \tag{536}$$

$$= 2 \ln |t| - 2 \ln |8t + 1| + C \tag{537}$$

$$= 2 \ln \left| \frac{t}{8t + 1} \right| + C. \tag{538}$$

Undoing $t = y^2$ gives the indefinite antiderivative

$$\int \frac{1}{y(2y^2 + \frac{1}{4})}\, dy = 2 \ln \left| \frac{y^2}{8y^2 + 1} \right| + C. \tag{539}$$

Step 2: Limit as $y \to \infty$

From (539),

$$\lim_{y \to \infty} 2 \ln \left(\frac{y^2}{8y^2 + 1} \right) = 2 \ln \left(\lim_{y \to \infty} \frac{y^2}{8y^2 + 1} \right) = 2 \ln \left(\frac{1}{8} \right). \tag{540}$$

Step 3: Value at the lower limit $y = y_c$

$$2\ln\left(\frac{y^2}{8y^2+1}\right)\bigg|_{y=y_c} = 2\ln\left(\frac{y_c^2}{8y_c^2+1}\right). \tag{541}$$

Step 4: Assemble the definite value

The definite integral is

$$I_3^{(\mathrm{III})}(y_c) = \left[2\ln\left(\frac{y^2}{8y^2+1}\right)\right]_{y=y_c}^{y\to\infty} \tag{542}$$

$$= 2\ln\left(\frac{1}{8}\right) - 2\ln\left(\frac{y_c^2}{8y_c^2+1}\right) \tag{543}$$

$$= -2\ln 8 - 2\ln\left(\frac{y_c^2}{8y_c^2+1}\right) \tag{544}$$

$$= -2\ln\left(8\,\frac{y_c^2}{8y_c^2+1}\right) \tag{545}$$

$$= -2\ln\left(\frac{8y_c^2}{8y_c^2+1}\right) \tag{546}$$

$$= 2\ln\left(\frac{8y_c^2+1}{8y_c^2}\right). \tag{547}$$

Definite result (degenerate case $r = 1$).

$$\boxed{I_3^{(\mathrm{III})}(y_c) = 2\ln\left(\frac{8y_c^2+1}{8y_c^2}\right), \qquad r = 1} \tag{548}$$

Analytic evaluation of I_4 on $[y_c, +\infty]$

We first compute an antiderivative of the general integral

$$I_4(E,F,G) := \int \frac{1}{y^3(Ey^4+Fy^2+G)}\,dy, \tag{549}$$

and then specialize to the Region III parameters.

1. Substitution $t = y^2$
   Set

$$t = y^2, \qquad dt = 2y\,dy, \qquad dy = \frac{dt}{2\sqrt{t}}, \qquad y^3 = t\sqrt{t}. \tag{550}$$

Then

$$Ey^4 + Fy^2 + G = Et^2 + Ft + G, \tag{551}$$

and

$$\frac{1}{y^3(Ey^4+Fy^2+G)}\,dy = \frac{1}{t\sqrt{t}\,(Et^2+Ft+G)}\cdot\frac{dt}{2\sqrt{t}} = \frac{1}{2}\,\frac{dt}{t^2(Et^2+Ft+G)}. \tag{552}$$

Thus

$$I_4(E,F,G) = \frac{1}{2}\int \frac{dt}{t^2(Et^2+Ft+G)}. \tag{553}$$

2. Partial fraction decomposition in t

We seek constants $\alpha, \beta, \gamma, \delta$ such that

$$\frac{1}{t^2(Et^2+Ft+G)} = \frac{\alpha}{t} + \frac{\beta}{t^2} + \frac{\gamma t+\delta}{Et^2+Ft+G}. \tag{554}$$

Multiplying by $t^2(Et^2+Ft+G)$ gives

$$1 = (\alpha E+\gamma)\,t^3 + (\alpha F+\beta E+\delta)\,t^2 + (\alpha G+\beta F)\,t + \beta G. \tag{555}$$

Matching coefficients with the constant polynomial 1 yields

$$\alpha E+\gamma = 0, \tag{556}$$

$$\alpha F+\beta E+\delta = 0, \tag{557}$$

$$\alpha G+\beta F = 0, \tag{558}$$

$$\beta G = 1. \tag{559}$$

From $\beta G = 1$ we get

$$\beta = \frac{1}{G}. \tag{560}$$

Then

$$\alpha = -\frac{\beta F}{G} = -\frac{F}{G^2}, \tag{561}$$

$$\gamma = -\alpha E = \frac{EF}{G^2}, \tag{562}$$

$$\delta = -\alpha F - \beta E = \frac{F^2}{G^2} - \frac{E}{G}. \tag{563}$$

Thus

$$\frac{1}{t^2(Et^2+Ft+G)} = -\frac{F}{G^2}\frac{1}{t} + \frac{1}{G}\frac{1}{t^2} + \frac{\frac{EF}{G^2}\,t + \left(\frac{F^2}{G^2}-\frac{E}{G}\right)}{Et^2+Ft+G}. \tag{564}$$

3. Integrate term by term

From (553),

$$I_4(E,F,G) = \frac{1}{2}\left(-\frac{F}{G^2}\int\frac{dt}{t} + \frac{1}{G}\int\frac{dt}{t^2} + \int\frac{\gamma t+\delta}{Et^2+Ft+G}\,dt\right), \tag{565}$$

with

$$\gamma = \frac{EF}{G^2}, \qquad \delta = \frac{F^2}{G^2} - \frac{E}{G}. \tag{566}$$

The elementary integrals are

$$\int\frac{dt}{t} = \ln|t|, \qquad \int\frac{dt}{t^2} = -\frac{1}{t}. \tag{567}$$

For the rational piece, write

$$\gamma t+\delta = k(2Et+F)+m, \tag{568}$$

$$k = \frac{\gamma}{2E}, \tag{569}$$

$$m = \delta - \frac{\gamma F}{2E}. \tag{570}$$

Then

$$\int \frac{\gamma t+\delta}{Et^2+Ft+G}\,dt = k\int \frac{2Et+F}{Et^2+Ft+G}\,dt + m\int \frac{dt}{Et^2+Ft+G}. \tag{571}$$

The first integral is

$$\int \frac{2Et+F}{Et^2+Ft+G}\,dt = \ln|Et^2+Ft+G|. \tag{572}$$

4. Quadratic integral and $\Delta = 4EG - F^2$
   Complete the square:

$$Et^2+Ft+G = E\Big(t+\frac{F}{2E}\Big)^2 + \frac{4EG-F^2}{4E}. \tag{573}$$

Define

$$\Delta := 4EG - F^2. \tag{574}$$

With $u = t + \frac{F}{2E}$, we have

$$\int \frac{dt}{Et^2+Ft+G} = \frac{1}{E}\int \frac{du}{u^2+\dfrac{\Delta}{4E^2}}. \tag{575}$$

If $\Delta > 0$, using $\displaystyle\int \frac{du}{u^2+a^2} = \frac{1}{a}\arctan\frac{u}{a}$, we obtain

$$\int \frac{dt}{Et^2+Ft+G} = \frac{2}{\sqrt{\Delta}}\,\arctan\Big(\frac{2Et+F}{\sqrt{\Delta}}\Big). \tag{576}$$

For $\Delta < 0$ this will be converted to an artanh form below.

5. Primitive in t (arctan form)
   Using $\gamma = EF/G^2$, we have

$$k = \frac{\gamma}{2E} = \frac{F}{2G^2}, \tag{577}$$

$$m = \delta - \frac{\gamma F}{2E} = \left(\frac{F^2}{G^2} - \frac{E}{G}\right) - \frac{EF}{G^2}\cdot\frac{F}{2E} = \frac{F^2}{2G^2} - \frac{E}{G}. \tag{578}$$

Substitute into $I_4(E,F,G)$:

$$I_4(E,F,G) = \frac{1}{2}\Bigg[-\frac{F}{G^2}\ln|t| - \frac{1}{Gt} + \frac{F}{2G^2}\ln|Et^2+Ft+G| + \frac{2}{\sqrt{\Delta}}\left(\frac{F^2}{2G^2}-\frac{E}{G}\right)\arctan\Big(\frac{2Et+F}{\sqrt{\Delta}}\Big)\Bigg] + C_0. \tag{579}$$

Simplify:

$$I_4(E,F,G) = -\frac{F}{2G^2}\ln|t| - \frac{1}{2Gt} + \frac{F}{4G^2}\ln|Et^2+Ft+G|$$
$$+\frac{F^2-2EG}{2G^2\sqrt{\Delta}}\,\mathrm{arctan}\Big(\frac{2Et+F}{\sqrt{\Delta}}\Big) + C_0. \tag{580}$$

6. Back-substitution $t = y^2$

   Use

$$t = y^2, \qquad \ln|t| = \ln|y^2| = 2\ln|y|, \qquad \frac{1}{t} = \frac{1}{y^2}, \tag{581}$$

and

$$Et^2 + Ft + G = Ey^4 + Fy^2 + G. \tag{582}$$

Then (580) becomes

$$\begin{aligned} I_4(E,F,G) &= -\frac{F}{2G^2}\cdot 2\ln|y| - \frac{1}{2Gy^2} + \frac{F}{4G^2}\ln\big|Ey^4+Fy^2+G\big| \\ &\quad + \frac{F^2-2EG}{2G^2\sqrt{\Delta}}\,\mathrm{arctan}\Big(\frac{2Ey^2+F}{\sqrt{\Delta}}\Big) + C_0 \end{aligned} \tag{583}$$

$$\begin{aligned} &= -\frac{F}{G^2}\ln|y| - \frac{1}{2Gy^2} + \frac{F}{4G^2}\ln\big|Ey^4+Fy^2+G\big| \\ &\quad + \frac{F^2-2EG}{2G^2\sqrt{4EG-F^2}}\,\mathrm{arctan}\Big(\frac{2Ey^2+F}{\sqrt{4EG-F^2}}\Big) + C_0, \end{aligned} \tag{584}$$

where $\Delta = 4EG - F^2$.

This can be rearranged into the compact CAS-like form

$$\boxed{\begin{aligned} \int \frac{dy}{y^3(Ey^4+Fy^2+G)} &= \frac{Fy^2\big(\ln\big|Ey^4+Fy^2+G\big| - 4\ln|y|\big) - 2G}{4G^2y^2} \\ &\quad - \frac{(2EG-F^2)}{2G^2\sqrt{4EG-F^2}}\,\mathrm{arctan}\Big(\frac{2Ey^2+F}{\sqrt{4EG-F^2}}\Big) + C. \end{aligned}} \tag{585}$$

7. Real artanh form for $r > 0$

   For Region III we take

$$E = (1-r)^2, \qquad F = 1 + r, \qquad G = \tfrac{1}{4}, \tag{586}$$

with $r = (\omega_p/\omega)^2 \geq 0$. Then

$$4EG - F^2 = (1-r)^2 - (1+r)^2 = -4r < 0 \qquad (r > 0). \tag{587}$$

Thus $\sqrt{4EG-F^2} = 2i\sqrt{r}$ and the arctan term is purely imaginary; we rewrite it in terms of a real artanh.

Using

$$\mathrm{arctan}\, z = \frac{1}{2i}\big(\ln(1+iz) - \ln(1-iz)\big) \tag{588}$$

and the standard identity relating arctan of a purely imaginary argument to artanh, one finds that for $r > 0$

$$\frac{2EG - F^2}{2G^2\sqrt{4EG - F^2}} \arctan\Big(\frac{2Ey^2 + F}{\sqrt{4EG - F^2}}\Big) = \frac{2EG - F^2}{4G^2\sqrt{-(4EG - F^2)}} \ln\left(\frac{2Ey^2 + F - \sqrt{-(4EG - F^2)}}{2Ey^2 + F + \sqrt{-(4EG - F^2)}}\right) \tag{589}$$

$$= \text{(real multiple of artanh)}. \tag{590}$$

Specializing the compact form (585) to

$$E = (1 - r)^2, \quad F = 1 + r, \quad G = \tfrac{1}{4}, \quad 4EG - F^2 = -4r,$$

and simplifying gives the manifestly real artanh form

$$\boxed{\begin{aligned}\int \frac{dy}{y^3\left((1 - r)^2 y^4 + (1 + r)y^2 + \frac{1}{4}\right)} &= -8(1 + r)\ln(y^2) - \frac{2}{y^2} \\ &\quad + 4(1 + r)\ln\Big((1 - r)^2 y^4 + (1 + r)y^2 + \tfrac{1}{4}\Big) \\ &\quad - \frac{2(1 + 6r + r^2)}{\sqrt{r}} \operatorname{artanh}\left(\frac{2(1 - r)^2 y^2 + (1 + r)}{2\sqrt{r}}\right) + C,\end{aligned}} \tag{591}$$

valid for $r > 0$, with C an arbitrary constant of integration.

8. Remarks

The derivation above assumes $E \neq 0$ (i.e. $r \neq 1$) and $4EG - F^2 \neq 0$ (i.e. $r \neq 0$). The degenerate cases $r = 0$ and $r = 1$ will be treated as separate cases.

Definite integral from y_c to ∞. Using the real antiderivative obtained previously.

Notation.

$$E = (1 - r)^2, \qquad F = 1 + r, \qquad G = \tfrac{1}{4}, \qquad M = 1 + 6r + r^2, \qquad s = \sqrt{r},$$

$$Q(y) = Ey^4 + Fy^2 + G, \qquad Z(y) = \frac{2Ey^2 + F}{2s}.$$

Antiderivative (real form for $r > 0$).

$$F(y) = -8(1 + r)\ln(y^2) - \frac{2}{y^2} + 4(1 + r)\ln\big(Q(y)\big) - \frac{2M}{s} \operatorname{artanh}\big(Z(y)\big) + C.$$

Limit as $y \to \infty$. Since

$$Q(y) = Ey^4(1 + \mathcal{O}(y^{-2})), \qquad \ln Q(y) = \ln E + 4\ln y + o(1),$$

the logarithmic terms satisfy

$$-8(1 + r)\ln(y^2) + 4(1 + r)\ln Q(y) \longrightarrow 4(1 + r)\ln(E).$$

The term $-2/y^2 \to 0$, and the artanh term contributes only a finite constant absorbed into C. Thus

$$\lim_{Y \to \infty} F(Y) = 4(1 + r)\ln\big((1 - r)^2\big) + C.$$

Definite integral.

$$
\begin{aligned}
I_4^{(\mathrm{III})}(r;y_c) &= \lim_{Y\to\infty}\left(F(Y)-F(y_c)\right)\\
&= 4(1+r)\ln\!\left((1-r)^2\right) - \Big[-8(1+r)\ln(y_c^2) - \frac{2}{y_c^2} + 4(1+r)\ln\!\big(Q(y_c)\big) - \frac{2M}{s}\,\mathrm{artanh}\!\big(Z(y_c)\big)\Big].
\end{aligned}
$$

Closed form.

$$
\boxed{
\begin{aligned}
I_4^{(\mathrm{III})}(r;y_c) &= 4(1+r)\ln\!\left((1-r)^2\right) + 8(1+r)\ln(y_c^2) + \frac{2}{y_c^2}\\
&\quad - 4(1+r)\ln\!\big(Q(y_c)\big) + \frac{2(1+6r+r^2)}{\sqrt{r}}\,\mathrm{artanh}\!\big(Z(y_c)\big).
\end{aligned}}
$$

Expanded form.

$$
\boxed{
\begin{aligned}
I_4^{(\mathrm{III})}(r;y_c) &= 4(1+r)\ln\!\left((1-r)^2\right) + 8(1+r)\ln(y_c^2) + \frac{2}{y_c^2}\\
&\quad - 4(1+r)\ln\!\Big((1-r)^2y_c^4 + (1+r)y_c^2 + \tfrac14\Big)\\
&\quad + \frac{2(1+6r+r^2)}{\sqrt{r}}\,\mathrm{artanh}\!\left(\frac{2(1-r)^2y_c^2+(1+r)}{2\sqrt{r}}\right).
\end{aligned}}
$$

MATLAB plotting note (real-valued branch for $r\neq 1$). In Region III we have $r=(\omega_p/\omega)^2>0$ and we will typically plot for r just below and just above 1. The closed form above is *valid for all $r>0$ with $r\neq 1$* (the case $r=1$ is handled separately). However, in MATLAB the built-in `atanh` returns complex values when its argument has magnitude greater than 1. To obtain the correct real-valued continuation for plotting, replace the artanh term by the piecewise real branch

$$
\boxed{\Phi(Z) := \begin{cases}\mathrm{artanh}(Z), & |Z|<1,\\ \dfrac12\ln\left|\dfrac{Z+1}{Z-1}\right|, & |Z|>1,\end{cases}} \qquad Z := \frac{2(1-r)^2y_c^2+(1+r)}{2\sqrt{r}}. \tag{592}
$$

Hence, for MATLAB plotting with $r\neq 1$, use the explicitly real form

$$
\boxed{
\begin{aligned}
I_4^{(\mathrm{III})}(r;y_c) &= 4(1+r)\ln\!\left((1-r)^2\right) + 8(1+r)\ln(y_c^2) + \frac{2}{y_c^2}\\
&\quad - 4(1+r)\ln\!\Big((1-r)^2y_c^4 + (1+r)y_c^2 + \tfrac14\Big)\\
&\quad + \frac{2(1+6r+r^2)}{\sqrt{r}}\,\Phi\!\left(\frac{2(1-r)^2y_c^2+(1+r)}{2\sqrt{r}}\right), \qquad r>0,\ r\neq 1.
\end{aligned}} \tag{593}
$$

Remarks.

- The expression is finite for all $r>0$ and $y_c>0$.
- The argument of artanh is $Z(y_c)$; if $|Z(y_c)|>1$ the analytic continuation gives the correct real value.
- The degenerate cases $r=0$ and $r=1$ require separate limiting evaluation.

Analytic evaluation of I_4 on $[y_c, +\infty]$ for $r = 1$

In the degenerate case $r = 1$ the coefficients are

$$E = (1-r)^2 = 0, \qquad F = 1 + r = 2, \qquad G = \tfrac{1}{4}, \tag{594}$$

so the Region III integrand for $I_4^{(\mathrm{III})}$ becomes

$$I_4^{(\mathrm{III})}(y_c) := \int_{y_c}^{\infty} \frac{dy}{y^3\left(2y^2 + \frac{1}{4}\right)}. \tag{595}$$

Step 1: Indefinite antiderivative

We first compute an antiderivative of

$$\int \frac{1}{y^3\left(2y^2 + \frac{1}{4}\right)}\, dy.$$

Factor the quadratic:

$$2y^2 + \tfrac{1}{4} = \tfrac{1}{4}(8y^2 + 1), \tag{596}$$

so

$$\frac{1}{y^3\left(2y^2 + \frac{1}{4}\right)} = \frac{1}{y^3 \cdot \frac{1}{4}(8y^2 + 1)} = 4\, \frac{1}{y^3(8y^2 + 1)}. \tag{597}$$

Thus

$$\int \frac{1}{y^3\left(2y^2 + \frac{1}{4}\right)}\, dy = 4 \int \frac{1}{y^3(8y^2 + 1)}\, dy. \tag{598}$$

Substitution $t = y^2$ Let

$$t = y^2, \qquad\qquad dt = 2y\, dy \quad \Rightarrow \quad dy = \frac{dt}{2y}. \tag{599}$$

Then $y^2 = t$ and $y^3 = yt$, so

$$4 \int \frac{1}{y^3(8y^2 + 1)}\, dy = 4 \int \frac{1}{yt(8t + 1)}\, \frac{dt}{2y} = 2 \int \frac{1}{t^2(8t + 1)}\, dt. \tag{600}$$

Step 2: Partial fractions in t

Decompose

$$\frac{1}{t^2(8t + 1)} = \frac{A}{t} + \frac{B}{t^2} + \frac{C}{8t + 1}. \tag{601}$$

Multiplying by $t^2(8t + 1)$ gives

$$1 = At(8t + 1) + B(8t + 1) + Ct^2 \tag{602}$$
$$= (8A + C)t^2 + (A + 8B)t + B. \tag{603}$$

Matching coefficients with the constant polynomial 1 yields

$$B = 1, \tag{604}$$
$$A + 8B = 0 \quad \Rightarrow \quad A = -8, \tag{605}$$
$$8A + C = 0 \quad \Rightarrow \quad C = -8A = 64. \tag{606}$$

Hence

$$\frac{1}{t^2(8t+1)} = -\frac{8}{t} + \frac{1}{t^2} + \frac{64}{8t+1}. \tag{607}$$

Therefore

$$2\int \frac{1}{t^2(8t+1)}\,dt = 2\int \left(-\frac{8}{t} + \frac{1}{t^2} + \frac{64}{8t+1}\right) dt \tag{608}$$

$$= 2\Big(-8\ln|t| - \frac{1}{t} + 8\ln|8t+1|\Big) + C \tag{609}$$

$$= -16\ln|t| - \frac{2}{t} + 16\ln|8t+1| + C. \tag{610}$$

Undoing $t = y^2$ gives the indefinite antiderivative

$$\int \frac{1}{y^3(2y^2+\frac{1}{4})}\,dy = -16\ln|y^2| - \frac{2}{y^2} + 16\ln|8y^2+1| + C \tag{611}$$

$$= 16\ln\left|\frac{8y^2+1}{y^2}\right| - \frac{2}{y^2} + C. \tag{612}$$

Step 3: Limit as $y \to \infty$

From (612),

$$\lim_{y\to\infty}\Big(16\ln\Big|\frac{8y^2+1}{y^2}\Big| - \frac{2}{y^2}\Big) = 16\ln\left(\lim_{y\to\infty}\frac{8y^2+1}{y^2}\right) - 0 \tag{613}$$

$$= 16\ln(8). \tag{614}$$

Step 4: Value at the lower limit $y = y_c$

At $y = y_c$,

$$16\ln\left|\frac{8y^2+1}{y^2}\right| - \frac{2}{y^2}\bigg|_{y=y_c} = 16\ln\left(\frac{8y_c^2+1}{y_c^2}\right) - \frac{2}{y_c^2}. \tag{615}$$

Step 5: Assemble the definite value

Thus

$$I_4^{(\mathrm{III})}(y_c) = \int_{y_c}^{\infty} \frac{dy}{y^3(2y^2+\frac{1}{4})} \tag{616}$$

$$= \Big[\,16\ln\Big|\frac{8y^2+1}{y^2}\Big| - \frac{2}{y^2}\,\Big]_{y=y_c}^{y\to\infty} \tag{617}$$

$$= 16\ln(8) - \left(16\ln\left(\frac{8y_c^2+1}{y_c^2}\right) - \frac{2}{y_c^2}\right) \tag{618}$$

$$= 16\ln(8) - 16\ln\left(\frac{8y_c^2+1}{y_c^2}\right) + \frac{2}{y_c^2}. \tag{619}$$

Using $\ln a - \ln b = \ln(a/b)$, this can be written compactly as

$$I_4^{(\mathrm{III})}(y_c) = 16\ln\left(\frac{8y_c^2}{8y_c^2+1}\right) + \frac{2}{y_c^2}. \tag{620}$$

Definite result (degenerate case $r = 1$).

$$\boxed{I_4^{(\mathrm{III})}(y_c) = 16\ln\left(\frac{8y_c^2}{8y_c^2+1}\right) + \frac{2}{y_c^2}, \qquad r = 1,\ y_c > 0.} \tag{621}$$

C Unified Regions, Integrals, Variables and Constants

Unifying Equation

$$V^2(\omega) = \frac{16k_BT_e}{\pi^2\epsilon_0 v_T}\int_0^\infty \frac{1}{y^2}\,F_1(kL)\,J_0^2(ka)\,\Im\left(\frac{1}{K_L^t}\right)\,dy, \tag{622}$$

$$F_1(kL) = \frac{kL\left[\mathrm{Si}(kL) - \frac{1}{2}\mathrm{Si}(2kL)\right] - 2\sin^4\left(\frac{kL}{2}\right)}{(kL)^2}, \quad k = \frac{\omega}{\sqrt{2}\,y\,v_T} \tag{623}$$

$$\Im\left(\frac{1}{K_L^t}\right) = -\frac{\sqrt{2}\,r\,y^3}{(1-r)^2y^4 + (1+r)y^2 + \frac{1}{4}}, \quad r = \left(\frac{\omega_p}{\omega}\right)^2 \tag{624}$$

$$V^2(\omega) \approx \frac{16k_BT_e}{\pi^2\epsilon_0 v_T}\left[\int_0^{y_{c,B}} \mathcal{I}_{\mathrm{I}}(y)\,dy + \int_{y_{c,B}}^{y_c} \mathcal{I}_{\mathrm{II}}(y)\,dy + \int_{y_c}^{\infty} \mathcal{I}_{\mathrm{III}}(y)\,dy\right]. \tag{625}$$

Region-wise Envelope Integrals

Region I: $0 < y < y_{c,B}$

$$I_{\mathrm{I}}^{(\mathrm{env})}(\omega) := \int_0^{y_{c,B}} \mathcal{I}_{\mathrm{I}}(y)\,dy, \tag{626}$$

$$I_{\mathrm{I}}^{(\mathrm{env})}(\omega) = -r\,\frac{2v_T}{\pi\,\omega a}\left[B_1\,I_1(y_{c,B}) - B_2\,I_2(y_{c,B})\right]. \tag{627}$$

Region II: $y_{c,B} < y < y_c$

$$I_{\mathrm{II}}^{(\mathrm{env})}(\omega) := \int_{y_{c,B}}^{y_c} \mathcal{I}_{\mathrm{II}}(y)\,dy, \tag{628}$$

$$I_{\mathrm{II}}^{(\mathrm{env})}(\omega) = -\sqrt{2}\,r\,\Big[-B_2\,I_5(y_{c,B},y_c) + B_1\,I_6(y_{c,B},y_c) + B_2C_2\,I_7(y_{c,B},y_c) \tag{629}$$

$$-B_1C_2\,I_8(y_{c,B},y_c) - B_2C_4\,I_9(y_{c,B},y_c) + B_1C_4\,I_{10}(y_{c,B},y_c)\Big]. \tag{630}$$

Region III: $y > y_c$

$$I_{\mathrm{III}}^{(\mathrm{env})}(\omega) := \int_{y_c}^{\infty} \mathcal{I}_{\mathrm{III}}(y)\,dy, \tag{631}$$

$$I_{\mathrm{III}}^{(\mathrm{env})}(\omega) = -\sqrt{2}\,r\left[D_2\,I_3^{(\mathrm{III})}(y_c) - (D_4 + C_2D_2)\,I_4^{(\mathrm{III})}(y_c)\right]. \tag{632}$$

Specific Integral Equations

Region I integrals I_1 and I_2

For $r \neq 1$,

$$I_1^{(\mathrm{I})}(y_{c,B}; r\neq 1) = \frac{1}{4(1-r)^2}\left[\ln\left|\,4(1-r)^2y_{c,B}^4 + 4(1+r)y_{c,B}^2 + 1\,\right| + \frac{1+r}{\sqrt{r}}\ \mathrm{artanh}\big(\xi(r,y_{c,B})\big)\right], \tag{633}$$

$$\xi(r,y_{c,B}) = -\frac{4\sqrt{r}\,y_{c,B}^2}{2(r+1)y_{c,B}^2 + 1}. \tag{634}$$

For $r = 1$,

$$I_1^{(\mathrm{I})}(y_{c,B}; r = 1) = \frac{y_{c,B}^2}{4} - \frac{1}{32} \ln\bigl(8y_{c,B}^2 + 1\bigr). \tag{635}$$

For $r \neq 1$,

$$I_2^{(\mathrm{I})}(y_{c,B}; r \neq 1) = \frac{1}{(1-r)^2} \Biggl\{ y_{c,B} + \frac{(\sqrt{r}-1)^2}{8\sqrt{r}(\sqrt{r}+1)^2 \sqrt{-\lambda_1(r)}} \arctan\Biggl(\frac{y_{c,B}}{\sqrt{-\lambda_1(r)}} \Biggr) \tag{636}$$

$$- \frac{(\sqrt{r}+1)^2}{8\sqrt{r}(\sqrt{r}-1)^2 \sqrt{-\lambda_2(r)}} \arctan\Biggl(\frac{y_{c,B}}{\sqrt{-\lambda_2(r)}} \Biggr) \Biggr\}, \tag{637}$$

and for $r = 1$,

$$I_2^{(\mathrm{I})}(y_{c,B}; r = 1) = \frac{y_{c,B}^3}{6} - \frac{y_{c,B}}{16} + \frac{1}{16\sqrt{8}} \arctan\bigl(\sqrt{8}\, y_{c,B}\bigr). \tag{638}$$

Region III integrals $I_3^{(\mathrm{III})}$ and $I_4^{(\mathrm{III})}$

For $r \neq 1$,

$$\begin{aligned} I_3^{(\mathrm{III})}(y_c; r \neq 1) &= \ln\Biggl(\frac{(1-r)^2 y_c^4 + (1+r) y_c^2 + \frac{1}{4}}{(1-r)^2} \Biggr) - 4 \ln y_c \\ &\quad + \frac{1+r}{2\sqrt{r}} \ln\Biggl(\frac{2(1-r)^2 y_c^2 + (1+r) - 2\sqrt{r}}{2(1-r)^2 y_c^2 + (1+r) + 2\sqrt{r}} \Biggr). \end{aligned} \tag{639}$$

For $r = 1$,

$$I_3^{(\mathrm{III})}(y_c; r = 1) = 2 \ln\Biggl(\frac{8y_c^2 + 1}{8y_c^2} \Biggr). \tag{640}$$

For $r \neq 1$,

$$\begin{aligned} I_4^{(\mathrm{III})}(y_c; r \neq 1) &= 4(1+r) \ln\bigl((1-r)^2\bigr) + 8(1+r) \ln(y_c^2) + \frac{2}{y_c^2} \\ &\quad - 4(1+r) \ln\Bigl((1-r)^2 y_c^4 + (1+r) y_c^2 + \tfrac{1}{4}\Bigr) \\ &\quad + \frac{2(1+6r+r^2)}{\sqrt{r}} \operatorname{artanh}\Biggl(\frac{2(1-r)^2 y_c^2 + (1+r)}{2\sqrt{r}} \Biggr), \end{aligned} \tag{641}$$

where, for real-valued MATLAB plotting when the argument exceeds unity, use the real continuation

$$\operatorname{artanh}(Z) \;\mapsto\; \begin{cases} \operatorname{artanh}(Z), & |Z| < 1, \\ \dfrac{1}{2} \ln\biggl| \dfrac{Z+1}{Z-1} \biggr|, & |Z| > 1, \end{cases} \qquad Z = \frac{2(1-r)^2 y_c^2 + (1+r)}{2\sqrt{r}}.$$

For $r = 1$,

$$I_4^{(\mathrm{III})}(y_c; r = 1) = 16 \ln\Biggl(\frac{8y_c^2}{8y_c^2 + 1} \Biggr) + \frac{2}{y_c^2}. \tag{642}$$

Region II integrals I_5–I_{10}

I_5 **on** $[y_{c,B}, y_c]$ For $r \neq 1$,

$$I_5(y_{c,B}, y_c; r \neq 1) = \frac{1}{4(1-r)^2} \ln\left(\frac{(1-r)^2 y_c^4 + (1+r) y_c^2 + \frac{1}{4}}{(1-r)^2 y_{c,B}^4 + (1+r) y_{c,B}^2 + \frac{1}{4}} \right)$$
$$- \frac{1+r}{8(1-r)^2\sqrt{r}} \ln\left(\frac{\left(2(1-r)^2 y_c^2 + (1+r) - 2\sqrt{r}\right)\left(2(1-r)^2 y_{c,B}^2 + (1+r) + 2\sqrt{r}\right)}{\left(2(1-r)^2 y_c^2 + (1+r) + 2\sqrt{r}\right)\left(2(1-r)^2 y_{c,B}^2 + (1+r) - 2\sqrt{r}\right)} \right). \tag{643}$$

For $r = 1$,

$$I_5(y_{c,B}, y_c; r = 1) = \frac{1}{4}(y_c^2 - y_{c,B}^2) - \frac{1}{32} \ln\left(\frac{8y_c^2 + 1}{8y_{c,B}^2 + 1} \right). \tag{644}$$

I_6 **on** $[y_{c,B}, y_c]$ For $r \neq 1$, let

$$u_{1,2} = \frac{-(1+r) \pm 2\sqrt{r}}{2(1-r)^2}, \qquad a_j := \sqrt{-u_j} > 0, \quad j = 1, 2. \tag{645}$$

Then

$$\begin{aligned} I_6(y_{c,B}, y_c; r \neq 1) = &- \frac{a_1}{2\sqrt{r}} \left[\arctan\left(\frac{y_c}{a_1}\right) - \arctan\left(\frac{y_{c,B}}{a_1}\right) \right] \\ &+ \frac{a_2}{2\sqrt{r}} \left[\arctan\left(\frac{y_c}{a_2}\right) - \arctan\left(\frac{y_{c,B}}{a_2}\right) \right]. \end{aligned} \tag{646}$$

For $r = 1$,

$$I_6(y_{c,B}, y_c; r = 1) = \frac{1}{2}(y_c - y_{c,B}) - \frac{\sqrt{2}}{8} \left[\arctan(2\sqrt{2}\, y_c) - \arctan(2\sqrt{2}\, y_{c,B}) \right]. \tag{647}$$

I_7 **on** $[y_{c,B}, y_c]$ For $r \neq 1$, define

$$N(y) := 2(1-r)^2 y^2 + (1+r). \tag{648}$$

Then

$$I_7(y_{c,B}, y_c; r \neq 1) = \frac{1}{2\sqrt{r}} \left[\operatorname{artanh}\left(\frac{2\sqrt{r}}{N(y_{c,B})} \right) - \operatorname{artanh}\left(\frac{2\sqrt{r}}{N(y_c)} \right) \right]. \tag{649}$$

Equivalently,

$$I_7(y_{c,B}, y_c; r \neq 1) = \frac{1}{4\sqrt{r}} \ln\left(\frac{\left(N(y_c) - 2\sqrt{r}\right)\left(N(y_{c,B}) + 2\sqrt{r}\right)}{\left(N(y_c) + 2\sqrt{r}\right)\left(N(y_{c,B}) - 2\sqrt{r}\right)} \right). \tag{650}$$

For $r = 1$,

$$I_7(y_{c,B}, y_c; r = 1) = \frac{1}{4} \ln\left(\frac{8y_c^2 + 1}{8y_{c,B}^2 + 1} \right), \qquad 0 < y_{c,B} < y_c. \tag{651}$$

I_8 **on** $[y_{c,B}, y_c]$ For $r \neq 1$,

$$\begin{aligned} I_8(y_{c,B}, y_c; r \neq 1) = \frac{1}{2\sqrt{r}} \Bigg\{ &\frac{1}{a_1} \left[\arctan\left(\frac{y_c}{a_1}\right) - \arctan\left(\frac{y_{c,B}}{a_1}\right) \right] \\ &- \frac{1}{a_2} \left[\arctan\left(\frac{y_c}{a_2}\right) - \arctan\left(\frac{y_{c,B}}{a_2}\right) \right] \Bigg\}, \end{aligned} \tag{652}$$

where a_1, a_2 are as above.

For $r = 1$,

$$I_8(y_{c,B}, y_c; r = 1) = \sqrt{2} \left[\arctan(2\sqrt{2}\, y_c) - \arctan(2\sqrt{2}\, y_{c,B}) \right]. \tag{653}$$

I_9 **on** $[y_{c,B}, y_c]$ For $r \neq 1$, define

$$D(y) := (1-r)^2 y^4 + (1+r) y^2 + \tfrac{1}{4}, \qquad N(y) := 2(1-r)^2 y^2 + (1+r). \tag{654}$$

A convenient log-only form is

$$\begin{aligned} I_9(y_{c,B}, y_c; r \neq 1) = 4 \ln\left(\frac{y_c}{y_{c,B}}\right) + \ln\left(\frac{D(y_{c,B})}{D(y_c)}\right) \\ - \frac{1+r}{2\sqrt{r}} \ln\left(\frac{\big(N(y_c) - 2\sqrt{r}\big)\big(N(y_{c,B}) + 2\sqrt{r}\big)}{\big(N(y_c) + 2\sqrt{r}\big)\big(N(y_{c,B}) - 2\sqrt{r}\big)}\right). \end{aligned} \tag{655}$$

Equivalently, in artanh form,

$$\begin{aligned} I_9(y_{c,B}, y_c; r \neq 1) = 4 \ln\left(\frac{y_c}{y_{c,B}}\right) + \ln\left(\frac{D(y_{c,B})}{D(y_c)}\right) \\ + \frac{1+r}{\sqrt{r}} \left[\operatorname{artanh}\left(\frac{2\sqrt{r}}{N(y_c)}\right) - \operatorname{artanh}\left(\frac{2\sqrt{r}}{N(y_{c,B})}\right)\right]. \end{aligned} \tag{656}$$

For $r = 1$,

$$I_9(y_{c,B}, y_c; r = 1) = 2 \ln\left(\frac{y_c^2 (8 y_{c,B}^2 + 1)}{y_{c,B}^2 (8 y_c^2 + 1)}\right), \qquad 0 < y_{c,B} < y_c. \tag{657}$$

I_{10} **on** $[y_{c,B}, y_c]$ For $r \neq 1$, with $u_{1,2}, a_j$ as above,

$$\begin{aligned} I_{10}(y_{c,B}, y_c; r \neq 1) = 4 \left(\frac{1}{y_{c,B}} - \frac{1}{y_c}\right) \\ - \frac{1}{2\sqrt{r}\, a_1^3} \left[\arctan\left(\frac{y_c}{a_1}\right) - \arctan\left(\frac{y_{c,B}}{a_1}\right)\right] \\ + \frac{1}{2\sqrt{r}\, a_2^3} \left[\arctan\left(\frac{y_c}{a_2}\right) - \arctan\left(\frac{y_{c,B}}{a_2}\right)\right]. \end{aligned} \tag{658}$$

For $r = 1$,

$$\begin{aligned} I_{10}(y_{c,B}, y_c; r = 1) = 4 \left(\frac{1}{y_{c,B}} - \frac{1}{y_c}\right) - 8\sqrt{2} \left[\arctan(2\sqrt{2}\, y_c) - \arctan(2\sqrt{2}\, y_{c,B})\right], \\ 0 < y_{c,B} < y_c. \end{aligned} \tag{659}$$

Variables

$$B_1 := \frac{\pi \sqrt{2}\, v_T}{4 \omega L}, \qquad B_2 := \frac{3 v_T^2}{2 \omega^2 L^2}, \tag{660}$$

$$D_2 := \frac{\omega^2 L^2}{48 v_T^2}, \qquad D_4 := \frac{\omega^4 L^4}{960 v_T^4}, \qquad D_6 := \frac{\omega^6 L^6}{35840 v_T^6}, \tag{661}$$

$$C_2 := \frac{\omega^2 a^2}{4 v_T^2}, \qquad C_4 := \frac{3 \omega^4 a^4}{128 v_T^4}, \tag{662}$$

$$E_B := \frac{2\sqrt{2}\, v_T}{\pi\, \omega a}. \tag{663}$$

$$r = \left(\frac{\omega_p}{\omega}\right)^2, \tag{664}$$

$$\lambda_1(r) = \frac{-(1+r)+2\sqrt{r}}{2(1-r)^2}, \qquad \lambda_2(r) = \frac{-(1+r)-2\sqrt{r}}{2(1-r)^2}, \tag{665}$$

$$u_{1,2}(r) = \frac{-(1+r)\pm 2\sqrt{r}}{2(1-r)^2}, \qquad a_j(r) = \sqrt{-u_j(r)} > 0, \quad j = 1, 2. \tag{666}$$

Cutoffs and Constants

$$y_{c,B} = \frac{\omega a}{\sqrt{2}\, v_T\, x_{c,B}}, \qquad 3\pi x_{c,B}^5 - 16\pi x_{c,B}^3 + 32\pi x_{c,B} - 32 = 0, \qquad x_{c,B} \approx 0.33704639, \tag{667}$$

$$y_c = \frac{\omega L}{\sqrt{2}\, v_T\, x_c}, \qquad x_c^6 - 10x_c^4 + 60\pi x_c - 180 = 0, \qquad x_c \approx 1.00326820. \tag{668}$$

Additional Empirical Equations

$$v_T = \sqrt{\frac{k_B T_e}{m_e}}. \tag{669}$$

$$n_e = 5.77 \times 10^{11}\,\mathrm{m}^{-3}, \tag{670}$$
$$T_e = 1700\,\mathrm{K}, \tag{671}$$
$$a = 2 \times 10^{-4}\,\mathrm{m}, \tag{672}$$
$$k_B = 1.380649 \times 10^{-23}\,\mathrm{J\,K^{-1}}, \tag{673}$$
$$\epsilon_0 = 8.8541878128 \times 10^{-12}\,\mathrm{F\,m^{-1}}, \tag{674}$$
$$L \in [0.1,\ 10]\,\mathrm{m}. \tag{675}$$

D MATLAB Code Implementation

D.1 Stable evaluation of logarithms and arctangents

Catastrophic cancellation at large y_c. For frequencies $f \gtrsim 10^7$ Hz (especially when $L = 3$ m), the cutoff $y_c \propto \omega L$ becomes large (typically $y_c \sim 10^3$ or more). In this regime, several closed-form expressions contain differences of large, nearly equal logarithmic terms whose true remainder is small. For example, in Region III the analytic form for I_3 contains the structure

$$\ln\left(\frac{D}{(1-r)^2}\right) - 4\ln y_c, \tag{676}$$

where both terms scale as $4\ln y_c$ for large y_c. In double precision, evaluating them separately and subtracting causes loss of significant digits (catastrophic cancellation), leading to noisy values, sign flips, and spurious oscillations in the analytic spectrum. The issue is amplified for larger L because y_c is larger at the same frequency.

Physical origin of the instability. The numerical instability arises from the mathematical structure of the Region III integrals, which describe the long-wavelength (small-k, large-y) tail of the QTN spectrum. In this asymptotic regime, the integrand decays as y^{-1} and y^{-3}, and the definite integrals from y_c to ∞ naturally involve logarithmic terms that grow with y_c. The closed-form antiderivatives, when evaluated at the limits, produce expressions like

$$[\ln(D(y)) - 4\ln y]_{y_c}^{\infty} = \lim_{y\to\infty} [\ln(D(y)) - 4\ln y] - [\ln(D(y_c)) - 4\ln y_c]\,, \tag{677}$$

where $D(y) = (1-r)^2y^4 + (1+r)y^2 + 1/4$. For large y, $D(y) \approx (1-r)^2y^4$, so $\ln(D(y)) - 4\ln y \to \ln(1-r)^2$, which is independent of y. However, the lower-limit evaluation $\ln(D(y_c)) - 4\ln y_c$ involves subtracting two terms that are both $O(\ln y_c)$ but nearly equal when $y_c \gg 1$. This is the textbook definition of catastrophic cancellation: the true value is small (of order unity or smaller), but the intermediate quantities being subtracted are both $O(\ln y_c) \sim 7$ for $y_c = 10^3$.

Why only I_3, I_4, and I_9 require stabilization. Not all integrals suffer from this issue. The basis integrals fall into three categories:

1. **Region I integrals (I_1, I_2):** These are definite integrals from 0 to $y_{c,B}$. Since $y_{c,B} \ll y_c$ (typically $y_{c,B} \sim 0.1$ even at high frequencies because $a \ll L$), the logarithmic terms remain small and cancellation is not significant. The expressions are numerically stable as written.
2. **Region II integrals (I_5–I_{10}):** These are finite-range integrals from $y_{c,B}$ to y_c. The closed forms involve differences of arctangents or logarithmic ratios evaluated at two finite limits. While both limits grow with frequency, the *differences* (e.g., $\arctan(y_c/a) - \arctan(y_{c,B}/a)$) remain well-conditioned because arctangent saturates at $\pi/2$. For the logarithmic terms, expressions like

$$\ln\left(\frac{D(y_c)}{D(y_{c,B})}\right) \tag{678}$$

   do not involve large cancellations because the numerator and denominator scale similarly. However, I_9 is an exception: it includes a term $4\ln(y_c/y_{c,B}) + \ln(D(y_{c,B})/D(y_c))$, where the two logarithms are constructed to cancel the dominant y^4 contribution from $D(y)$, leaving only the subleading terms. This cancellation is prone to precision loss when both y_c and $y_{c,B}$ are large.
3. **Region III integrals (I_3, I_4):** These are improper integrals from y_c to ∞. The upper-limit evaluation at $y \to \infty$ yields simple constants (e.g., $\ln(1-r)^2$), but the lower-limit evaluation at y_c produces logarithmic terms that grow with y_c. The final expression subtracts these large terms, causing catastrophic cancellation when $y_c \gg 1$.

In summary, only the improper integrals (I_3, I_4) and the special finite-range integral I_9 (which contains an implicit large-y cancellation structure) require algebraic reorganization.

Stabilization principle. The remedy is to algebraically combine cancelling terms *before* evaluation and to use numerically stable log transforms (notably $\ln(1+x)$ for small x). Accordingly, the implementation rewrites the problematic logarithmic groups into a form involving

$$\ln\big(1+t(y)\big) \quad \text{evaluated via} \quad \texttt{log1p}(t(y)), \tag{679}$$

which preserves precision when $t(y) \ll 1$ (the large-y regime). The key algebraic insight is that

$$\ln(D(y)) - 4\ln y = \ln\left(\frac{D(y)}{y^4}\right) = \ln\left((1-r)^2 + \frac{1+r}{y^2} + \frac{1}{4y^4}\right) = \ln\left(1 + \frac{1+r}{(1-r)^2 y^2} + \frac{1}{4(1-r)^2 y^4}\right) + \ln(1-r)^2, \tag{680}$$

so the large terms $\ln y^4$ and $\ln((1-r)^2 y^4)$ cancel *analytically*, leaving only a small argument for the logarithm when $y \gg 1$.

(a) `compute_I3`: stable evaluation of Region III logarithms. The original `compute_I3` formula subtracts $4\ln y_c$ from a large log term. In the updated form, these are combined to yield a single logarithm of a quantity close to unity:

$$\ln\left(\frac{D}{(1-r)^2}\right) - 4\ln y_c = \ln\left(\frac{D}{(1-r)^2 y_c^4}\right) = \ln\left(1 + \frac{1+r}{(1-r)^2 y_c^2} + \frac{1}{4(1-r)^2 y_c^4}\right), \tag{681}$$

which is evaluated using `log1p` to maintain accuracy for large y_c. The remaining log-ratio term is also rewritten in a stable $\ln(1+\cdots)$ form using the identity

$$\ln\left(\frac{N-2\sqrt{r}}{N+2\sqrt{r}}\right) = \ln\left(1 - \frac{4\sqrt{r}}{N+2\sqrt{r}}\right), \tag{682}$$

where $N = 2(1-r)^2 y_c^2 + (1+r)$. For large y_c, $N \gg 2\sqrt{r}$, so the argument of `log1p` is small.

(b) `compute_I4`: stable cancellation of large log groups. The original `compute_I4` contained multiple large log terms (including $\ln\big((1-r)^2\big)$, $\ln(y_c^2)$, and $\ln(D)$) that nearly cancel at high y_c. These terms are algebraically merged into a single $\ln(1+t)$ structure, eliminating the subtraction of large numbers:

$$\begin{aligned} &4(1+r)\ln\big((1-r)^2\big) + 8(1+r)\ln(y_c^2) - 4(1+r)\ln(D) \\ &\quad = -4(1+r)\ln\left(\frac{D}{(1-r)^2 y_c^4}\right) = -4(1+r)\ln(1+t), \end{aligned} \tag{683}$$

where $t = \frac{1+r}{(1-r)^2 y_c^2} + \frac{1}{4(1-r)^2 y_c^4}$. This is again evaluated via `log1p`. For the $r=1$ degenerate case, an additional stable representation prevents cancellation in $\ln\big((8y_c^2)/(8y_c^2+1)\big)$ by rewriting it as

$$\ln\left(\frac{8y_c^2}{8y_c^2+1}\right) = -\ln\left(1 + \frac{1}{8y_c^2}\right), \tag{684}$$

which is evaluated as $-\texttt{log1p}(1/(8y_c^2))$. When y_c is extremely large ($y_c \gtrsim 10^6$), even $1/(8y_c^2)$ may underflow; in this case, the asymptotic expansion $\ln(1+x) \approx x - x^2/2$ can be used, but this is typically unnecessary for ionospheric frequencies.

(c) `compute_I9`: stable difference of close log terms across limits. The original `compute_I9` includes a term

$$4\ln\left(\frac{y_c}{y_{c,B}}\right) + \ln\left(\frac{D(y_{c,B})}{D(y_c)}\right), \tag{685}$$

which also experiences cancellation when both cutoffs are large. The updated version rescales $D(y)$ by $(1-r)^2y^4$ so that the large contributions cancel analytically *before* evaluation, giving

$$\ln\bigl(1+t(y_{c,B})\bigr)-\ln\bigl(1+t(y_c)\bigr), \tag{686}$$

where $t(y)=\frac{1+r}{(1-r)^2y^2}+\frac{1}{4(1-r)^2y^4}$. Both logarithms are computed using `log1p`. The remaining log-ratio terms involving $N(y)$ are similarly rewritten into stable `log1p` forms. The key is that each `log1p` argument is $O(1/y^2)$ or smaller, so no large intermediate values are generated.

Practical outcome. After these stabilizations, the analytic curve no longer exhibits non-physical oscillations or spikes at high frequency. The improved agreement is not a change in the underlying physics or asymptotic model; it is purely a correction to the numerical evaluation of the closed forms in a regime where straightforward double-precision arithmetic loses accuracy due to cancellation. Figure 1 shows that the stabilized implementation produces smooth, monotonic agreement with the numerical quadrature out to 10^8 Hz, confirming that the algebraic reorganization preserves the mathematical correctness of the solution while dramatically improving numerical robustness.

E Explicit MATLAB Code

Listing 1: Analytical and Numerical Plotting Function with Nested Numerical Evaluation

```matlab
clear; clc; close all;

k_B = 1.380649e-23;
epsilon_0 = 8.8541878128e-12;
m_e = 9.10938356e-31;
n_e = 5.77e11;
T_e = 1700;
a = 2e-4;
L_values = [0.3, 3.0];

v_T = sqrt(k_B * T_e / m_e);
omega_p = sqrt(n_e * (1.602176634e-19)^2 / (epsilon_0 * m_e));

x_cB = 0.33704639;
x_c = 1.00326820;

f_vec = logspace(log10(1e6), log10(5e7), 100);
omega_vec = 2*pi*f_vec;

V2_numeric_all = zeros(length(L_values), length(omega_vec));
V2_analytic_all = zeros(length(L_values), length(omega_vec));

for L_idx = 1:length(L_values)
    L = L_values(L_idx);
    fprintf('\n=== Computing for L = %.1f m ===\n', L);

    V2_numeric = zeros(size(omega_vec));
    V2_analytic = zeros(size(omega_vec));

    fprintf('Computing solutions for %d frequency points...\n', length(omega_vec));

    for idx = 1:length(omega_vec)
        omega = omega_vec(idx);
        r = (omega_p/omega)^2;

        y_cB = (omega*a)/(sqrt(2)*v_T*x_cB);
        y_c = (omega*L)/(sqrt(2)*v_T*x_c);

        B1 = (pi*sqrt(2)*v_T)/(4*omega*L);
        B2 = (3*v_T^2)/(2*omega^2*L^2);
        C2 = (omega^2*a^2)/(4*v_T^2);
        C4 = (3*omega^4*a^4)/(128*v_T^4);
        D2 = (omega^2*L^2)/(48*v_T^2);
        D4 = (omega^4*L^4)/(960*v_T^4);

        y_max = max(50, 5*y_c);

        try
            integrand = @(y) compute_integrand_vectorized(y, omega, omega_p, v_T, a, L);
            V2_numeric(idx) = (16*k_B*T_e)/(pi^2*epsilon_0*v_T) * ...
                quadgk(integrand, 1e-6, y_max, 'RelTol', 1e-4, 'AbsTol', 1e-10, ...
```

```
                       'Waypoints', [y_cB, y_c]);
        catch
            warning('Numeric integration failed at omega/omega_p = %.3f', omega/omega_p);
            V2_numeric(idx) = NaN;
        end

        I_I = compute_region_I(y_cB, r, B1, B2);
        I_II = compute_region_II(y_cB, y_c, r, B1, B2, C2, C4);
        I_III = compute_region_III(y_c, r, D2, D4, C2);

        V2_analytic(idx) = (16*k_B*T_e)/(pi^2*epsilon_0*v_T) * (I_I + I_II + I_III);

        if mod(idx, 10) == 0
            fprintf('  Progress: %d/%d completed\n', idx, length(omega_vec));
        end
    end

    V2_numeric_all(L_idx, :) = V2_numeric;
    V2_analytic_all(L_idx, :) = V2_analytic;

    fprintf('Computation complete for L = %.1f m!\n', L);
end

figure('Position', [100 100 900 700]);

colors = {'b', 'm'};

loglog(f_vec, abs(V2_numeric_all(1, :)), '-', ...
       'Color', colors{1}, 'LineWidth', 2.5, ...
       'DisplayName', sprintf('Numeric (L=%.1f m)', L_values(1)));
hold on;
loglog(f_vec, abs(V2_analytic_all(1, :)), '--', ...
       'Color', colors{1}, 'LineWidth', 2.5, ...
       'DisplayName', sprintf('Analytic (L=%.1f m)', L_values(1)));
loglog(f_vec, abs(V2_numeric_all(2, :)), '-', ...
       'Color', colors{2}, 'LineWidth', 2.5, ...
       'DisplayName', sprintf('Numeric (L=%.1f m)', L_values(2)));
loglog(f_vec, abs(V2_analytic_all(2, :)), '--', ...
       'Color', colors{2}, 'LineWidth', 2.5, ...
       'DisplayName', sprintf('Analytic (L=%.1f m)', L_values(2)));
hold off;

grid on;
xlabel('Frequency (Hz)', 'FontSize', 14, 'Interpreter', 'tex');
ylabel('|V^2(\omega)| [V^2]', 'FontSize', 14);
title('Numeric vs Analytical Solution Comparison', 'FontSize', 16);
legend('Location', 'best', 'FontSize', 12);
xlim([1e6 5e7]);
ylim([1e-19 1e-13]);
set(gca, 'FontSize', 12);

figure('Position', [150 150 900 700]);

rel_error_1 = abs(V2_numeric_all(1, :) - V2_analytic_all(1, :)) ./ abs(V2_numeric_all(1, :));
```

```
semilogx(f_vec, rel_error_1 * 100, '-', ...
         'Color', colors{1}, 'LineWidth', 2, ...
         'DisplayName', sprintf('L = %.1f m', L_values(1)));
hold on;
rel_error_2 = abs(V2_numeric_all(2, :) - V2_analytic_all(2, :)) ./ abs(V2_numeric_all(2, :));
semilogx(f_vec, rel_error_2 * 100, '-', ...
         'Color', colors{2}, 'LineWidth', 2, ...
         'DisplayName', sprintf('L = %.1f m', L_values(2)));
hold off;

grid on;
xlabel('Frequency (Hz)', 'FontSize', 14, 'Interpreter', 'tex');
ylabel('Relative Error [%]', 'FontSize', 14);
title('Relative Error: |Numeric - Analytic| / |Numeric|', 'FontSize', 14);
legend('Location', 'best', 'FontSize', 12);
set(gca, 'FontSize', 12);


function val = compute_integrand_vectorized(y, omega, omega_p, v_T, a, L)
    k = omega./(sqrt(2).*y.*v_T);
    r = (omega_p/omega)^2;

    kL = k.*L;
    Si_kL = sinint(kL);
    Si_2kL = sinint(2*kL);

    F1 = (kL.*(Si_kL - 0.5.*Si_2kL) - 2.*sin(kL./2).^4) ./ (kL).^2;
    J0_ka = besselj(0, k.*a);
    Im_inv_kL = -sqrt(2).*r.*y.^3 ./ ((1-r)^2.*y.^4 + (1+r).*y.^2 + 0.25);

    val = (1./y.^2) .* F1 .* J0_ka.^2 .* Im_inv_kL;
end

function I = compute_region_I(y_cB, r, B1, B2)
    I1 = compute_I1(y_cB, r);
    I2 = compute_I2(y_cB, r);
    I = -r * (B1*I1 - B2*I2);
end

function I = compute_region_II(y_cB, y_c, r, B1, B2, C2, C4)
    I5 = compute_I5(y_cB, y_c, r);
    I6 = compute_I6(y_cB, y_c, r);
    I7 = compute_I7(y_cB, y_c, r);
    I8 = compute_I8(y_cB, y_c, r);
    I9 = compute_I9(y_cB, y_c, r);
    I10 = compute_I10(y_cB, y_c, r);

    I = -sqrt(2)*r * (-B2*I5 + B1*I6 + B2*C2*I7 - B1*C2*I8 - B2*C4*I9 + B1*C4*I10);
end

function I = compute_region_III(y_c, r, D2, D4, C2)
    I3 = compute_I3(y_c, r);
    I4 = compute_I4(y_c, r);
```

```
    I = -sqrt(2)*r * (D2*I3 - (D4 + C2*D2)*I4);
end

function val = compute_I1(y_cB, r)
    if abs(r - 1) < 1e-10
        val = y_cB^2/4 - (1/32)*log(8*y_cB^2 + 1);
    else
        xi = -4*sqrt(r)*y_cB^2 / (2*(r+1)*y_cB^2 + 1);
        val = (1/(4*(1-r)^2)) * (log(abs(4*(1-r)^2*y_cB^4 + 4*(1+r)*
            y_cB^2 + 1)) + ...
              ((1+r)/sqrt(r)) * atanh(xi));
    end
end

function val = compute_I2(y_cB, r)
    if abs(r - 1) < 1e-10
        val = y_cB^3/6 - y_cB/16 + (1/(16*sqrt(8)))*atan(sqrt(8)*y_cB);
    else
        lambda1 = (-(1+r) + 2*sqrt(r))/(2*(1-r)^2);
        lambda2 = (-(1+r) - 2*sqrt(r))/(2*(1-r)^2);
        val = (1/(1-r)^2) * (y_cB + ...
              ((sqrt(r)-1)^2/(8*sqrt(r)*(sqrt(r)+1)^2*sqrt(-lambda1)))*
                 atan(y_cB/sqrt(-lambda1)) - ...
              ((sqrt(r)+1)^2/(8*sqrt(r)*(sqrt(r)-1)^2*sqrt(-lambda2)))*
                 atan(y_cB/sqrt(-lambda2)));
    end
end

function val = compute_I3(y_c, r)
    if abs(r - 1) < 1e-10
        t = 1./(8*y_c.^2);
        val = 2*log1p(t);
        return;
    end

    E = (1-r)^2;   F = 1+r;   G = 0.25;
    s = sqrt(r);

    t = F./(E*y_c.^2) + G./(E*y_c.^4);
    termA = log1p(t);

    N = 2*E*y_c.^2 + F;

    termB = (F./(2*s)) .* log1p( -4*s./(N + 2*s) );

    val = termA + termB;
end


function val = compute_I4(y_c, r)
    if abs(r - 1) < 1e-10
        t = 1./(8*y_c.^2);

        if t < 1e-6
            val = 1./(8*y_c.^4) - 1./(12*y_c.^6);
        else
            val = -16*log1p(t) + 2./y_c.^2;
        end
```

```
        return;
    end

    E = (1-r)^2;  F = 1+r;  G = 0.25;
    s = sqrt(r);

    t = F./(E*y_c.^2) + G./(E*y_c.^4);
    logpart = -4*F .* log1p(t);

    N = 2*E*y_c.^2 + F;
    Z = N./(2*s);

    artanh_val = 0.5 * log1p( 2./(Z - 1) );

    val = 2./y_c.^2 + logpart + (2*(1+6*r+r.^2)./s).*artanh_val;
end


function val = compute_I5(y_cB, y_c, r)
    if abs(r - 1) < 1e-10
        val = 0.25*(y_c^2 - y_cB^2) - (1/32)*log((8*y_c^2+1)/(8*y_cB^2+1));
    else
        D_c = (1-r)^2*y_c^4 + (1+r)*y_c^2 + 0.25;
        D_cB = (1-r)^2*y_cB^4 + (1+r)*y_cB^2 + 0.25;
        N_c = 2*(1-r)^2*y_c^2 + (1+r);
        N_cB = 2*(1-r)^2*y_cB^2 + (1+r);
        val = (1/(4*(1-r)^2))*log(D_c/D_cB) - ...
              ((1+r)/(8*(1-r)^2*sqrt(r)))*log(((N_c-2*sqrt(r))*(N_cB+2*sqrt(r)))/((N_c+2*sqrt(r))*(N_cB-2*sqrt(r))));
    end
end

function val = compute_I6(y_cB, y_c, r)
    if abs(r - 1) < 1e-10
        val = 0.5*(y_c - y_cB) - (sqrt(2)/8)*(atan(2*sqrt(2)*y_c) - atan(2*sqrt(2)*y_cB));
    else
        u1 = (-(1+r) + 2*sqrt(r))/(2*(1-r)^2);
        u2 = (-(1+r) - 2*sqrt(r))/(2*(1-r)^2);
        a1 = sqrt(-u1);
        a2 = sqrt(-u2);
        val = -(a1/(2*sqrt(r)))*(atan(y_c/a1) - atan(y_cB/a1)) + ...
               (a2/(2*sqrt(r)))*(atan(y_c/a2) - atan(y_cB/a2));
    end
end

function val = compute_I7(y_cB, y_c, r)
    if abs(r - 1) < 1e-10
        val = 0.25*log((8*y_c^2+1)/(8*y_cB^2+1));
    else
        N_c = 2*(1-r)^2*y_c^2 + (1+r);
        N_cB = 2*(1-r)^2*y_cB^2 + (1+r);
        val = (1/(4*sqrt(r)))*log(((N_c-2*sqrt(r))*(N_cB+2*sqrt(r)))/((N_c+2*sqrt(r))*(N_cB-2*sqrt(r))));
    end
end
```

```
function val = compute_I8(y_cB, y_c, r)
    if abs(r - 1) < 1e-10
        val = sqrt(2)*(atan(2*sqrt(2)*y_c) - atan(2*sqrt(2)*y_cB));
    else
        u1 = (-(1+r) + 2*sqrt(r))/(2*(1-r)^2);
        u2 = (-(1+r) - 2*sqrt(r))/(2*(1-r)^2);
        a1 = sqrt(-u1);
        a2 = sqrt(-u2);
        val = (1/(2*sqrt(r)))*(1/a1*(atan(y_c/a1)-atan(y_cB/a1)) - ...
                               1/a2*(atan(y_c/a2)-atan(y_cB/a2)));
    end
end

function val = compute_I9(y_cB, y_c, r)
    if abs(r - 1) < 1e-10
        val = 2*log((y_c.^2.*(8*y_cB.^2+1))./(y_cB.^2.*(8*y_c.^2+1)));
        return;
    end

    E = (1-r)^2;  F = 1+r;  G = 0.25;
    s = sqrt(r);

    tB = F./(E*y_cB.^2) + G./(E*y_cB.^4);
    t  = F./(E*y_c.^2)  + G./(E*y_c.^4);
    part1 = log1p(tB) - log1p(t);

    N  = 2*E*y_c.^2  + F;
    NB = 2*E*y_cB.^2 + F;

    part2 = log1p( -4*s./(N  + 2*s) ) + log1p(  4*s./(NB - 2*s) );

    val = part1 - (F./(2*s)).*part2;
end


function val = compute_I10(y_cB, y_c, r)
    if abs(r - 1) < 1e-10
        val = 4*(1/y_cB - 1/y_c) - 8*sqrt(2)*(atan(2*sqrt(2)*y_c) -
            atan(2*sqrt(2)*y_cB));
    else
        u1 = (-(1+r) + 2*sqrt(r))/(2*(1-r)^2);
        u2 = (-(1+r) - 2*sqrt(r))/(2*(1-r)^2);
        a1 = sqrt(-u1);
        a2 = sqrt(-u2);
        val = 4*(1/y_cB - 1/y_c) - ...
              (1/(2*sqrt(r)*a1^3))*(atan(y_c/a1) - atan(y_cB/a1)) + ...
              (1/(2*sqrt(r)*a2^3))*(atan(y_c/a2) - atan(y_cB/a2));
    end
end
```

References

[1] N. Meyer-Vernet. On natural noises detected by antennas in plasmas. *Journal of Geophysical Research*, 84(A9):5373–5377, 1979. doi: 10.1029/JA084iA09p05373.

[2] P. Couturier, S. Hoang, N. Meyer-Vernet, and J. L. Steinberg. Quasi-thermal noise in a stable plasma at rest: Theory and observations from ISEE 3. *Journal of Geophysical Research*, 86(A13):11127–11138, 1981. doi: 10.1029/JA086iA13p11127.

[3] M. Maksimovic, S. Hoang, N. Meyer-Vernet, M. Moncuquet, J.-L. Bougeret, J. L. Phillips, and P. Canu. Solar wind electron parameters from quasi-thermal noise spectroscopy and comparison with other measurements on Ulysses. *Journal of Geophysical Research*, 100 (A10):19881–19891, 1995. doi: 10.1029/95JA01550.

[4] K. Issautier, N. Meyer-Vernet, M. Moncuquet, S. Hoang, and D. J. McComas. Quasi-thermal noise in a drifting plasma: Theory and application to solar wind diagnostic on Ulysses. *Journal of Geophysical Research*, 104(A4):6691–6704, 1999. doi: 10.1029/1998JA900165.

[5] M. Moncuquet, A. Lecacheux, N. Meyer-Vernet, B. Cecconi, and W. S. Kurth. Quasi-thermal noise spectroscopy in the inner magnetosphere of Saturn with Cassini/RPWS: Electron temperatures and density. *Geophysical Research Letters*, 32:L20S02, 2005. doi: 10.1029/2005GL022508.

[6] G. Le Chat, K. Issautier, N. Meyer-Vernet, and S. Hoang. Large-scale variation of solar wind electron properties from quasi-thermal noise spectroscopy: Ulysses measurements. *Solar Physics*, 271(1–2):141–148, 2011. doi: 10.1007/s11207-011-9797-3.

[7] Ronald Maj and Iver H. Cairns. Quasi-thermal noise and shot noise spectroscopy on a CubeSat in Earth's ionosphere. *Journal of Geophysical Research: Space Physics*, 122:1–15, 2017. doi: 10.1002/2016JA023832.

[8] D. Bilitza. The International Reference Ionosphere—status 2013. *Advances in Space Research*, 55(8):1914–1927, 2015. doi: 10.1016/j.asr.2014.07.032.

[9] Burton D. Fried and Samuel D. Conte. *The Plasma Dispersion Function.* Academic Press, New York, 1961.

[10] Eugene Chon. Plasma dispersion function approximations for space physics applications. *Journal of Plasma Physics*, 86(3):905860301, 2020. (Note: This is a placeholder entry - please provide the correct citation for the Lorentzian/Padé approximation reference from Eugene Chon's SSP project mentioned by Prof. Cairns).

[11] M. M. Martinović, A. Zaslavsky, M. Maksimović, and S. Šegan. Electrostatic thermal noise in a weakly ionized collisional plasma. *Radio Science*, 52:70–77, 2017. doi: 10.1002/2016RS006189.

[12] H. H. Kuehl. Resistance of a short antenna in a warm plasma. *Radio Science*, 1:971–976, 1966. doi: 10.1002/rds196619971.